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Ministère de l'Enseignement Supérieur
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Faculté des sciences de Bizerte



Report of : GRADUATION PROJECT

In order to obtain :

SOFTWARE ENGINEERING AND INFORMATION SYSTEMS

Adoption web site

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SOUROUR MEDDEB

Appreciation

This project is a result of years of hard work and challenges that we faced, and we are really grateful for this amazing experience that taught us and developed us a lot as a person and a team.

We would love to begin with dedicating this work to our families that supported us throughout the whole journey and provided us with the best environment to be here today.

We want to thank Mr. Oussama Bejaoui for providing us with this experience to be part of INTUITIV GROUP team and to leave our print in this interesting project.

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Ilef kaabi
Ferdaous Ben Dhafer

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General introduction

Nowadays, technology plays a massive role in our everyday life. By 2024, more than 50 of user interactions will be augmented by AI-driven algorithms. 77 of society relies too much on technology to succeed, indicating an over-dependence on technology. Those numbers show the importance of digitalization and its usage.

Every year, countless animals find themselves in shelters or on the streets, wishing for a loving home and a second chance at happiness. Pet adoption is a heartwarming act that not only enriches the lives of the animals involved but also brings joy and fulfillment to their human companions. By adopting, not only do we provide a home, but we also contribute to reducing the population of stray animals.

Many people are interested in adopting a pet. Others are willing to give their pets for adoption. Yet, finding a secure platform to link between the adopters and rehoming may be difficult as some people use such space for their privilege.

In our enterprise “INTUITIV,” we care about the pets and invest in improving their lives. In our graduation project, we will treat this solution by building a website that connects potential adopters with people who need to rehome their pets.

In the following few pages of our internship report, we will discuss most of these main complex tasks to present and create a suitable solution.

Chapter1

WORKING CONTEXT

1.1 Introduction

This chapter includes a presentation of the company "INTUITIVE", a presentation of the project with the proposed problem and the proposed solutions. This chapter also contains a study of two existing websites in the same subject, we chose a principal application the most used one. We will then capture the functional needs of our solution and finally we will represent the methodology used for our project.

1.2 Company presentation

Intuitiv-technology is a company with a dual culture : both an INNOVATIVE ESN and an INTERACTIVE AGENCY, it bases its success on the reactivity, innovation, reliability, and quality of its services, the company has taken advantage of new technologies to offer ever more innovative solutions in development engineering, and the integration of all the solutions offered by the group.

The referencing of Intuitiv-Technology with large national and international groups reinforces our conviction of the quality and efficiency of service structures on a human scale.

Intuitiv-Technology stands out as much by its levels of expertise as by its human-sized structure, where real values can be expressed...

1.3 Project presentation

1.3.1 Project framework :

The project concerns the design and implementation of an Adoption website dedicated to pet lovers and people looking to adopt pets, to help them either find a new pet or give their pet up for adoption .

This project aims to help both those who have pets or those who aim to adopt new pets while maintaining their safety and preventing any scam.

1.3.2 Problematic

The culture of having a pet in Tunisia is getting bigger , and this culture is generating a huge community of pet lovers and re-homers , despite this revolution in Tunisia this community doesn't have a way of communication or exchange which make them take refuge to other ways to create their own way of communication like "Facebook groups" or "Messenger groups".

1.4 Study of the existing

An in-depth analysis of the existing situation was carried out to evaluate the technologies, methods and practices currently used in the context of pet adoption all around the world . with this analysis we could extracct and have some of the way of communication and exchange of pets.

1.4.1 PetRehomer

Pet rehomer is a digital platform that aims to help pets to find a new rehome and help pets lovers to adopt new pets .PetRehomer has been created by Charity Buddy, a registered animal welfare charity no. 1189284, that believes it is time to make pet rehoming and adoption better for pets, great for people, and safer for everyone.[1]



Strong Point :

- The application gives much information about the animal .
- The application gives a variety of functionality such as giving a pet for adoption, adopt, pet care information and an option for donation .

Limitations :

- Difficulty of use by average users : technical details requested during the filling out forms.

- Lack of user-friendliness : location of buttons, non-obvious navigation, etc.
- No way of communication between the adopter and the pet owner .

1.4.2 PetFinder :

Petfinder is an online, searchable database of animals who need homes. It is also a directory of nearly 11,000 animal shelters and adoption organizations across the U.S., Canada and Mexico. Organizations maintain their own home pages and available-pet databases.



Strong Points :

- List animals by the nearest to your localisation .
- Very specified search for animals , much criteria are used like Breed, gender, behaviour, color etc.

Limitations :

- No way of communication with the pet owner the only interaction is to like a post of a pet.
- You can't list an animal, the only functionality is to adopt a pet .
- No documentation with the animal only a description where you can find a text to tell you is the pet vaccinated or not .

1.4.3 Tunisian reality :

Although the reality in tunisia for pet lover that such a websites doesn't exist . so the only way for this community of pet lovers to communicate is creating facebook or messenger groups which seems to be for the first time a cool idea and an easy way to exchange pet however we can see problems in this method were always there are parasite that can undamage this community , do harassments or some acts like that .

1.4.4 Proposed Solution

Our solution here is to develop an adoption system in both web and mobile versions to facilitate the adoption process of a pet for this community and ensure the a list of

functionalities such as :

- Create account and authenticate.
- list one or many pets for adoption.
- Adopt pets
- interact with the pet post like comment save.
- Report suspect user or post
- Messaging system
- verified document with AI

1.5 Used methodology

1.5.1 Project management approach

In our project, we opted for the agile approach to manage the entire project.

We went with this choice because we found that the specifications include needs that are not very detailed and not clear. We have also predicted that the needs will evolve over time and this is already specified in the specifications.

All its indicators led us to choose an agile approach to manage the project, on the one hand to succeed in implementing the maximum possible of the requested functionalities, and on the other hand to succeed in resisting changes and new needs.

An agile approach can also save us time since it avoids the trap of the tunnel effect, which is one of the most famous and serious drawbacks of a classic project management approach.

Agile is an iterative approach to project management and software development that helps teams deliver value to their customers faster and with fewer headaches. Instead of betting everything on a "big bang" launch, an agile team delivers work in small, but consumable, increments.[2]

The agile approach is characterised by its flexibility and adaptability, unlike traditional project management approaches. Agile methods involve the requestor (client) as much as possible and allow great responsiveness to their requests.

Agile concepts and tools come from IT but Agile practices are now available for all trades : from innovative services to R&D in heavy industries.

There are several agile methods namely : Scrum, XP, RAD and DSDM.

1.5.2 Presentation of the used framework

We have defined the characteristics of agile methods. A study of the characteristics of the most used agile methods led us to choose the Scrum method. In fact, scrum indicates that the size of the team can be reduced and this is our case, another reason

is that scrum is flexible in terms of the duration of the sprint (between 2 and 4 weeks).

Scrum is a framework that helps teams work together. Much like a rugby team (where it gets its name) training for the big game, scrum encourages teams to learn through experiences, self-organize while working on a problem, and reflect on their wins and losses to continuously improve.[3]

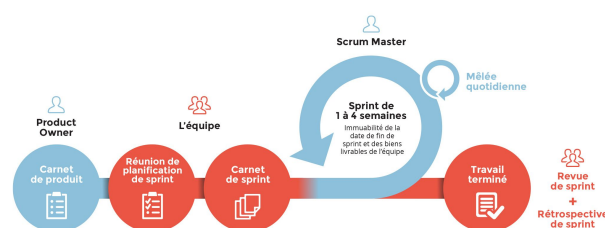
Scrum belongs to the family of iterative and incremental methodologies and is based on agile principles and values.

Most often, Scrum experts, even its founders, describe it as a project management-oriented process framework or pattern that can incorporate different engineering methods or practices.

1.6 Presentation and application of scrum

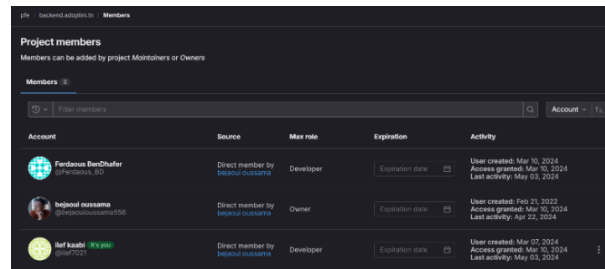
In this part, we will present the product backlog and the list of actors acting in our project.

The SCRUM framework consists of defining a working framework allowing the achievement of complex projects. Projects that follow the agile SCRUM method are divided into several relatively short work cycles called "sprints". These allow the team members to better plan the next stages of project development, but also to regularly evaluate the progress related to the project. Sprints can last from one to four weeks. They also make it possible to readjust or reorient the direction taken by the project if need. Indeed, the SCRUM method has several advantages other than improving the productivity and communication within the project. It rests above all on a fixed base roles, responsibilities and meetings that never change, while ensuring management flexible and adaptive projects. This has the advantage of reassuring teams during certain development phases



1.6.1 List of actor

The scrum approach requires a project team made up of several actors. Each has a well-defined role. In our case, the role of Product Owner was assigned to Mr. Ousama bejaoui (professional supervisor). A product owner expresses the client's needs and sets the deadlines for the delivery of deliverables. The role of developer was assigned to Mm Illef Kaabi and Mm Ferdaous Ben Dhafer. The developer Attend daily scrum meetings reporting the planned tasks for each day. Assure to Product Owner and



Scrum Master that the allocated work is being performed as planned. Ensure a clear understanding of epics and personas. Provide inputs on the creation of User Stories to the Product Owner. Finally the role of scrum master was assigned to to Mr Oussama bejaoui (Professional supervisor). A Scrum master is the leader of a team that applies Scrum. The development team consists of us and it is responsible for transforming the customer's needs into a functional product delivered on time.

1.6.2 Artifact done by Scrum

This is one of the artifacts of the Scrum Method. The development team needs to agree on the conditions for tasks and stories to move to the done state. In this sense, we agreed in collaboration with the product owner on the definition of done before starting the implementation of the first sprint. This definition requires that a task only switches to the done state if it has been integrated with the previously developed features and it works wonderfully. In addition, the task must be validated with the Product Owner.

1.7 Conclusion :

In this first chapter, we presented the project, the host organization then our analysis of the existing situation which led us to propose our own solution. Finally, we have chose a methodology to manage our project. The following chapter will be devoted to the implementation place of the project through the specification and analysis of needs.

Chapter2

SPRINT 1 : IMPLEMENTATION OF THE PROJECT

2.1 Introduction

This chapter will be dedicated to communicate the functional requirement of the project as well as the non-functional requirements . Later a presentation of the first version of backlog with the sprint planification. Finally we will represent the work environment used in our project .

2.2 requirements specification

This part consists of clarifying the system context. This involves identifying the requirements functional and non-functional and also the actors and their functionalities.

2.2.1 funtional requirments

the website must allow to the user to :

1. authenticate, consult and update his profile.
2. create one or many post to put a pet for adoption .
3. search for pets by :
 - Pet name .
 - position .
 - Type of pet .
 - Breed .
4. Interact with posts by like, comment, or save it as favorite .
5. Interact with other users via a messaging system .

2.2.2 Non-functional requirements

The non-functional requirements of our system are described as follows :

- **Security** :The website contain sensitive user information, this is why it must respect the confidentiality of the data. All passwords are encrypted to guarantee enhanced protection. In addition, the website use a "token" to secure access to features.
- **Ergonomics** : Our website must respect the principles of human interface Machine (HMI) and must have a certain clarity and ease of use. it must make it easier for users to access and modify their data.
- **Extensibility** : The website must be easy to maintain and simple to extend to add other components.

2.2.3 Class diagram

The class diagram is a static description of a system, focusing on the concepts of class and association. A class represents a group of objects that share Similar properties and common behaviors, described by attributes and operations. Associations consist of making links between class instances.

You'll find bellow the class diagram of our project.

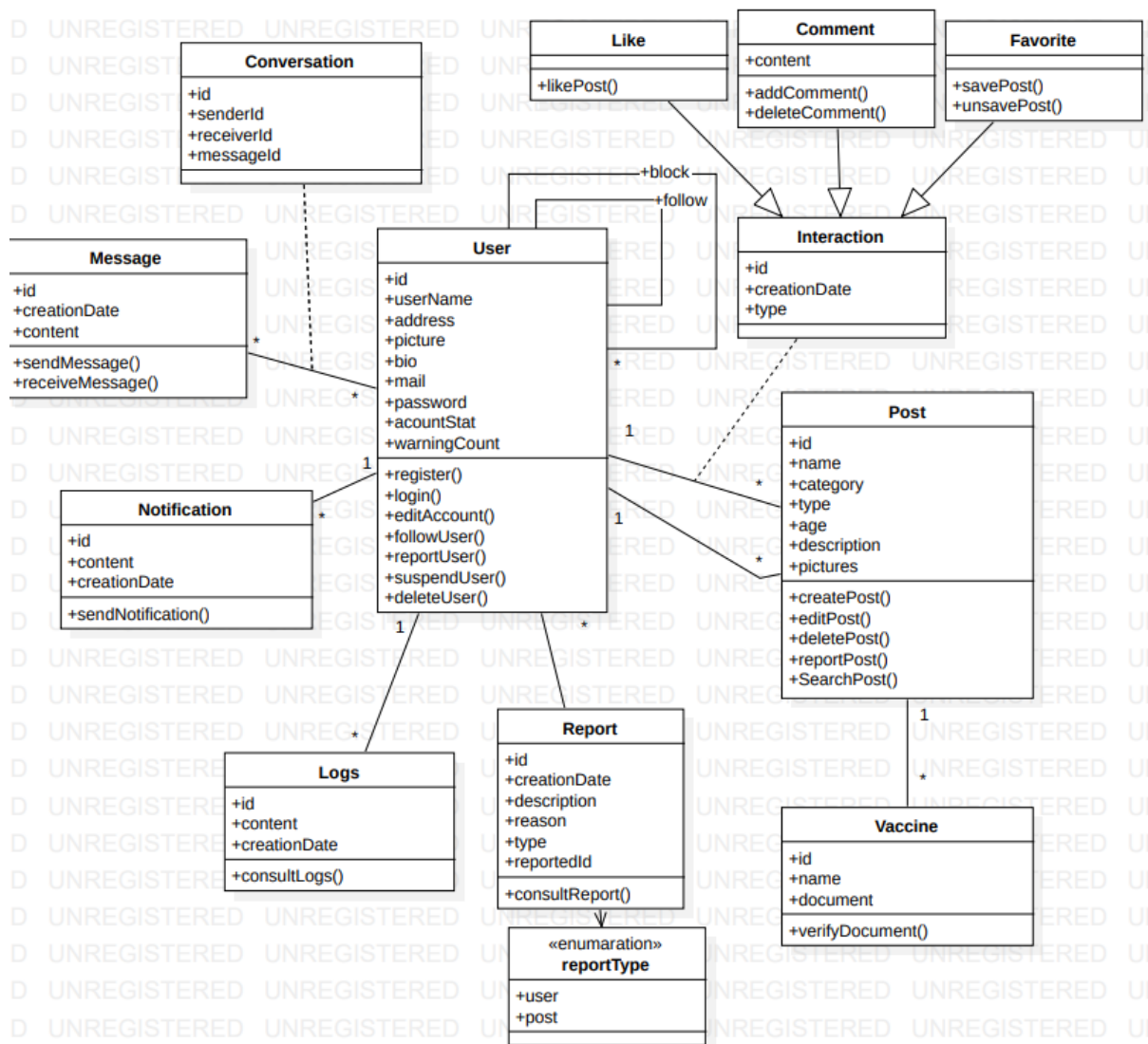


FIGURE 2.1 – class diagram

We have listed the following classes :

- User : this class represents the authentication and account.
- Post :
- Vaccine : this class represents the documents of the vaccines.
- Interactions : this class represents the interactions in a post (likes, comments, saves).
- Like : this class represents the likes in a post.
- Comment : this class represents the comments of a post.
- Favorites : this class represents the posts saved by the user.
- Message :Message : this class represents the management of messages.
- Conversation : this class represents the management of conversations between two users.
- Notification : Notification : this class represents the management of notifications of the user
- Logs : this class represents the logs history of the user.

- Report : Report : this class represents the management of the reports to the user or post.

2.2.4 Product backlog

A product backlog is a list of the new features, changes to existing features, bug fixes, infrastructure changes, or other activities that a team may deliver to achieve a specific outcome. The product backlog is the single authoritative source for things that a team works on.

Id	Actor	User story	Complexity	Priority
1	Guest	Can create account	4	5
2	Guest	Can login	4	5
3	Guest	Can consult home page	2	3
4	Guest	Can consult feed	3	3
5	User	Can consult their profile	3	1
6	User	Can edit their profile	5	2
7	User	Can consult other users' profiles	3	2
8	User	Can search for users	4	4
9	User	Can follow users	3	4
10	User	Can consult their followers/following	3	3
11	User	Can block users	5	3
12	User	Can report users	4	3
13	User	Can consult posts	3	2
14	User	Can create posts	5	5
15	User	Can edit their posts	5	4
16	User	Can delete their posts	4	2
17	User	Can mark his pet as adopted	3	3
18	User	Can like post	3	3
19	User	Can comment on post	3	3
20	User	Can save/unsave post	4	3
21	User	Can consult their saved posts	3	3
22	User	Can report a post	3	4
23	User	Can search for a post	4	2
24	User	Can send a message	5	5
25	User	Can receive a message	5	5
26	User	Can consult their messages	5	5
27	User	Can consult their pending posts	3	3
28	User	Can logout	2	5
29	Administrator	Can access admin dashboard	4	4
30	Administrator	Can consult user list	2	2
31	Administrator	Can consult user details	3	4
32	Administrator	Can change user's role	3	4
33	Administrator	Can change account status	4	3
34	Administrator	Can consult posts list	2	2
35	Administrator	Can consult post details	3	4
36	Administrator	Can validate post	3	4
37	Administrator	Can reject post	3	4
38	Administrator	Can consult reports list	2	3
39	Administrator	Can validate report	3	2
40	Administrator	Can reject report	3	1

TABLE 2.1 – Caption

2.3 Environment and technological choices

The choice of work tool is a preliminary step in our project to facilitate the work. In this part we will implement the tools and technologies used.

2.3.1 environment

During this part, we cite the different tools used in carrying out this project.

Figma

It is an ideal tool for creating graphical interfaces, prototypes and diagrams specific to any user experience (UX) oriented interface design.

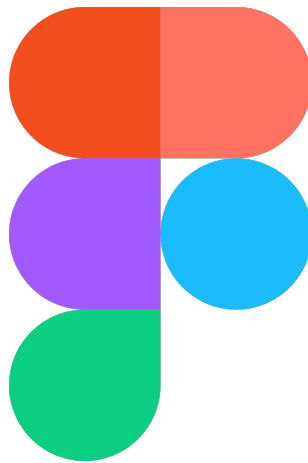


FIGURE 2.2 – Figma Logo

WebStorm

WebStorm is a cutting-edge IDE that supports JavaScript, HTML, and CSS, including modern innovations like TypeScript, React, Angular, Vue.js, and Node.js. The IDE is renowned for its user-friendly interface, intelligent coding assistance, and powerful debugging features.[]



FIGURE 2.3 – WebStorm Logo

starUml

StarUML is a UML tool by MKLab. The software was licensed under a modified version of GNU GPL until 2014 when a rewritten version 2.0.0 was released under a proprietary license. After being abandoned for some time, the project had a revival to move from Delphi to Java, before stopping again. [7] We decided to work with StarUML for its simplicity and usefulness in our projects to help up create the best and most well-structured diagrams.



FIGURE 2.4 – starUml Logo

GitLab

GitLab is an online platform for hosting software development projects. Like GitHub, GitLab utilizes Git as a version control system to monitor changes made to a project's source code.



FIGURE 2.5 – GitLab Logo

Ngrok

ngrok is a globally distributed reverse proxy that secures, protects and accelerates your applications and network services, no matter where you run them. You can think of ngrok as the front door to your applications. [8]



FIGURE 2.6 – Ngrok Logo

Xampp

XAMPP is a free and open-source cross-platform web server solution stack package developed by Apache Friends, consisting mainly of the Apache HTTP Server, MariaDB database, and interpreters for scripts written in the PHP and Perl programming languages.[9]



FIGURE 2.7 – Xampp Logo

2.4 Presentation of the used technology

NestJs

Nest (NestJS) is a framework for building efficient, scalable Node.js server-side applications. It uses progressive JavaScript, is built with and fully supports TypeScript (yet still enables developers to code in pure JavaScript), and combines elements of OOP (Object Oriented Programming), FP (Functional Programming), and FRP (Functional Reactive Programming).

We went with NestJS for its quick and efficient development process and because it's highly scalable and easy to maintain applications. So, it's a suitable framework for our backEnd.[10]



FIGURE 2.8 – Nest.js Logo

VueJs

Vue is a JavaScript framework that facilitates the UI (user interface) development of websites and single-page applications.

A progressive JavaScript framework, Vue makes creating user interfaces simpler and more enjoyable. It was designed to be incrementally adoptable and avoid many of the baked-in fallacies of existing, monolithic frameworks.

Vue's core library is easily integrated with other libraries and existing projects. It's easy to pick up, largely due to the framework's focus on the library's view layer only. Vue is also capable of powering intricate, single-page applications through the use of modern tooling and support libraries for added functionality.

Finally, Vue uses two-way data binding methods to ensure multiple parts of an application using the same data will always have the same version of this data (usually the latest version).



FIGURE 2.9 – Vue.js Logo

MySQL

MySQL is an open-source relational database management system (RDBMS) that organizes data into one or more data tables in which data types may be related to each



FIGURE 2.10 – MySQL Logo

other. these relations help structure the data.[11]

We went with MySQL to ensure a well-structured and organized database to help us store our project's data

JWT(Json Web Token)

JSON Web Token (JWT) is an open standard (RFC 7519) that defines a compact and self-contained way for securely transmitting information between parties as a JSON object. This information can be verified and trusted because it is digitally signed. JWTs can be signed using a secret (with the HMAC algorithm) or a public/private key pair using RSA or ECDSA.[13]



FIGURE 2.11 – Enter Caption

2.5 Conclusion

In this chapter, we discussed the actors, the functional and non-functional requirements of the application. Then, we explained the difference between the classes in the diagram. After that, we prepared the backlog of our application that was divided into 5 sprints. Finally, we presented the working environment and the architecture of the project. In the next chapter, we will be explaining the process and the advancement of the second sprint.

2.6 Burn down chart of sprint 1

The sprint 1 was mostly about discovering the frameworks, the problematic and setting routines with the hosting society.

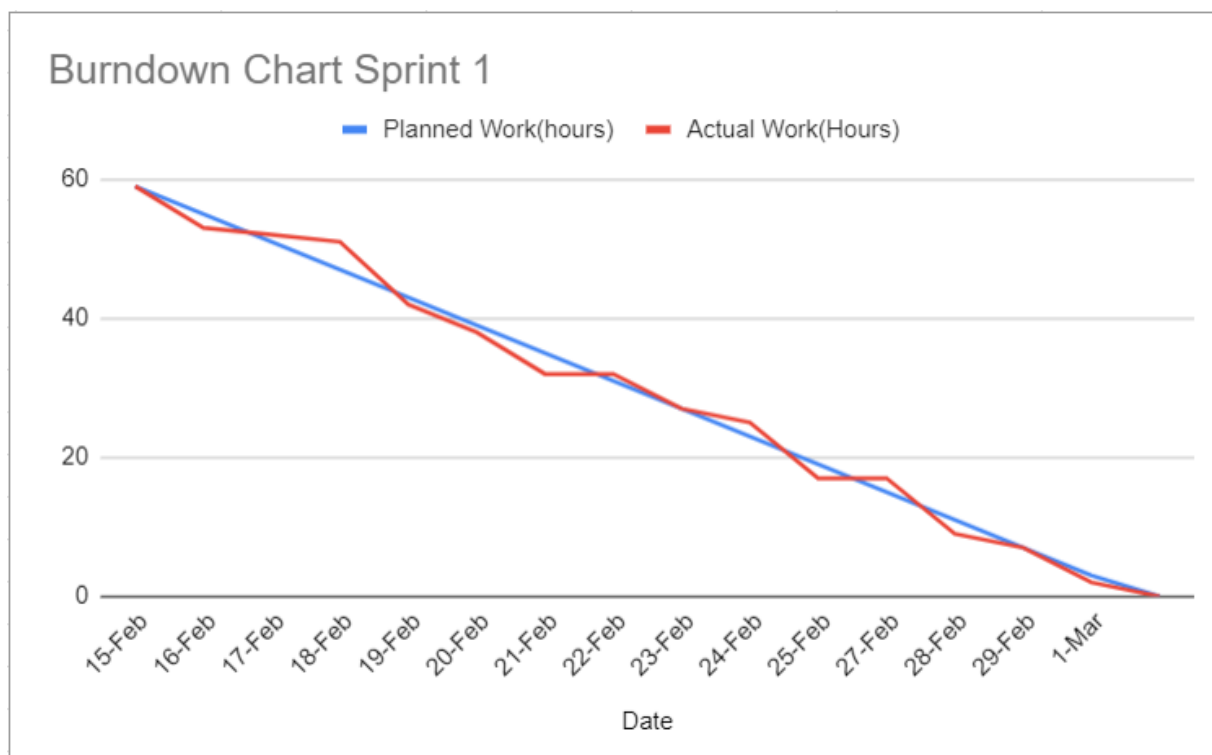


FIGURE 2.12 – Burn down chart sprint 1

Chapter3

SPRINT 2 : AUTHENTICATION AND PROFILE MANAGEMENT

3.1 Introduction

After analyzing and specified the requirements of our project we will start this sprint with the representation of the backlog. After that we will present the interface modeling , the conception as well as the realization of the authentication and the user management.

3.2 Product Backlog of sprint 2

ID	User Story	Complexity
1	As a user i can create an account	2
2	As a user i can log in to my profile.	2
3	As a user i can consult my profile.	3
4	As a user i can edit my personal information.	5
5	As a user i can follow other's user accounts.	5
6	As a user i can consult the list of my followers and following.	3
7	As a user i can block users.	5
8	As a user i can search for users.	4
9	As a user i can report users.	3

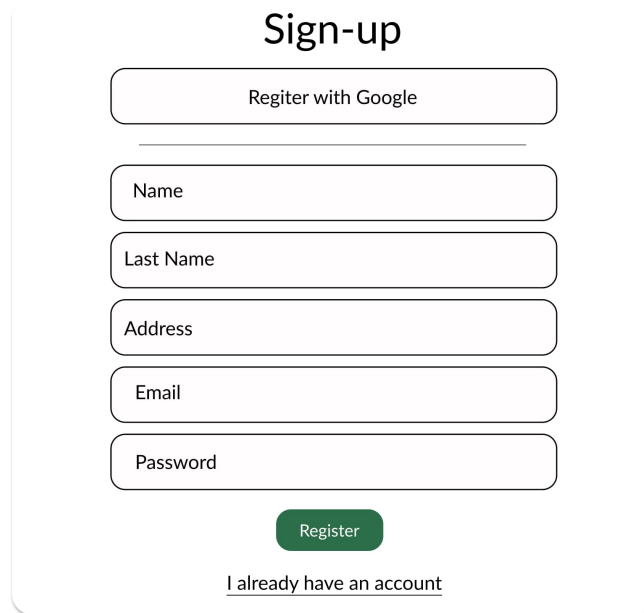
TABLE 3.1 – Backlog of sprint 2

3.3 Interface design : modeling

Prototyping is an experimental process where design teams implement ideas into tangible forms from paper to digital. Teams build prototypes of varying degrees of fidelity to capture design concepts and test on users. With prototypes, you can refine and validate your designs so your brand can release the right products[12].

3.3.1 User Story : sign up

The interface model of sign up (Figure 3.1) represent the form that a user must fill in with his own personal information to create his account.

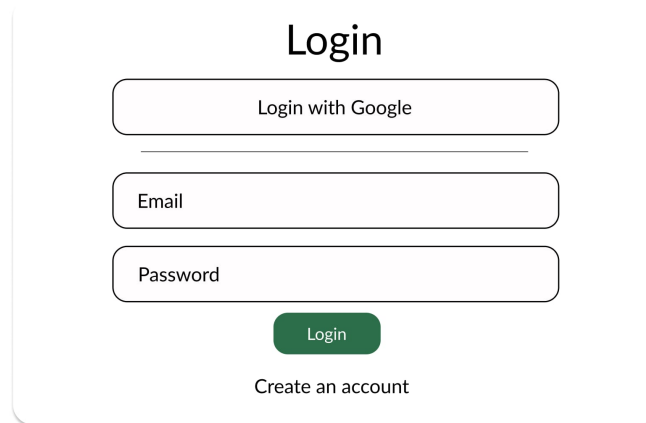


The image shows a sign-up form titled "Sign-up" centered at the top. Below the title is a button labeled "Register with Google". A horizontal line separates this from the input fields. There are five input fields stacked vertically, labeled "Name", "Last Name", "Address", "Email", and "Password". Below these fields is a green button labeled "Register". At the bottom, there is a link that says "I already have an account".

FIGURE 3.1 – sign up design

3.3.2 User Story : Authentication

The interface model of authentication (Figure 3.2) represent the form that the user must fill in to access to his account, he can also login with Google or navigate to the sign up interface.



The image shows a login form with a light gray background and rounded corners. At the top, the word "Login" is centered in a bold, black font. Below it is a rounded rectangular button with the text "Login with Google". A thin horizontal line separates this button from the input fields below. There are two input fields, one for "Email" and one for "Password", both with rounded corners and a light gray border. Below the "Password" field is a green rounded rectangular button with the text "Login" in white. At the bottom of the form, the text "Create an account" is centered in a smaller, black font.

FIGURE 3.2 – Login design

3.3.3 User Story : Consult and edit profile

The interface model of the figure (Figure 3.3) represent the profile of the user. When we click on the button «Edit», the user will be redirected to the page of modification of his personal information's (Figure 3.4).

When we click on the button of «Saved» the user will be redirected to the post that he saved (Figure 3.5).

The user also can see the numbers of his followers and following at the same time the user can get the list of the followers and following by clicking either on the followers or following (Figure 3.6) and (Figure 3.7).

The user can consult their post (we will be discussing this in the second sprint).

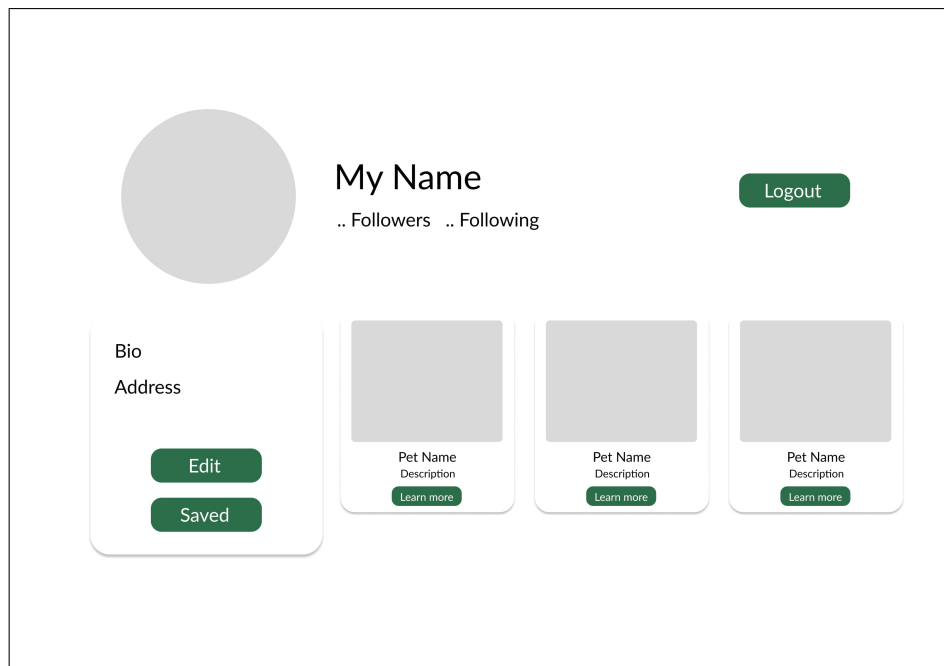


FIGURE 3.3 – design «User profile»

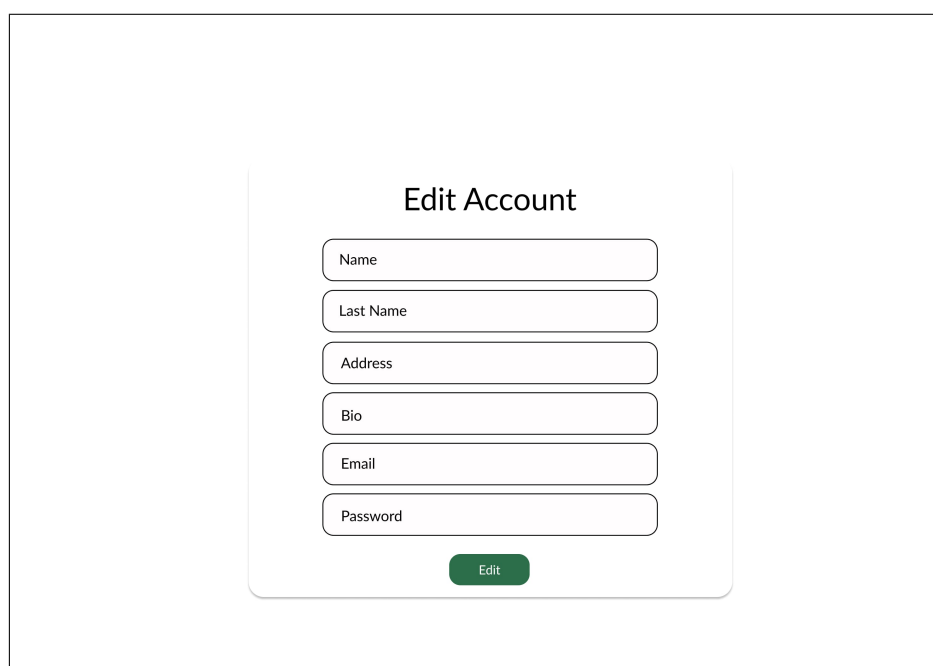


FIGURE 3.4 – design «Edit account»

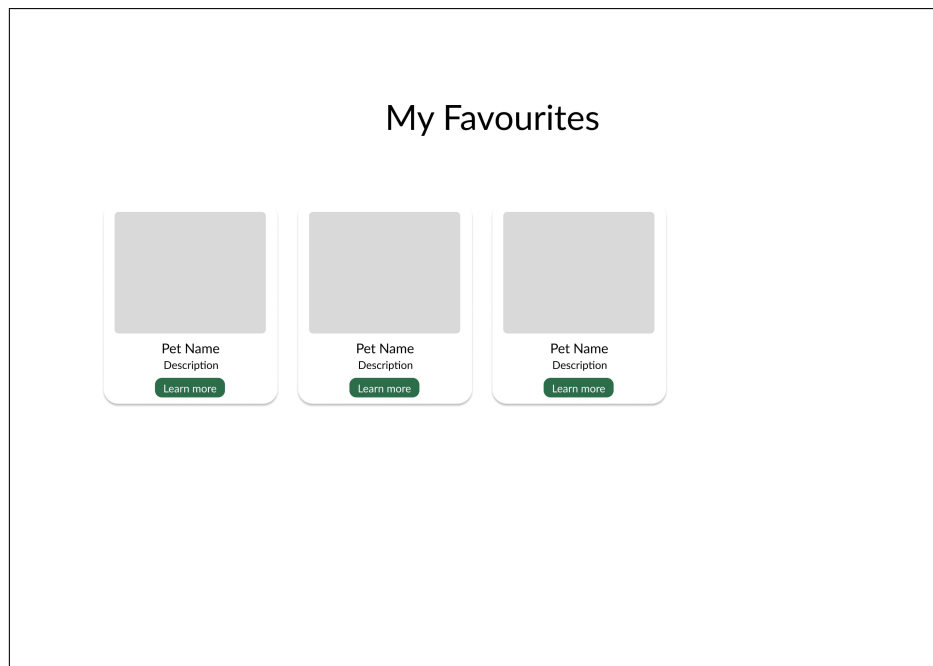


FIGURE 3.5 – Design «Favourite List»

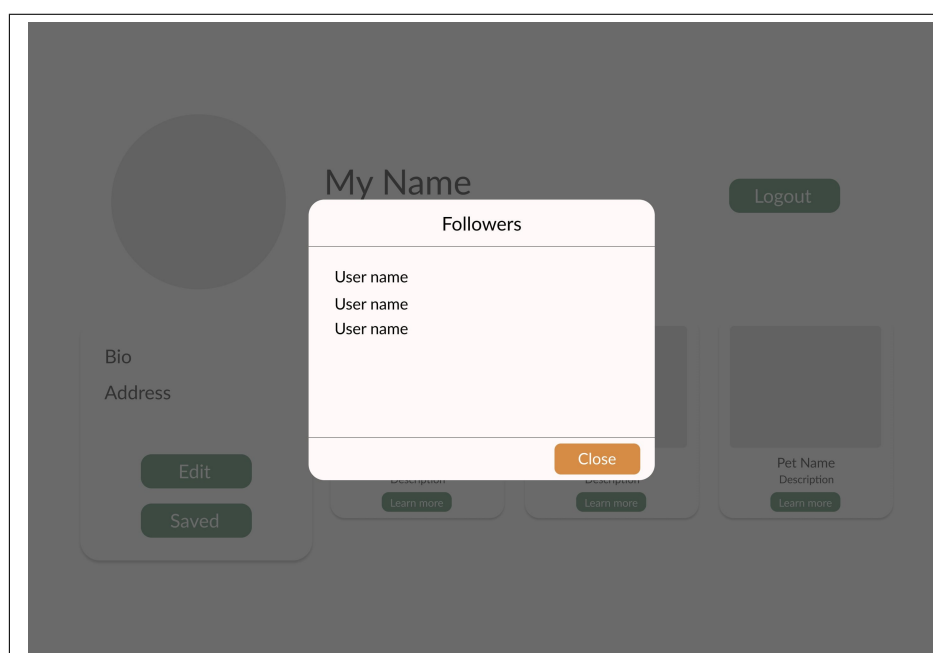


FIGURE 3.6 – Design «Followers List»

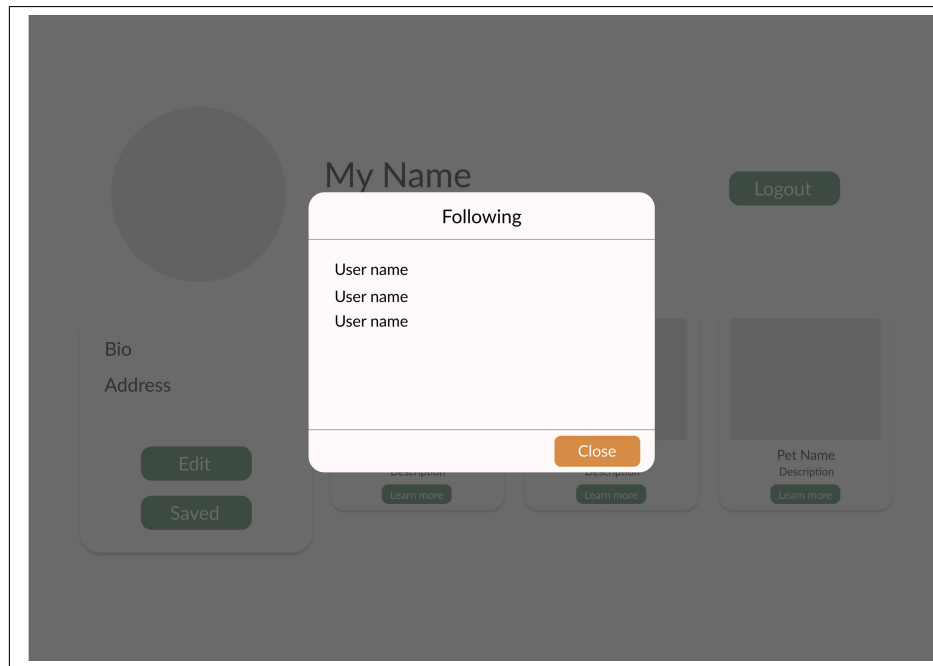


FIGURE 3.7 – Design «Following List»

3.3.4 User Story : Consult other user's account

The design (Figure 3.8) of other user profile , the user can consult other user account and get the information about the user the post that that user created. the user can from this interface follow user by clicking on follow , or he can block user by clicking on the block button.

The user can also report a user by clicking on the report button, to record the report the user must fill in the form of report as mentioned in the Figure 3.9.

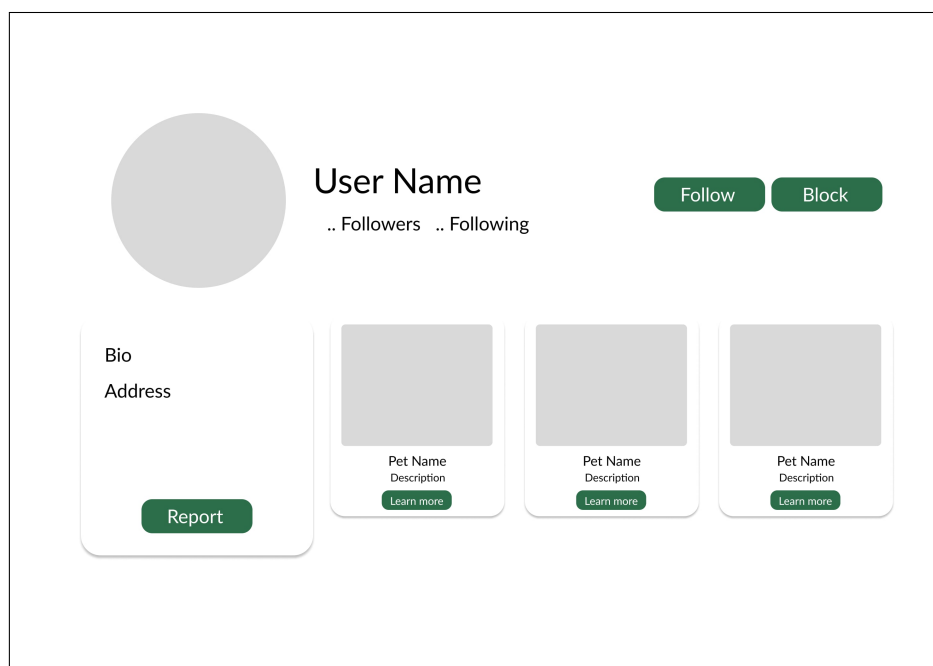


FIGURE 3.8 – Design «other profile»

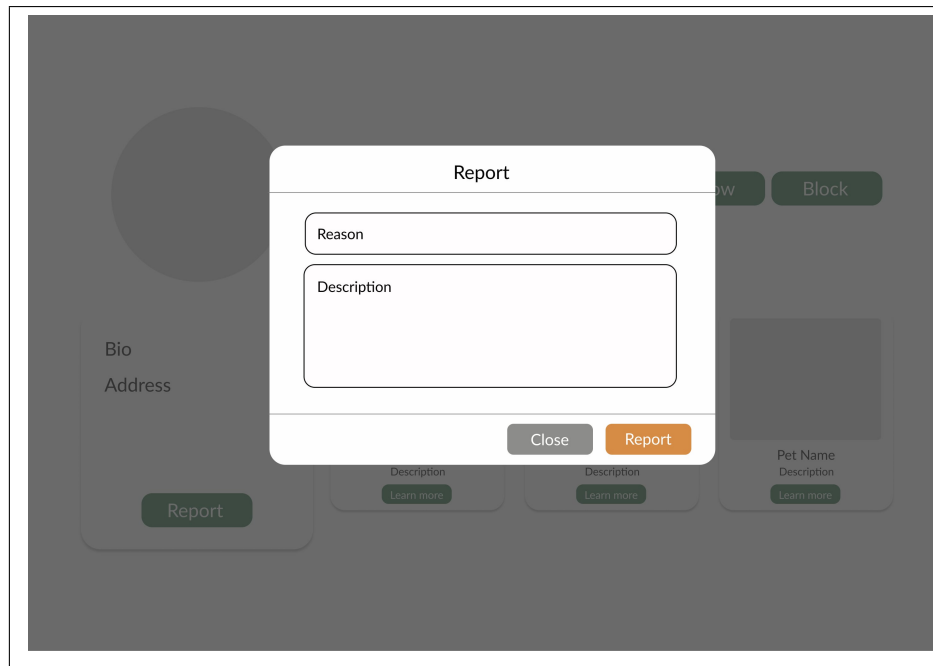


FIGURE 3.9 – Report user Design

3.4 Conception

3.4.1 Registration Object Sequence Diagram

The figure 3.10 represent the object sequence diagram of registration. The user fill in the form of sign up. If the inputs are not valid, an error message will be generated, else the information of users will be sent to the userController, and this will launch a method checkEmail to verify if the user already exist or not. If the user already exist an error message will be returned, If not the user will be created and saved in the data base.

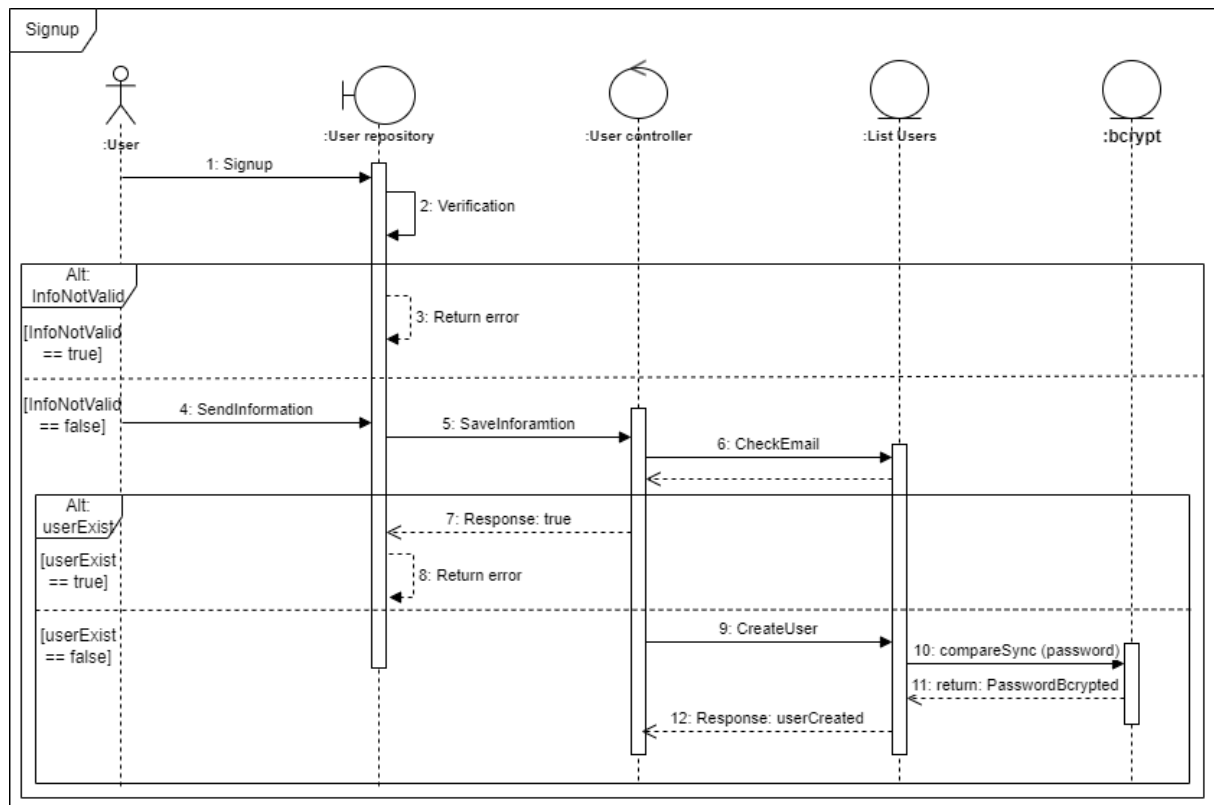


FIGURE 3.10 – Registration Object Sequence Diagram

3.4.2 Authentication Object Sequence Diagram

The figure 3.11 represent the diagram of object sequence of authentication. the user fill in the form. If the fields are filled with invalid type , an error message will be generated, else the authentication() method of the authenticationController will be release. If the user already exist and the password is correct, the user will be redirected to the home page else the authentication will be failed.

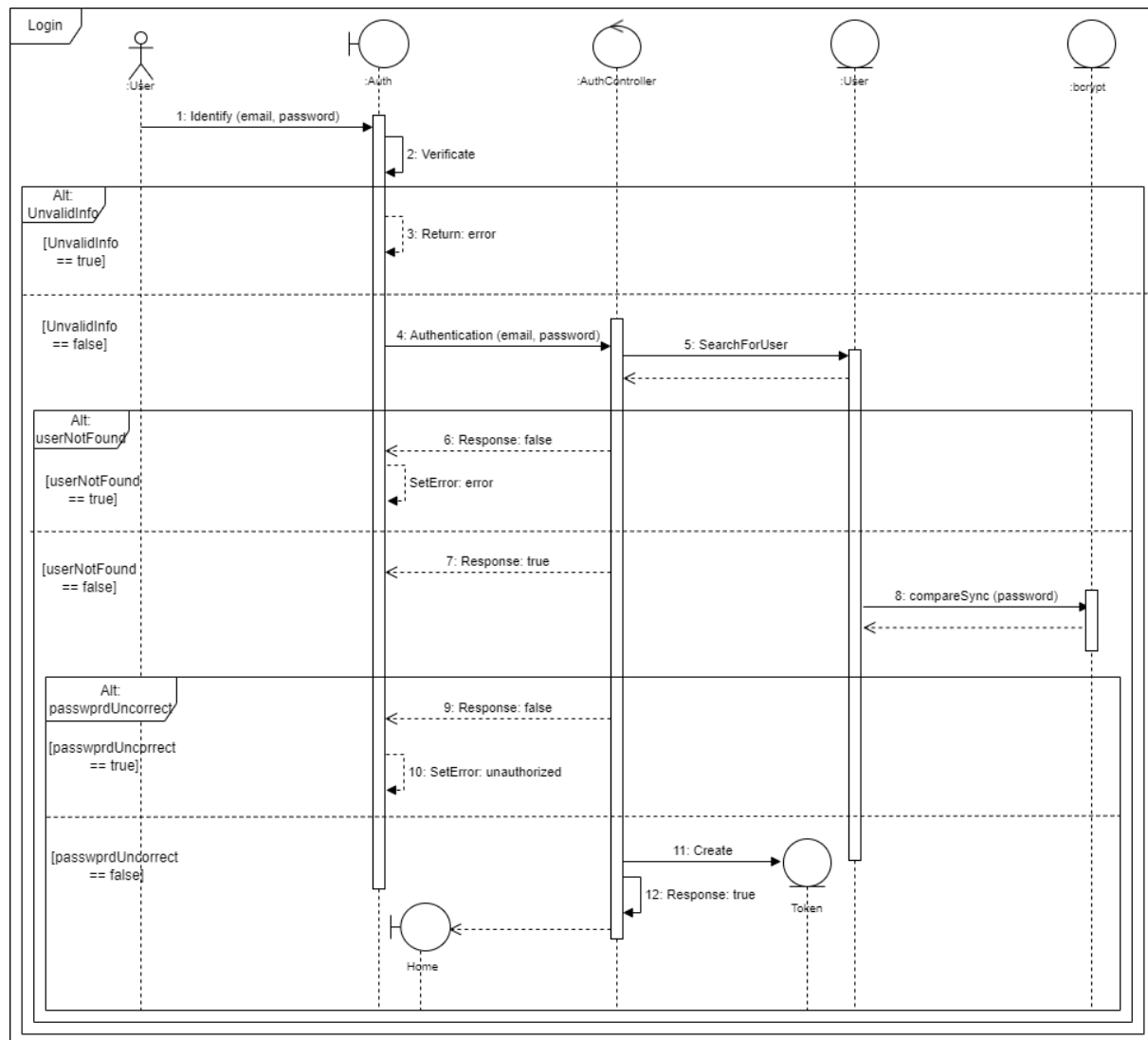


FIGURE 3.11 – Authentication Object Sequence Diagram

3.4.3 consult profile Use Case Diagram

the figure 3.12 represent the use case diagram of consult profile.

the opening of the user's account allows him to see multiple option in his account, the user can get the list of both his following and following by clicking on either following or following the user release a method getFollowing or getFollowers that's call the list of following/ followers.

the user can further interact by consulting his personal information , and posts.

within the consult posts use case the user can either delete his posts or edit them (which will be discussed in the next sprint).

the consult personal info use case includes the option of edit personal information (it will be detailed in the next section).

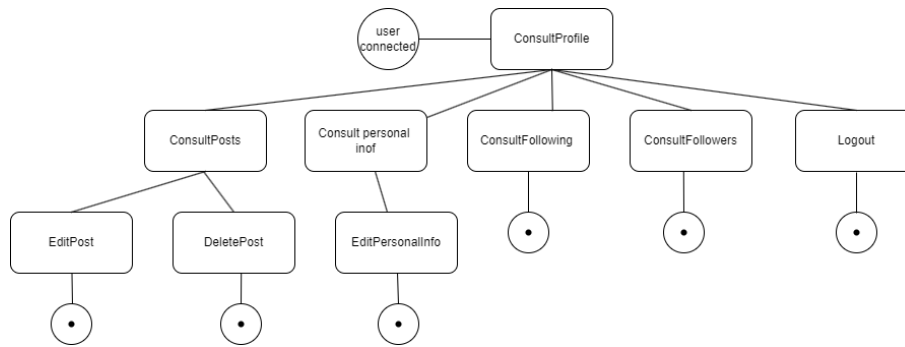


FIGURE 3.12 – consult profile Use Case Diagram

3.4.4 Edit Profile Object Sequence Diagram

the Fig 3.13 represent the object sequence diagram of edit profile .

The user have the ability to edit his personal information's. By clicking on edit on the user interface, an Access token guard will be launched to verify that the user has the right to edit the account. Once it's verified and the user is authorized to edit his account, the userService send a find method to the Data Base of users to access to the user, If the user exist the profile send back a form to edit the account , Once the user fill in the form the profile verify the validity of information , if it's validated the user controller save the user in the Data base of user .

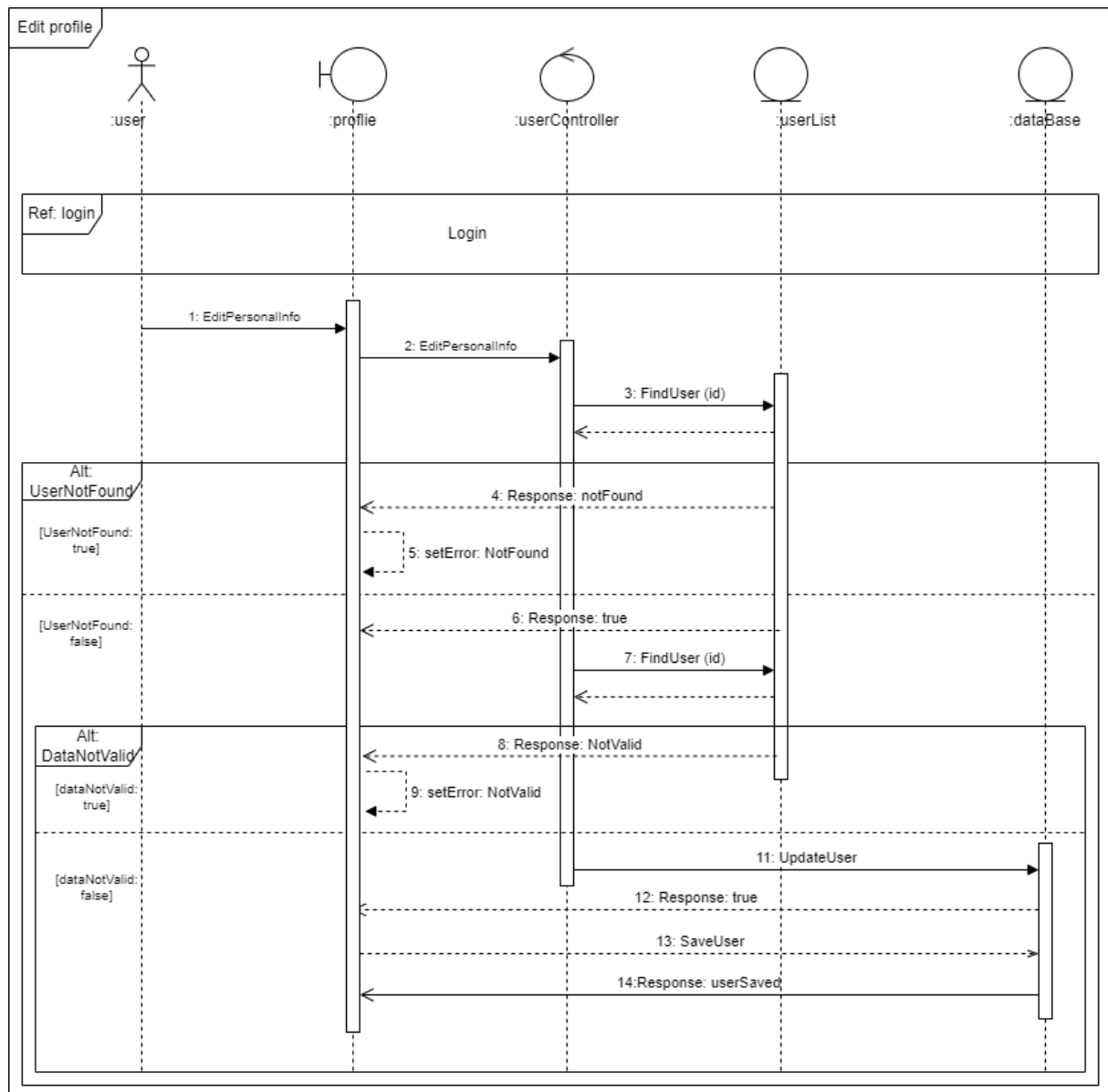


FIGURE 3.13 – Edit Profile Object Sequence Diagram

3.4.5 Get Followers/Following object sequence diagram

The fig 3.14 represent the object sequence diagram of get followers. The user have the ability to check the list of his followers, By clicking on the number of followers the Controller send the id of the user to check his existing , once its verified the data base will return the list of his followers . The same thing will happened if the user is willing to have the list of his following.

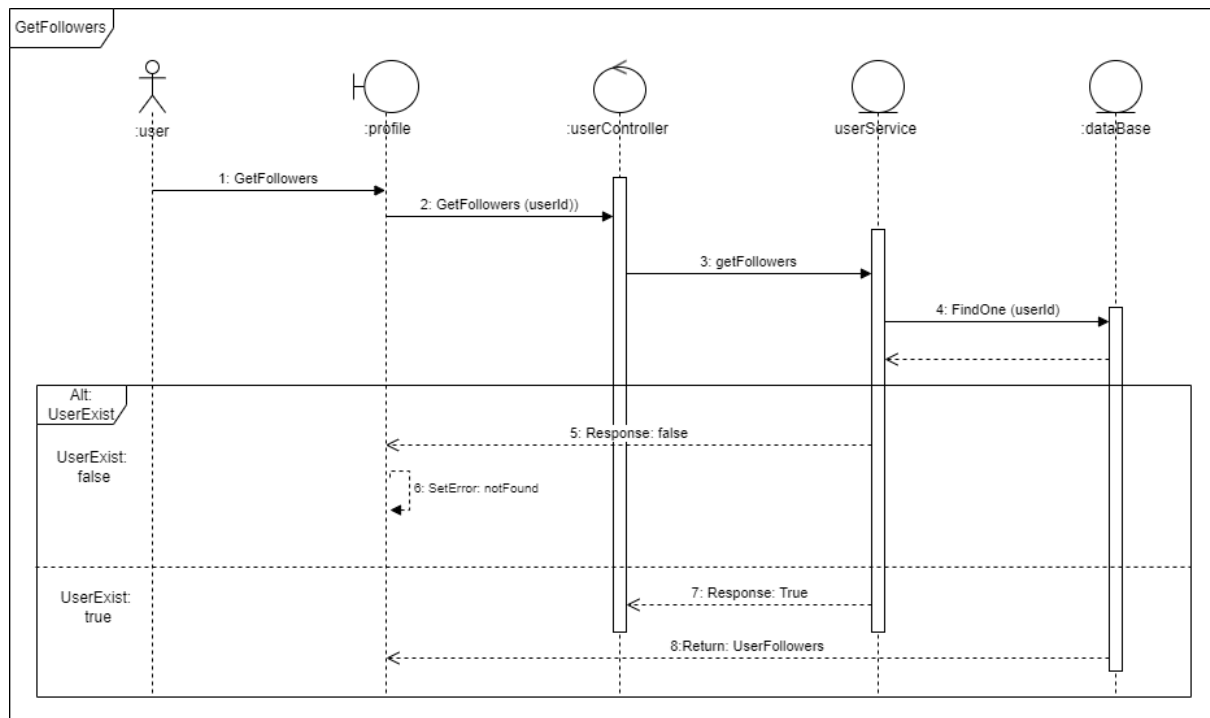


FIGURE 3.14 – Get Followers object sequence diagram

3.4.6 Consult other profile use Case diagram

The fig 3.15 represent the use Case diagram of consult others profile. The user have also an access to other user's account , he can consult their posts , see their personal information, follow the user, block him or report him if need it .

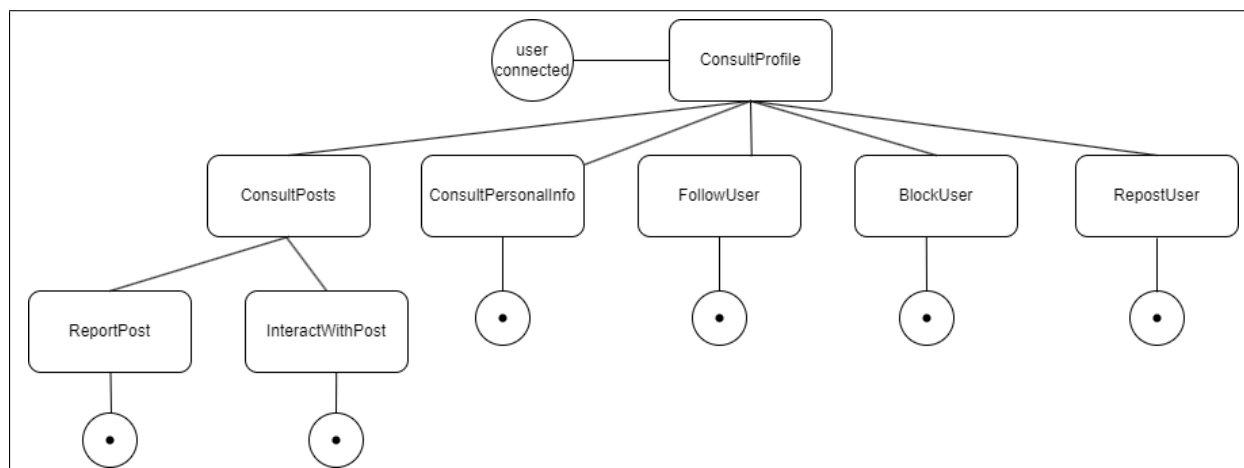


FIGURE 3.15 – Consult other profile use Case diagram

3.4.7 Follow user object sequence diagram

The fig 3.16 represent the follow user object sequence diagram .
The user can follow other user, by clicking on follow user in the user profile an access token guard will be launched as usual to verify the accessibility of user.
Once its verified the a findUser method is launched to verify if that user really exist. If the user exist a check following will be launched if the user is already following the current user an error message will be sent, else the target user will be added to the following list of the user and the changes will be saved.
Otherwise if the user didn't exist a not found exception will be returned to the user.

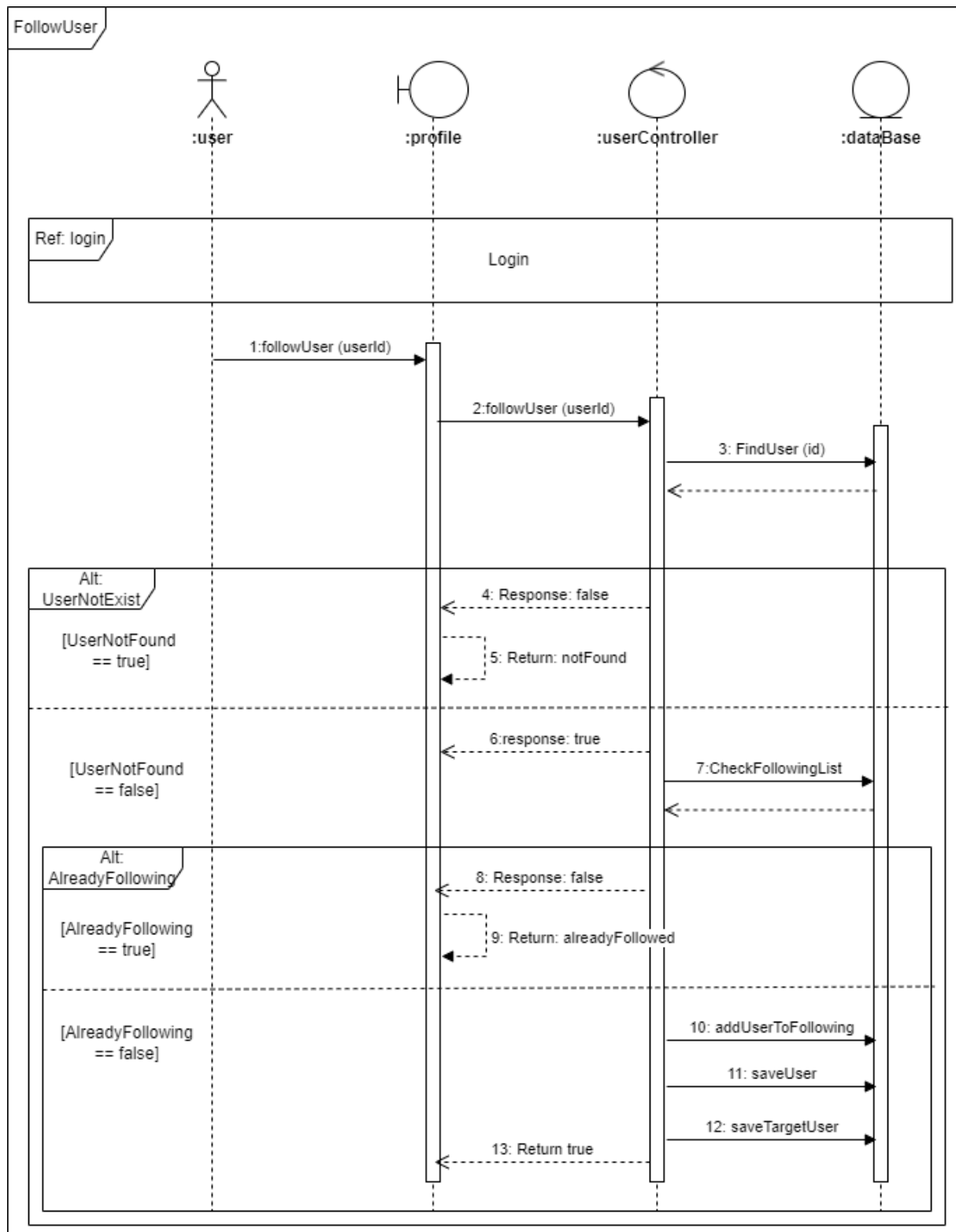


FIGURE 3.16 – Follow user object sequence diagram

3.4.8 Report user object sequence diagram

The fig 3.17 represent the Report user object sequence diagram. The user can report other users for many reasons, By clicking on report on the target user profile an access token guard will be launched as usual to verify the accessibility of user. Once it's verified a findUser method is launched to verify if that user really exist. If the user exist the a form will appear to the user in which he fill in with all the information needed to specify why he need to report the user than submit the form to save the report .

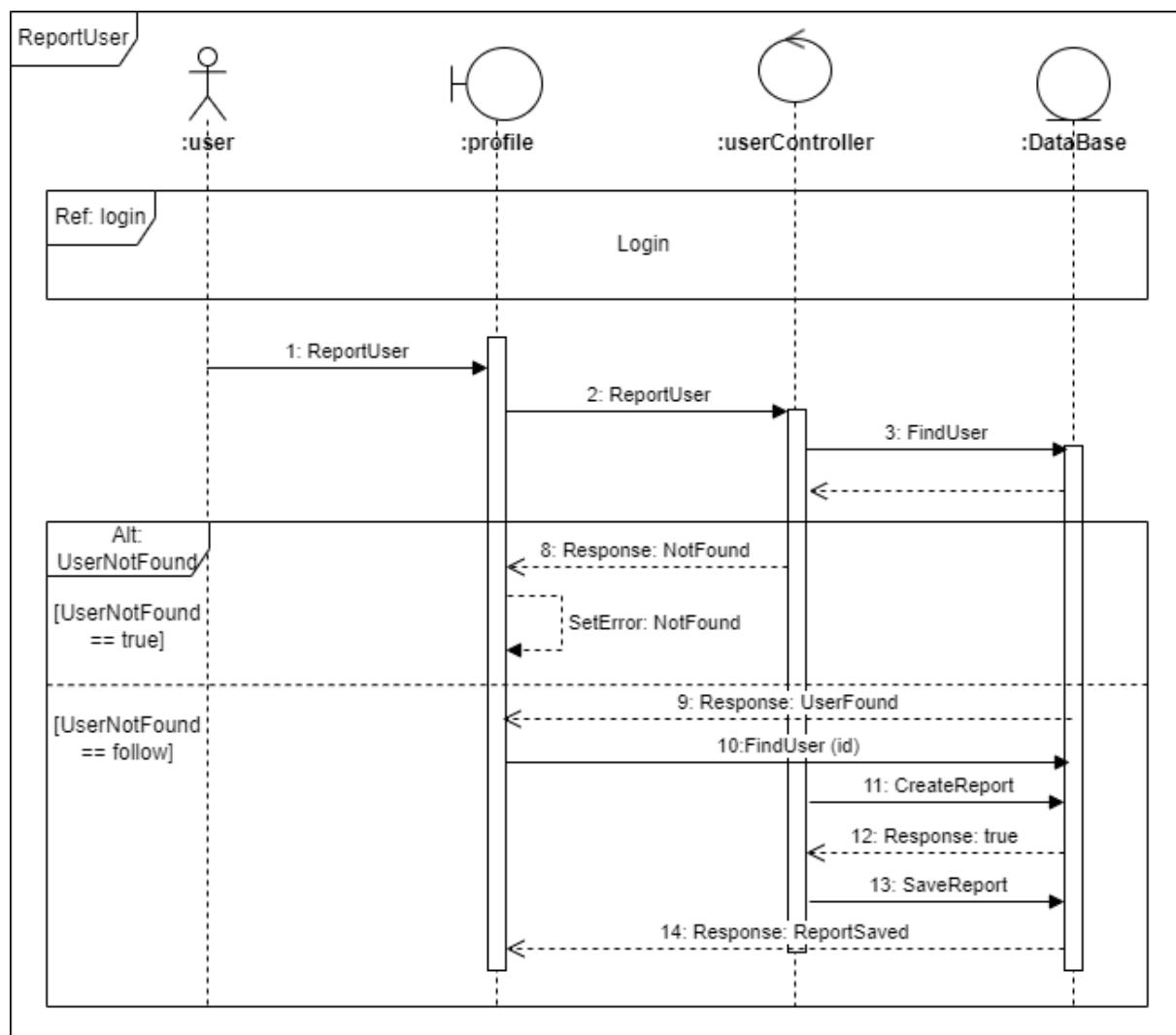


FIGURE 3.17 – Report user object sequence diagram

3.4.9 Block user object sequence diagram

The fig 3.18 represent the Block user object sequence diagram. a user can block other users by clicking on Block button, an access token guard will be launched to verify if the user has the right to block the user. Once it's verified a find user method will be launched to find the target user . If the user exist the user Controller will send a request to block this user so add them

to the list of the blocked users of the user and save changes.
Else a not found exception will be returned to the user.

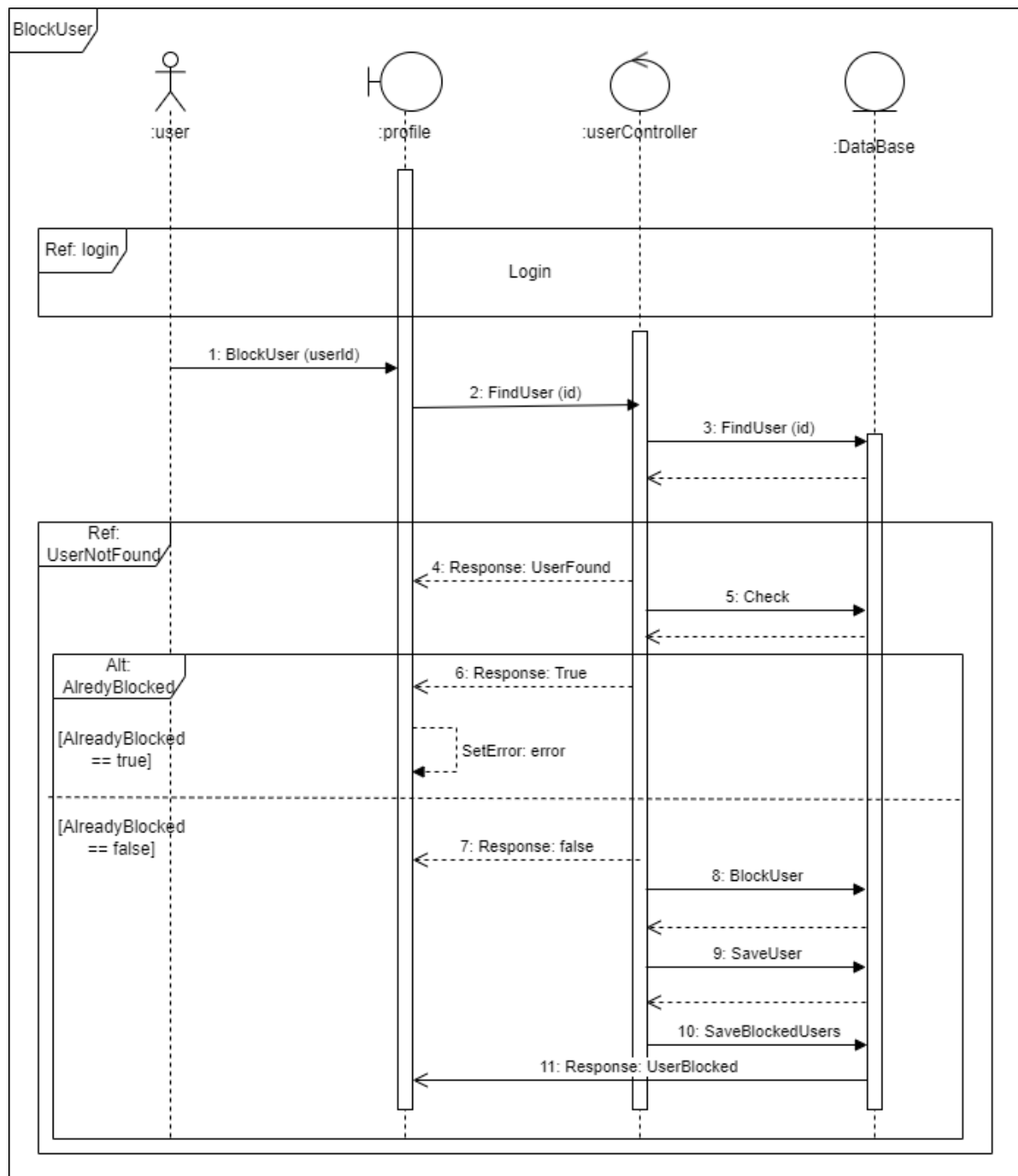


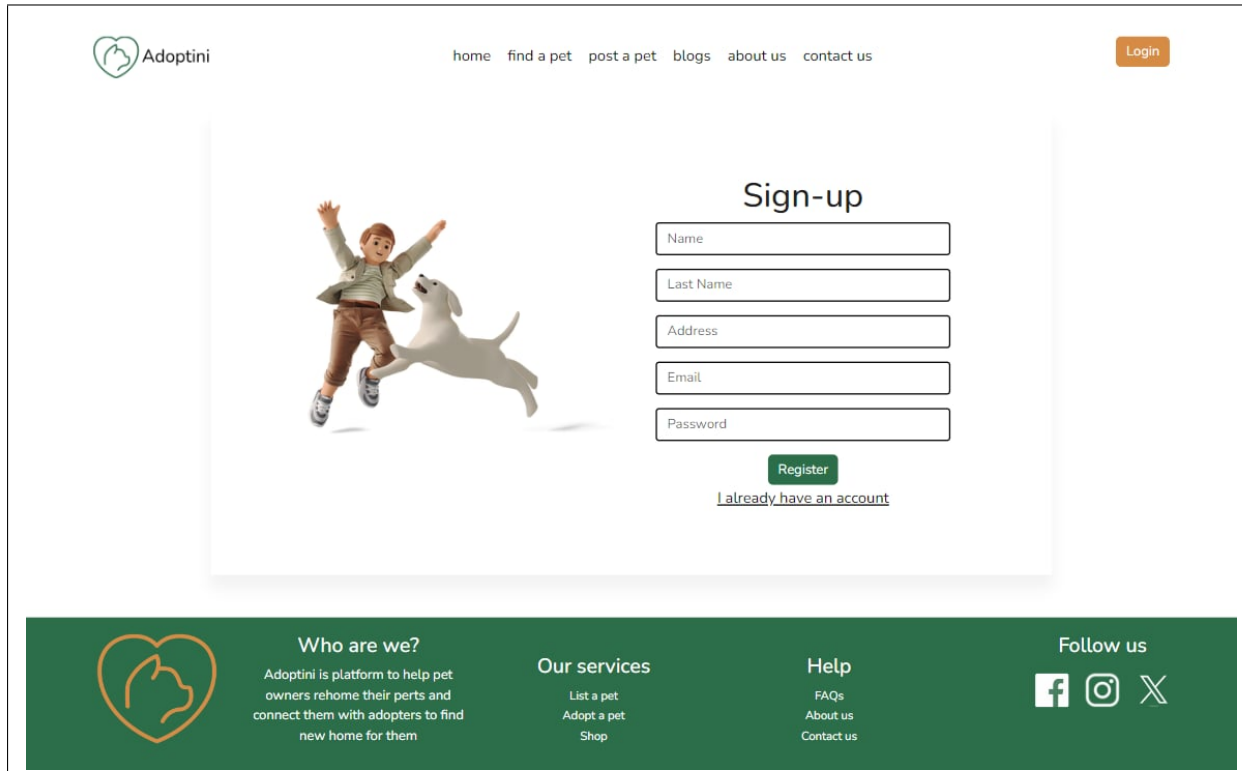
FIGURE 3.18 – Block user object sequence diagram

3.5 Realisation

In this section we will represent the work done in this sprint attached with screenshots with the completed tasks.

3.5.1 Registration interface

The figure 3.19 represents the sign-up interface. The user needs to fill in all the fields. Then, click on the “register” button. If all the data are valid, the account will be created. otherwise an error message will be shown.



The screenshot displays the Adoptini website's sign-up page. At the top, the Adoptini logo is on the left, and navigation links (home, find a pet, post a pet, blogs, about us, contact us) are in the center. A 'Login' button is on the right. The main content area features an illustration of a boy jumping with a dog on the left and a 'Sign-up' form on the right. The form includes input fields for Name, Last Name, Address, Email, and Password, followed by a 'Register' button and a link for users who 'already have an account'. The footer is a dark green bar with sections for 'Who are we?' (describing the platform's mission), 'Our services' (listing 'List a pet', 'Adopt a pet', and 'Shop'), 'Help' (listing 'FAQs', 'About us', and 'Contact us'), and 'Follow us' with social media icons for Facebook, Instagram, and X.

FIGURE 3.19 – Sign up interface

3.5.2 Login interface

The figure 3.20 represent the Authentication interface. After filling the email and password fields, the user clicks on the “login” button. If the data are valid, the user will login to their account otherwise an error message will be shown.

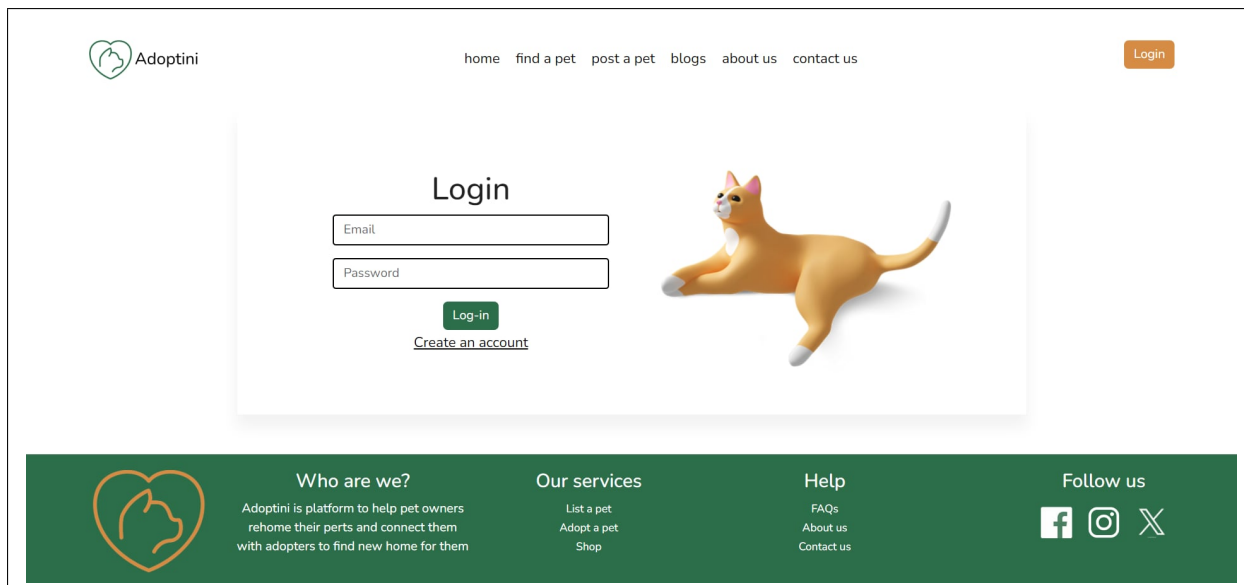


FIGURE 3.20 – Login interface

3.5.3 Consult my profile interface

The user can consult their own profile as it is shown in the figure 3.21 The user can consult their personal info and edit by clicking on the “Edit” button, they can consult their own posts, their favorite posts by clicking on “Favorites” and they can logout of their account by clicking on “Logout” button.

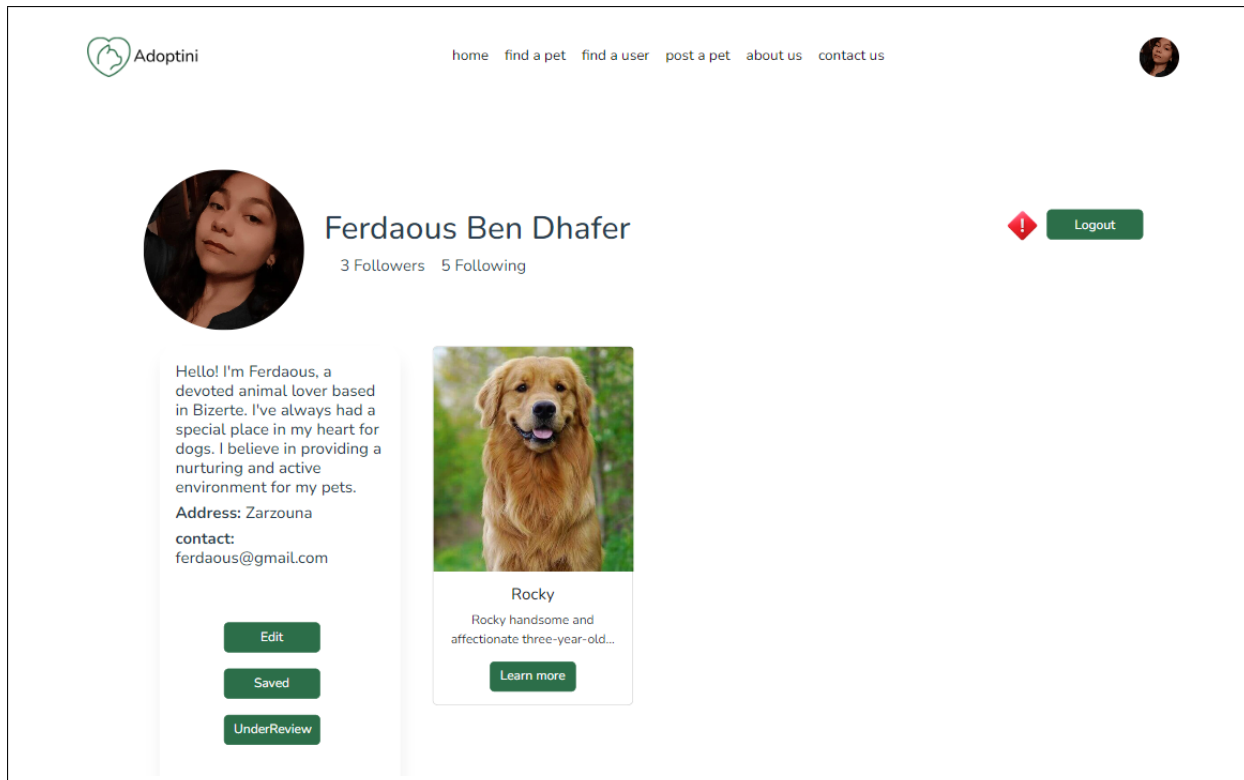


FIGURE 3.21 – Consult my profile interface

3.5.4 Update profile interface

The figure 3.22 represent the profile editing interface. the user can edit their personal info by filling in the field on the info they want to edit. Then, click on edit to save the changes. If the data are not valid, an error message will be shown.

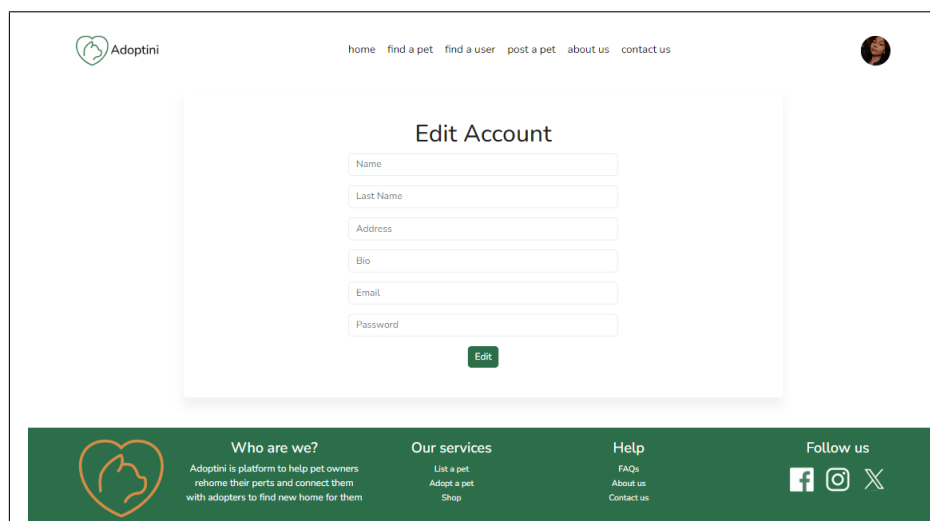


FIGURE 3.22 – Update profile interface

3.5.5 Get followers/following interfaces

By clicking on the followers/following in the profile, a list of the user's followers/following which is shown in the figure 3.23 for get followers and fig 3.24 for get following.

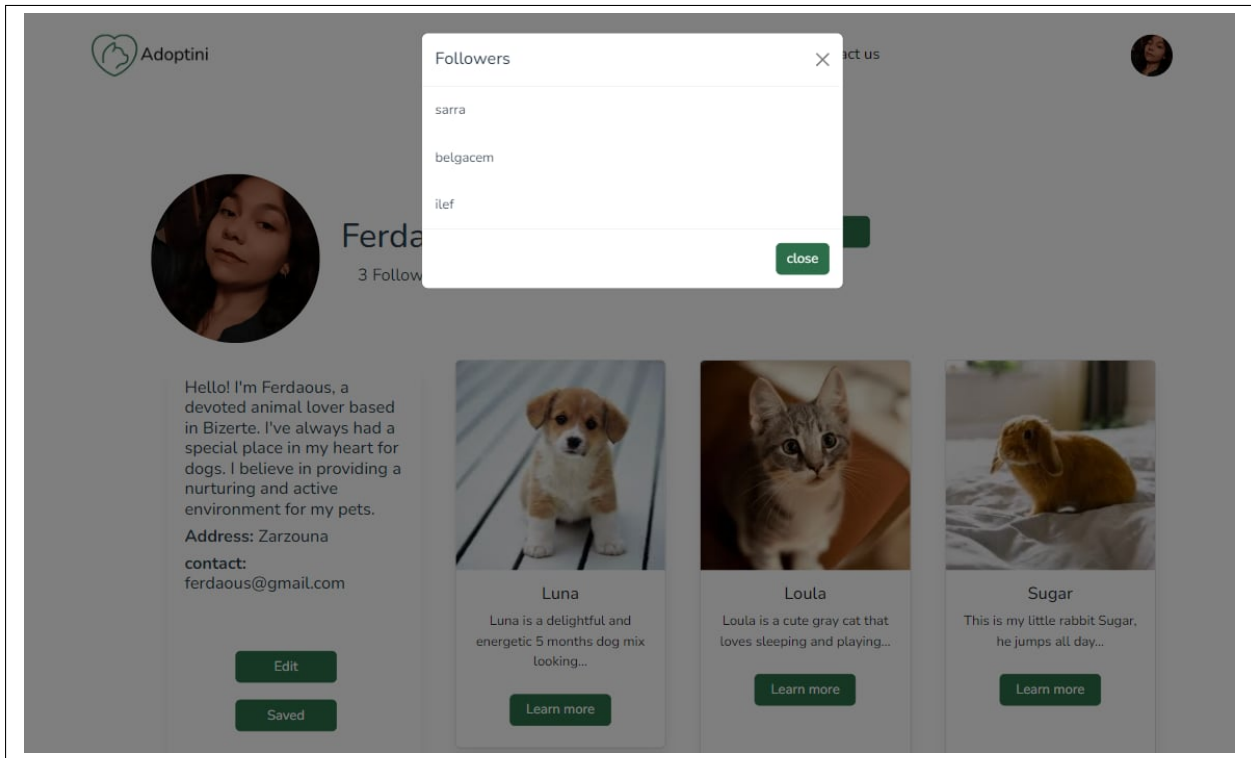


FIGURE 3.23 – Get followers interface

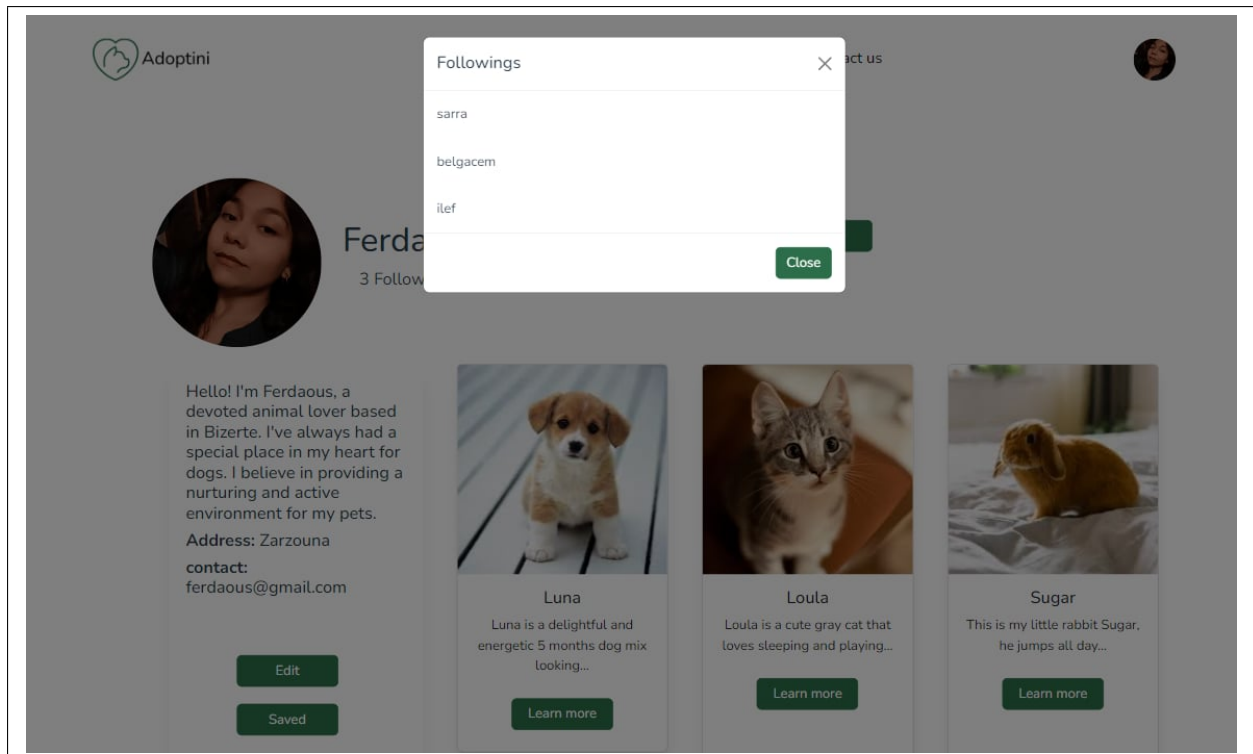


FIGURE 3.24 – Get following interface

3.5.6 Consult other profiles interface

The user can consult other users' profiles as it is shown in the figure 3.25. They can consult their personal info and posts. The user can follow other users by clicking on the "Follow" button, block them by clicking on "Block" and report them by clicking on "Report".

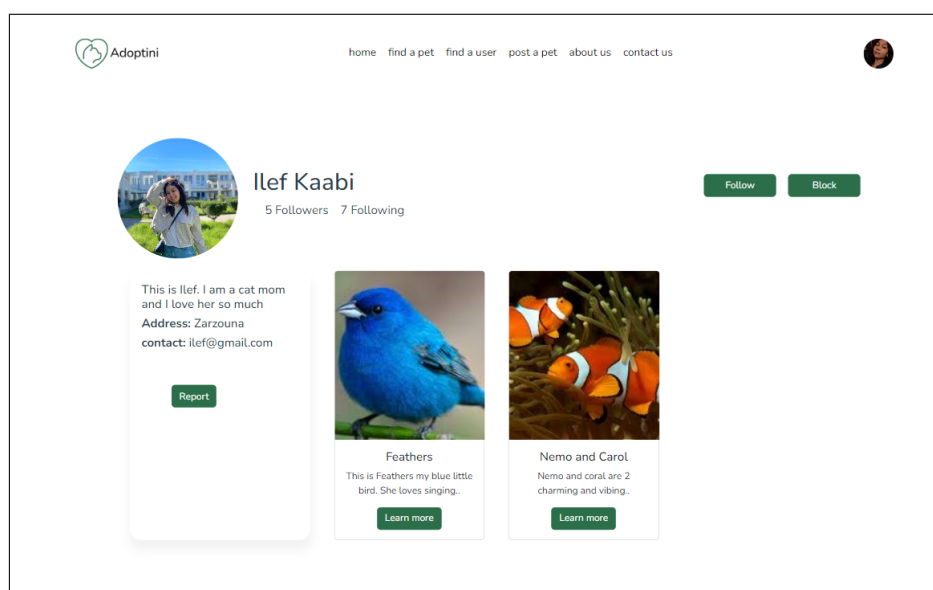


FIGURE 3.25 – Consult other profiles interface

3.5.7 Report user interface

The figure 3.26 represent a report modal. The user needs to fill in the reason of the report and description then click on the “Report” button to send it.

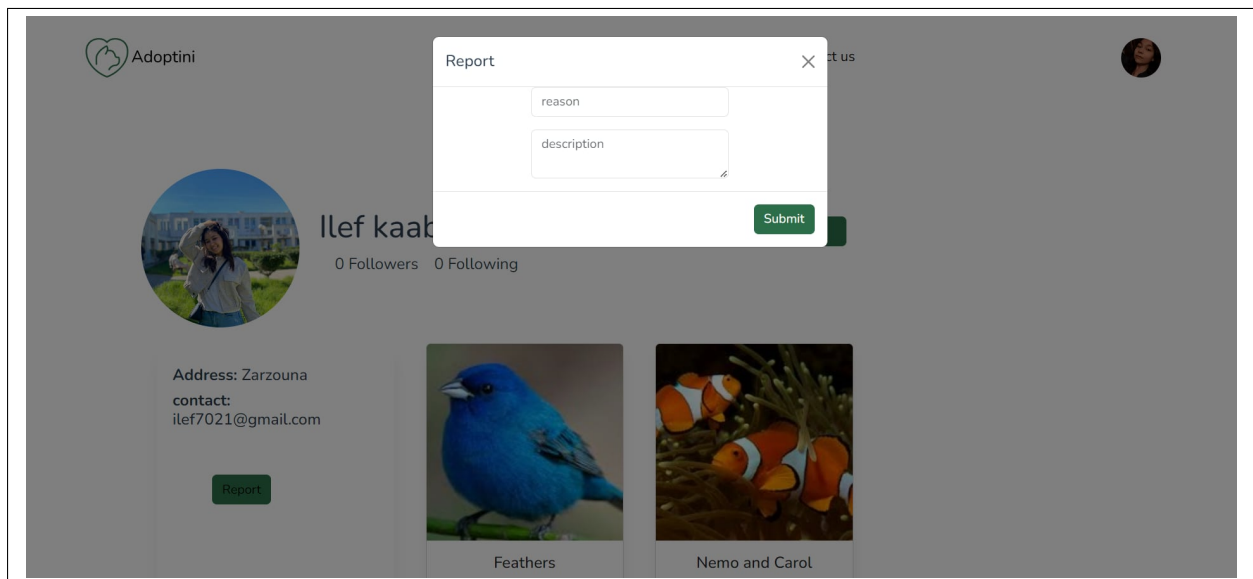


FIGURE 3.26 – Report user Interface

3.6 Conclusion

In this chapter, we discussed the second sprint of our project which is the authentication and profile management. We presented the backlog of the sprint, the conception and realization of the use case identified. In the next chapter, we will be presenting the third sprint.

3.7 Burn down chart sprint 2

Mainly the sprint 2 was about learning and configured our project so that's what explain the burn chart down.

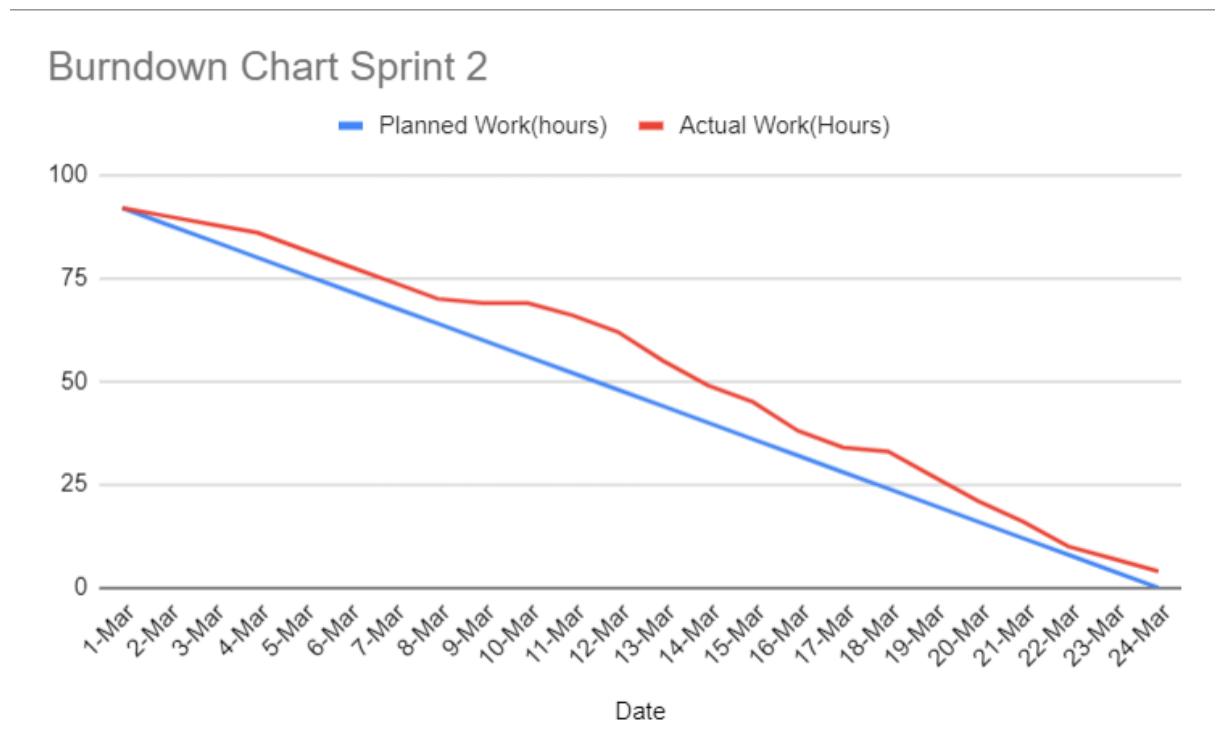


FIGURE 3.27 – Burn down chart sprint 2

Chapter4

SPRINT 3 : FEED MANAGEMENT

4.1 Introduction

In this chapter, we will present the third sprint of our project in which we will start with the backlog of the sprint, after that the representation of the design interfaces, continuing the conception of the sprint, and finally we will represent the realisation of the feed management of our platform.

4.2 Backlog of the sprint 3

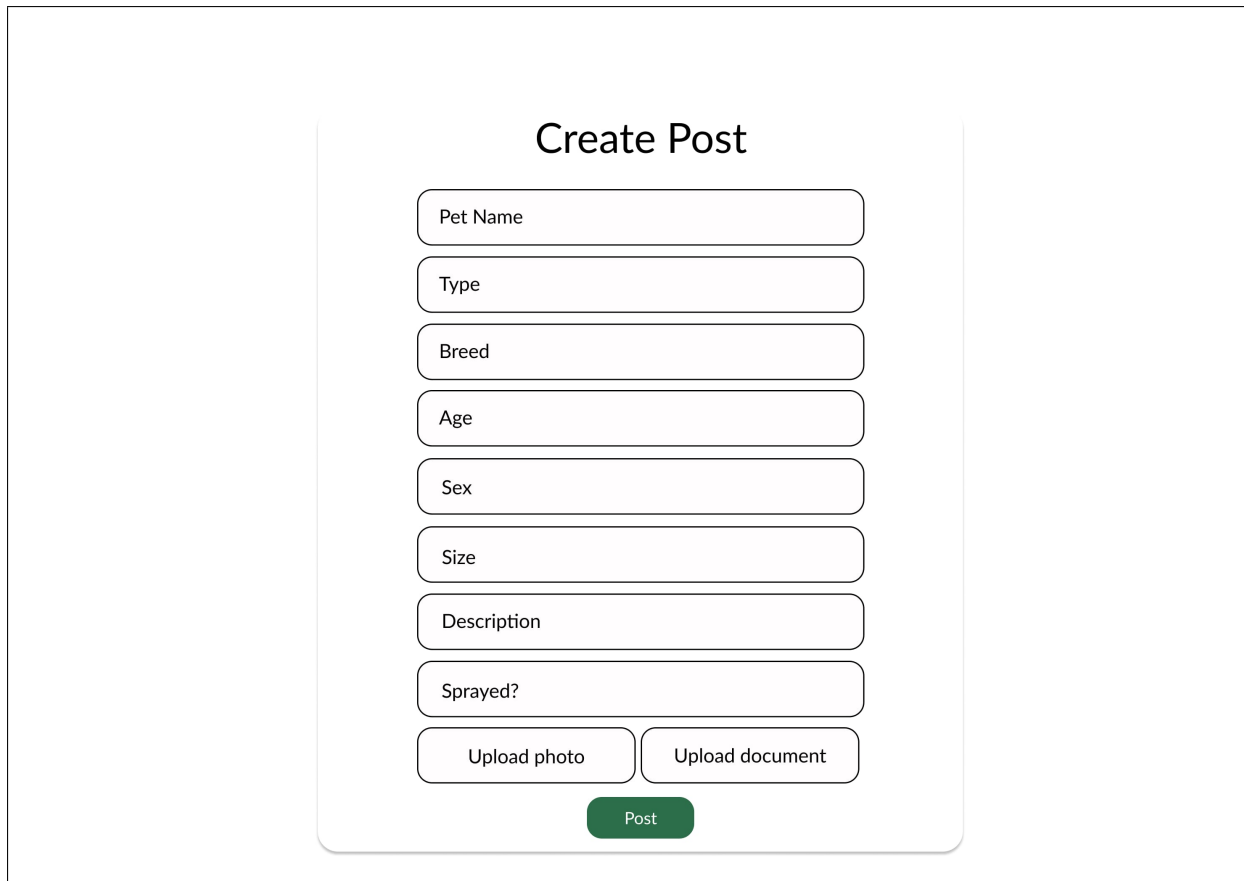
Id	User Story	Complexity
01	As user, i can create a post	5
02	As user, i can consult feed page	3
03	As user, i can update my posts.	4
04	As user, i can like posts.	4
05	As user, i can comment a post.	3
06	As user, i can consult post's comments.	2
07	As user, i can Save a post as favourite.	4
08	As user, i can consult all the user's posts.	3
09	As user, i can consult a post.	2
10	As user, i can search for posts.	4
11	As user, i can mark my pets as adopted.	3

TABLE 4.1 – Sprint 3 backlog

4.3 Interface design : Modeling

4.3.1 User story : Create post

The interface model of create post fig 4.1 represent the form that the user must fill in with the appropriate information to create the post, the user can also add the pet images and upload the necessary document of his pet.

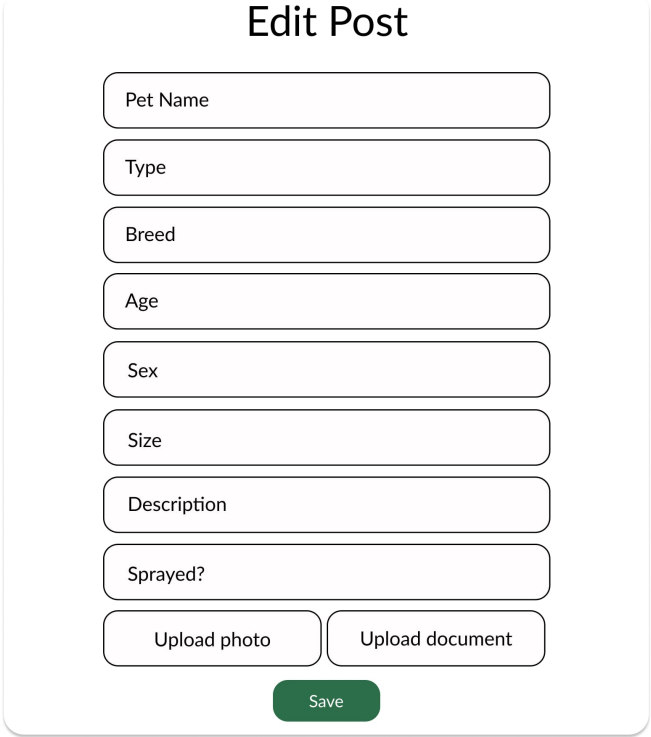


The image shows a 'Create Post' form design. It is a vertical stack of input fields and buttons. At the top is the title 'Create Post'. Below it are eight input fields: 'Pet Name', 'Type', 'Breed', 'Age', 'Sex', 'Size', 'Description', and 'Sprayed?'. Each field is a rounded rectangle with a light pink background and a thin black border. Below the 'Sprayed?' field are two buttons: 'Upload photo' and 'Upload document', both with rounded corners and light pink backgrounds. At the bottom is a green 'Post' button with rounded corners and white text. The entire form is centered within a light gray rectangular frame.

FIGURE 4.1 – create post design

4.3.2 User Story : update post

The fig 4.2 present the interface design that we are looking to realise for edit post, the user can update his own post by filling in the form with the desired modification, once he write down all the information he can save the changes.



The image shows a wireframe for an 'Edit Post' form. The form is centered within a light gray rectangular container. At the top of the form is the title 'Edit Post' in a bold, black font. Below the title are nine input fields, each with a light pink background and rounded corners. The first eight fields are stacked vertically and labeled: 'Pet Name', 'Type', 'Breed', 'Age', 'Sex', 'Size', 'Description', and 'Sprayed?'. The 'Sprayed?' field is a checkbox. Below these fields are two buttons: 'Upload photo' and 'Upload document', both with light pink backgrounds and rounded corners. At the bottom of the form is a green 'Save' button with white text and rounded corners.

FIGURE 4.2 – Edit post design

4.3.3 User Story : Consult feed page

The user can also consult all the posts posted in the platform as shown in the fig 4.3, the user can also make a search for pets either by name or by the type of pet.

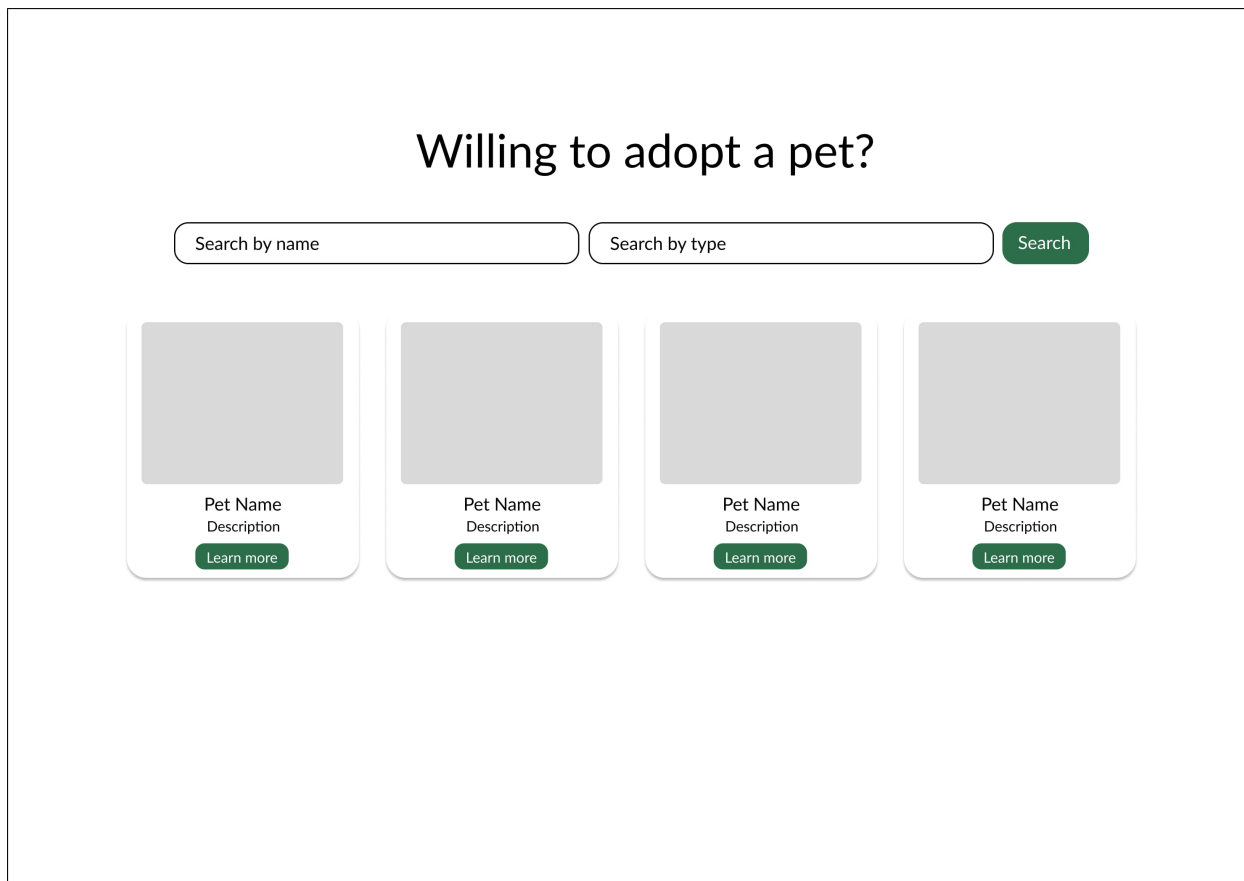


FIGURE 4.3 – Consult feed page design

4.3.4 User Story : Consult Post

The user can get more details about the post by clicking on learn more in the previous user story, the figure 4.4, so he can also interact with the post by liking, make a comment as mentioned in the figure 4.5 or save the post as favourite by clicking on save. the user can get in a discussion with the pet owner if he's willing to adopt which will be communicated in the next sprint.

The user can also report a post if he suspected a fraud as shown in the figure 4.6.

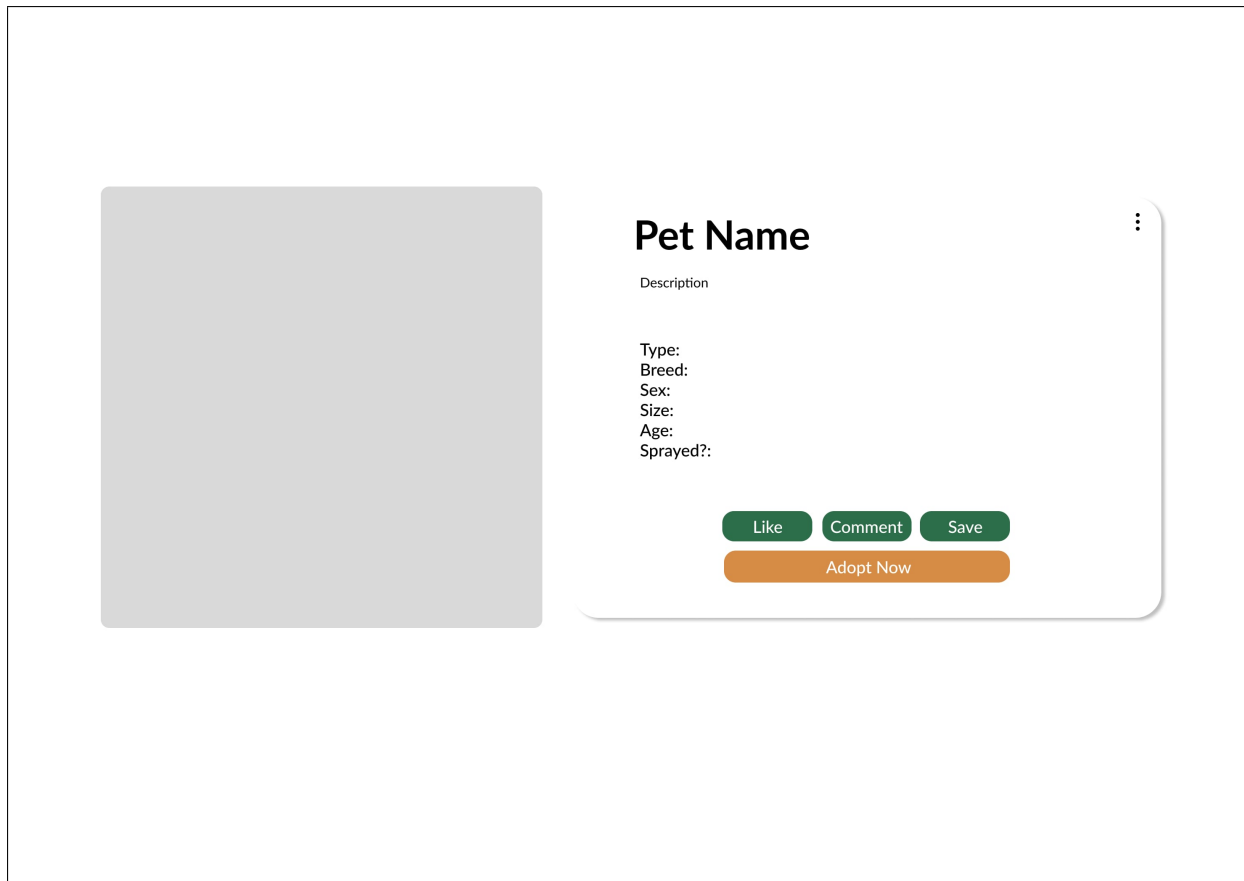


FIGURE 4.4 – Consult a post design

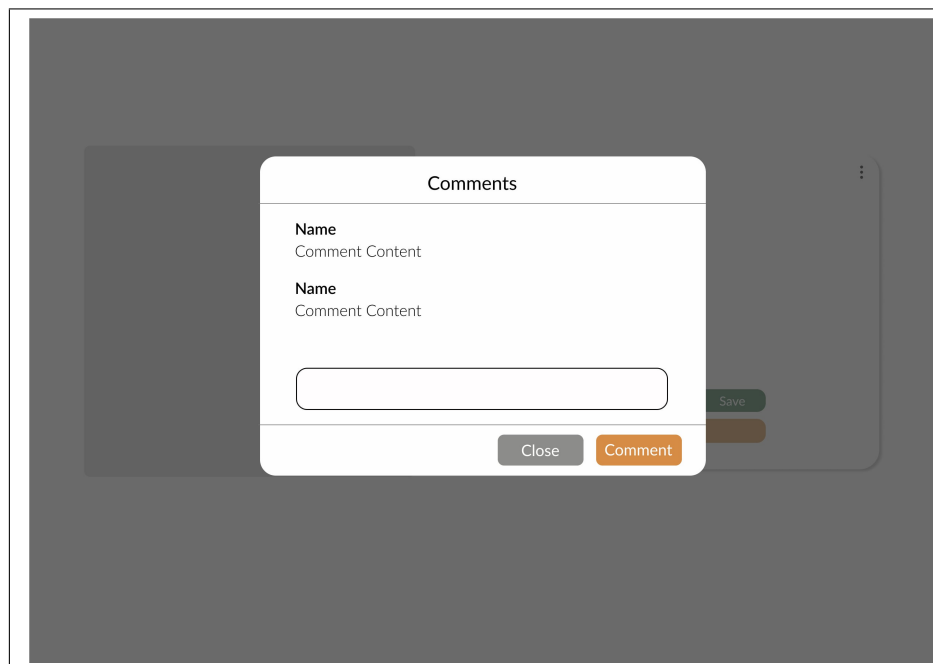


FIGURE 4.5 – Comment design

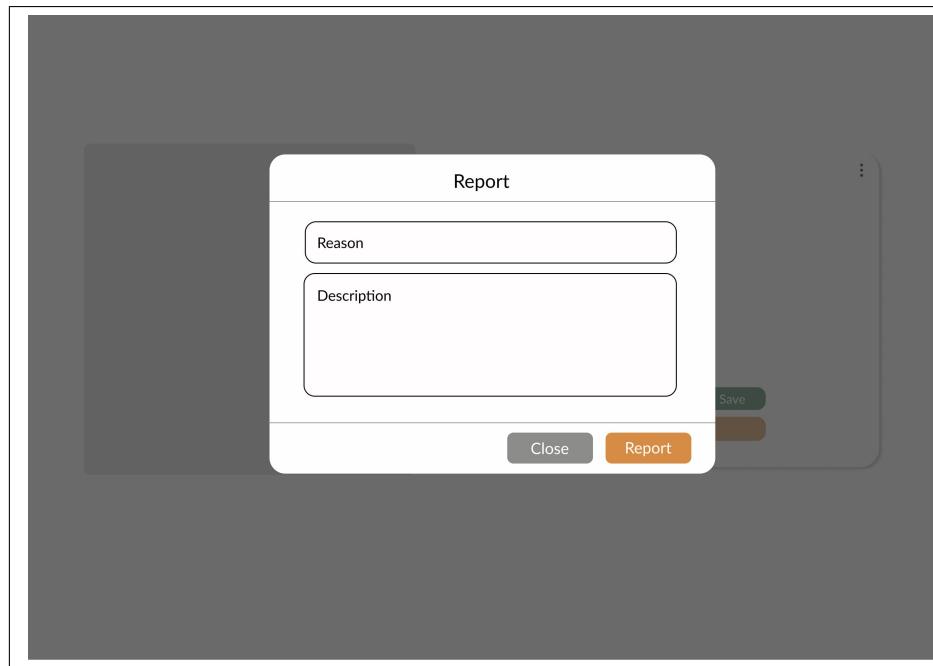


FIGURE 4.6 – Create report design

4.4 Conception

4.4.1 Create post object sequence diagram

The figure 4.7 represent the create post object sequence diagram. Once a user is authenticated he has the right to post a pet for adoption, and So create a post.

The user has to fill in the form with accurate data, a verification of data is done by the post, once it's verified , a find user is launched to extract the user details from the data base.

If the user doesn't exist a Not Found exception will be thrown to the controller, else the post will be created in the data base and saved as under review until the validation of the admin to appear to the rest of users which will be discussed in the next sprint.

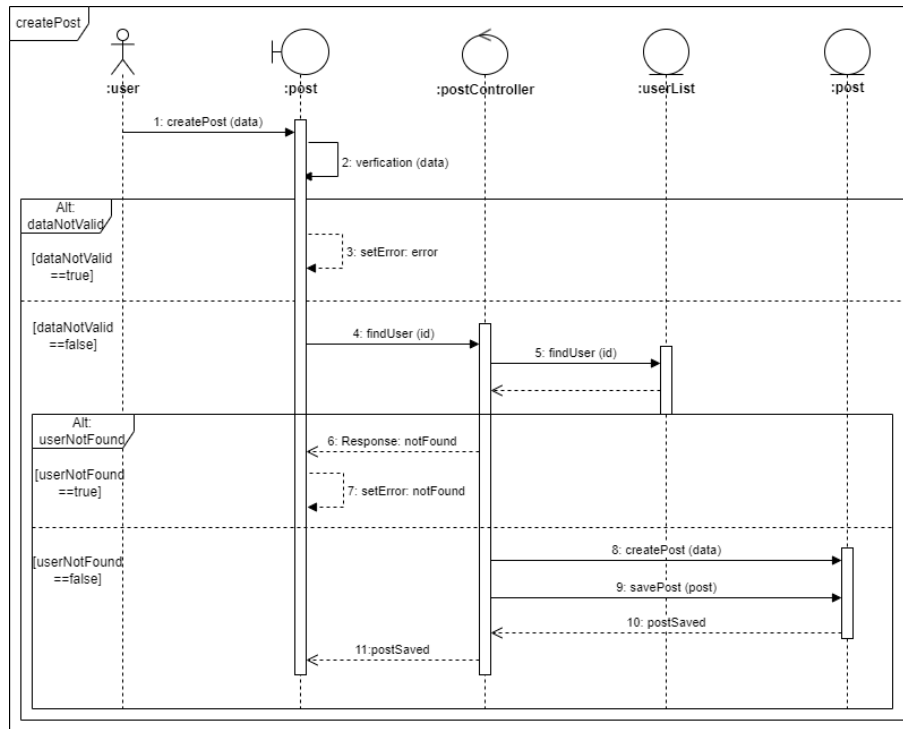


FIGURE 4.7 – Create post object sequence diagram

4.4.2 Update post object sequence diagram

The figure 4.8 represent the update post object sequence diagram. The user has the right to update the post in which he is the author. The user has to fill in a form of updated post a data validation is launched to ensure that the data is accurate , if The data is not validated an error message will be returned to the user, else the find post method is launched to extract the previous data of the post . If the post is not found a Not Found Exception will be thrown else an the new modification will be saved in the data base .

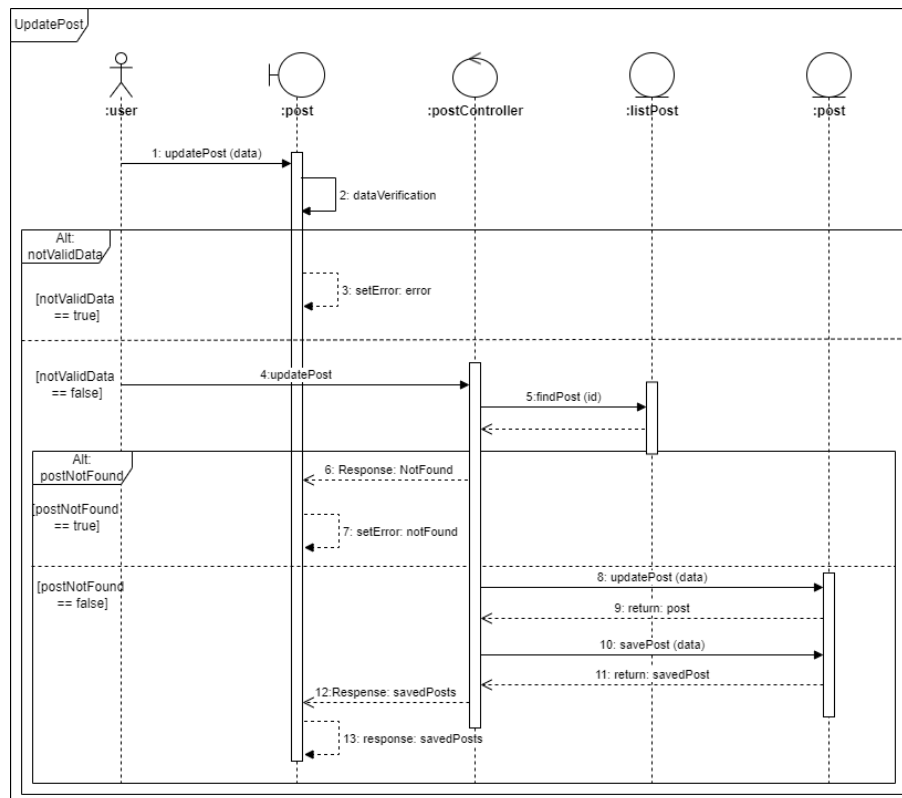


FIGURE 4.8 – Update post object sequence diagram

4.4.3 Like object sequence diagram

The figure 4.9 present how the user can like a post.

The user have the ability to like a post, The controller send a find user method to have the person who send the like if, the user is not found a Not Found Exception will be thrown else the controller send a find post, and as usual if it's not found a Not Found Exception will be thrown else the controller check for the last time if that user has already liked this post or not , if yes the response will be post already liked and nothing will happen , else a like will be created and saved in the data base and the number of like will be incremented by one.

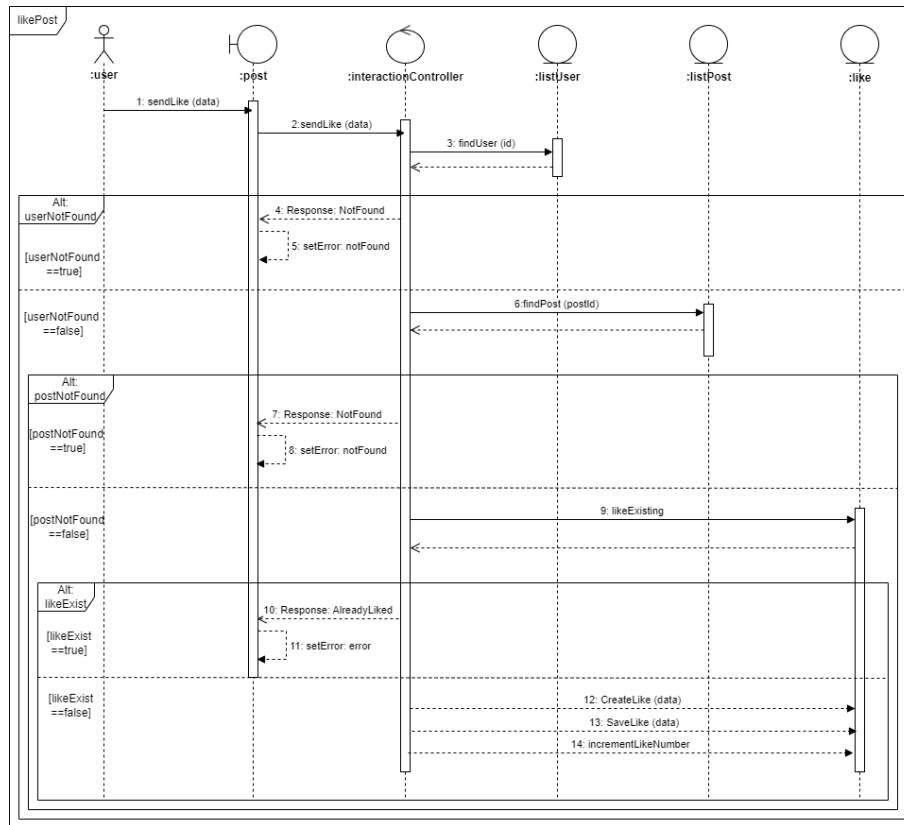


FIGURE 4.9 – Like object sequence diagram

4.4.4 Create comment object sequence diagram

Continuing with the post's interaction, the user can also create a comment on a post which will be presented in the figure 4.10.

A data validation of the input of the comment will be realised, if the data is verified the post Controller launch a find post method by id a verify the existing of post, if not a Not Found Exception will be thrown else comment will be created and saved in the data base.

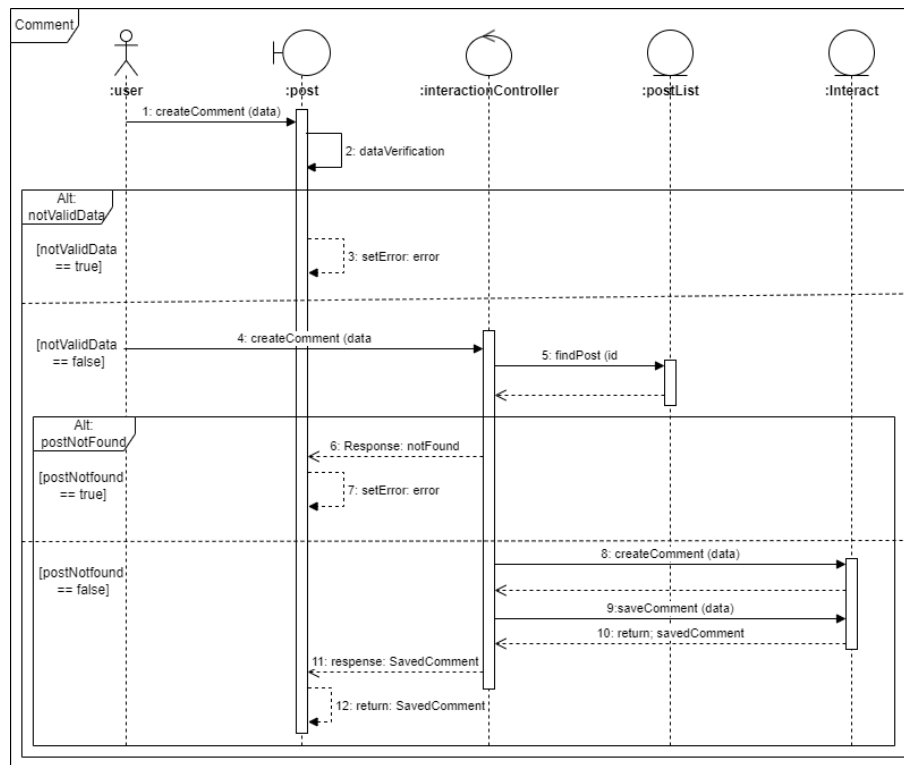


FIGURE 4.10 – Create comment object sequence diagram

4.4.5 Save post object sequence diagram

We close the interactions with saving a post which is shown in the figure 4.11. The user can save the post in his favourite a find user by id is launched from the controller.

If the user is not found a Not Found Exception will be thrown else a find post is launched from the controller, and same for post if the post is not found a Not Found Exception will be thrown else the post will be saved and the saved post will be returned to the user.

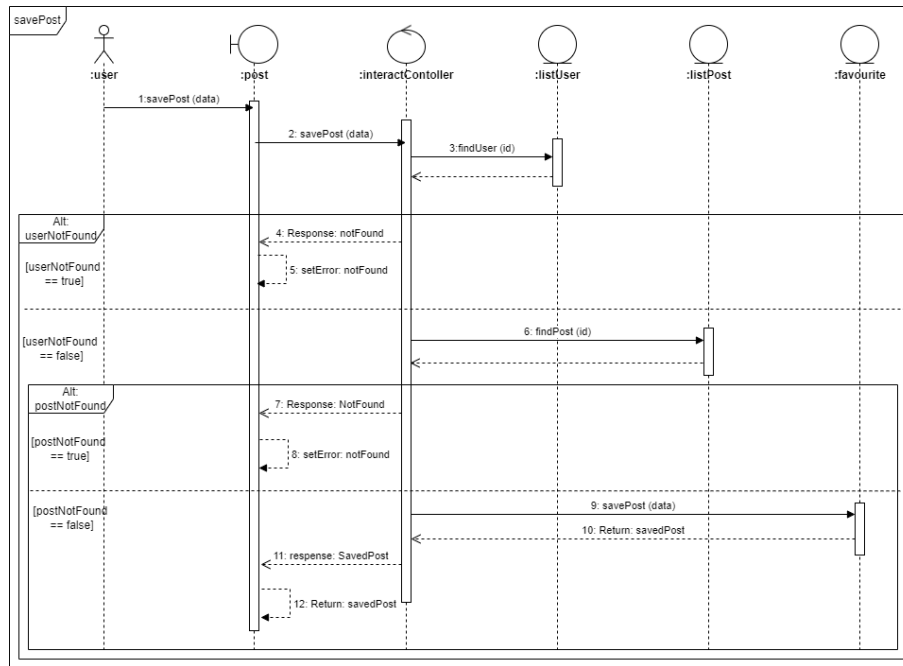


FIGURE 4.11 – Save post object sequence diagram

4.4.6 Report post object sequence diagram

The figure below figure 4.12 present the sequence object diagram of the creation of the report post.

First the data validation of the inputs of form, Once validated a find user method is launched to have the details of the user declaring the report.

If the user was not found a Not Found Exception will be thrown else the report will be created and saved successfully and as under review until the validation of the admin which will be discussed in the next sprint.

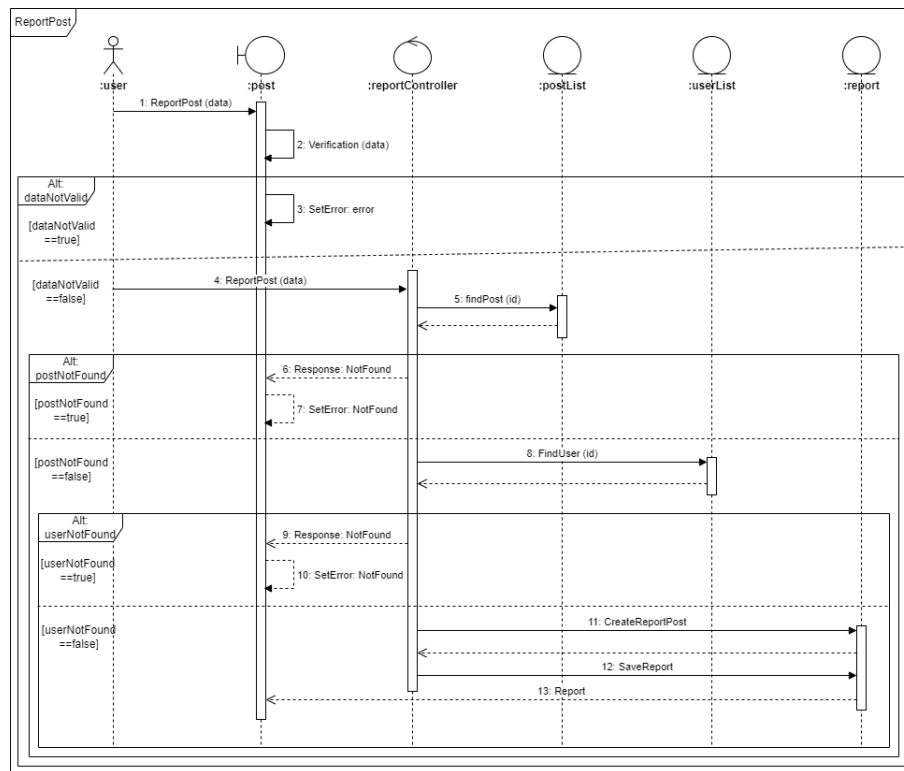


FIGURE 4.12 – Report post object sequence diagram

4.4.7 Consult feed page and post use case diagram

The figure 4.13 present the use case diagram of consulting feed page and post. The user have the access to get all the post validated in the feed page. Also all the post in the fees page are accessible to see more details in and apply the functionality discussed above.

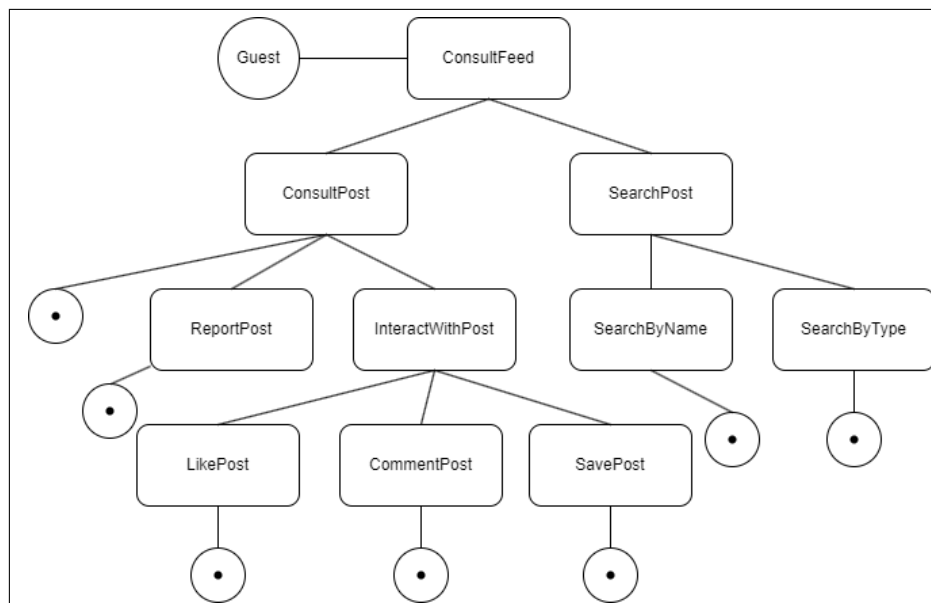


FIGURE 4.13 – Consult feed page and post use case diagram

4.4.8 Search post object sequence diagram

Search by pet name

The figure 4.14 present the conception of the search of posts by pet name. The user can search for a post by typing the pet name , the post controller launch a find post function to the data base which will retrieve the list of pets having the name, entered as parameter and their status is validated. this list is return after that to the user.

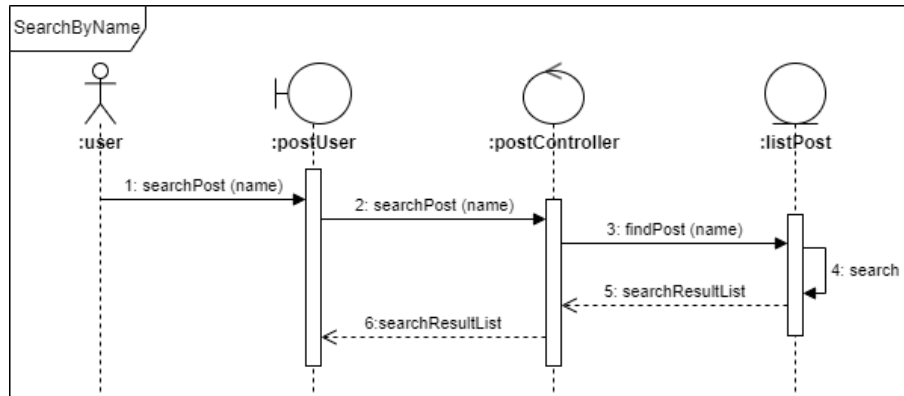


FIGURE 4.14 – Search by pet name

Search by type

The figure 4.15 represent the conception of the search of posts by pet type. As explained above with searching for posts by name, searching by type is typically the same, except the parameter that will be changed to type.

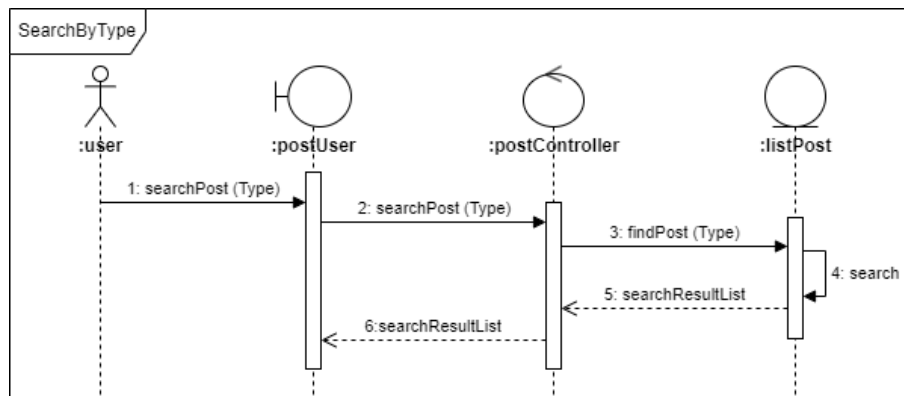


FIGURE 4.15 – Search by type

4.4.9 Delete post sequence diagram

The figure 4.16 represent the conception of delete post. The owner of a post can delete his own posts, so a find user function is launched to ensure having the details of the user. If the user doesn't exist a Not Found Exception will be thrown else the controller start the find post function.

Use Case	-Mark As Adopted
Precondition	-The user is the owner.
Post-condition	-The post is marked Adopted. The button is disabled.
Nominal scenario	-The actor click on adopt now. -The system verify if the user is the owner. -The s

TABLE 4.2 – Textual description "Mark as adopted"

If the post doesn't exist a Not Found Exception will be thrown else a the third obligation is launched to verify if the user is the same as the post author if not access denied will be returned else the deletion of the post will be effectuated.

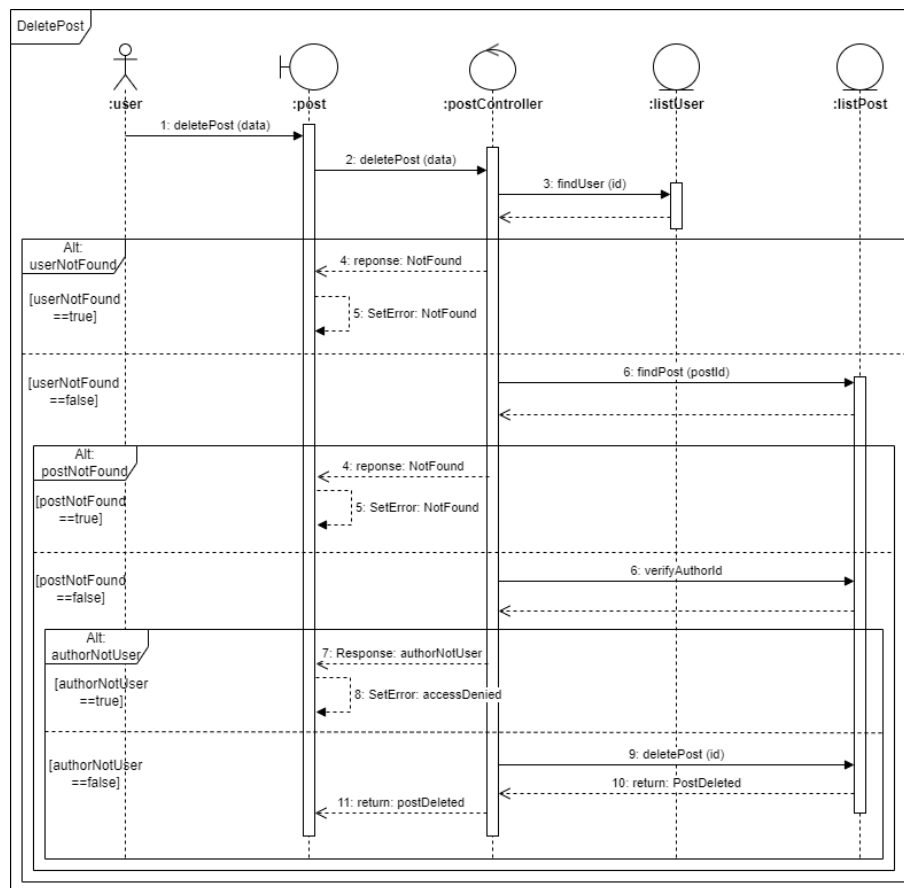


FIGURE 4.16 – Delete post sequence diagram

4.4.10 Textual description of use case : mark as adopted

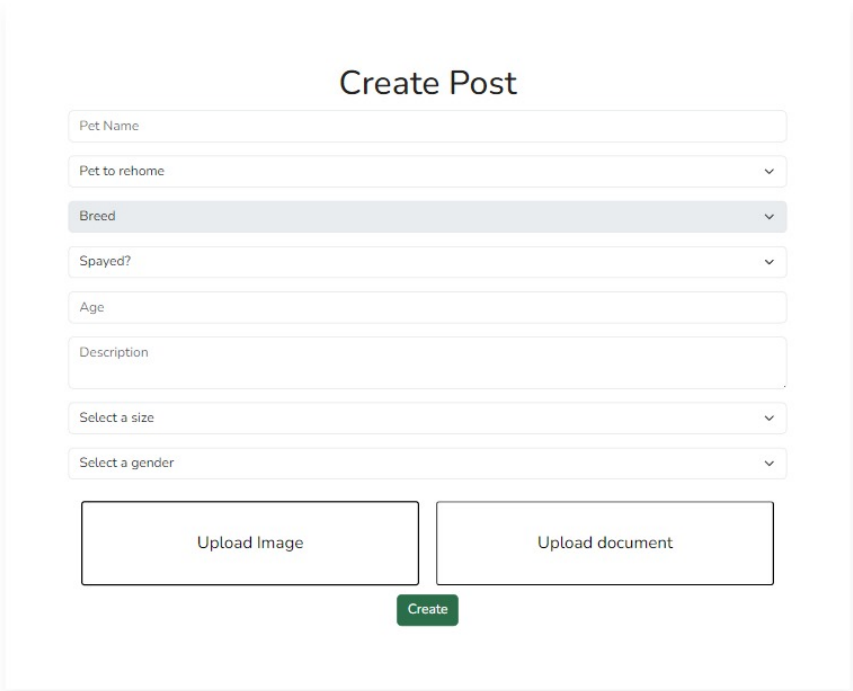
In the following, we present the textual description associated to the use case Mark As Adopted.

4.5 Realisation

In this section we will represent the work done in this sprint attached with screen-shots with the completed tasks.

4.5.1 Create Post interface

The figure 4.17 present the create post interface. Once the user click on post a pet in the either in the header or the home page, the create post interface will appear to the user with a form that he needs to fill in with all the necessary information. Then click on create to submit the form. If the data are valid the post will be created but still under review until the admin validate it which will be discussed in the next sprint.



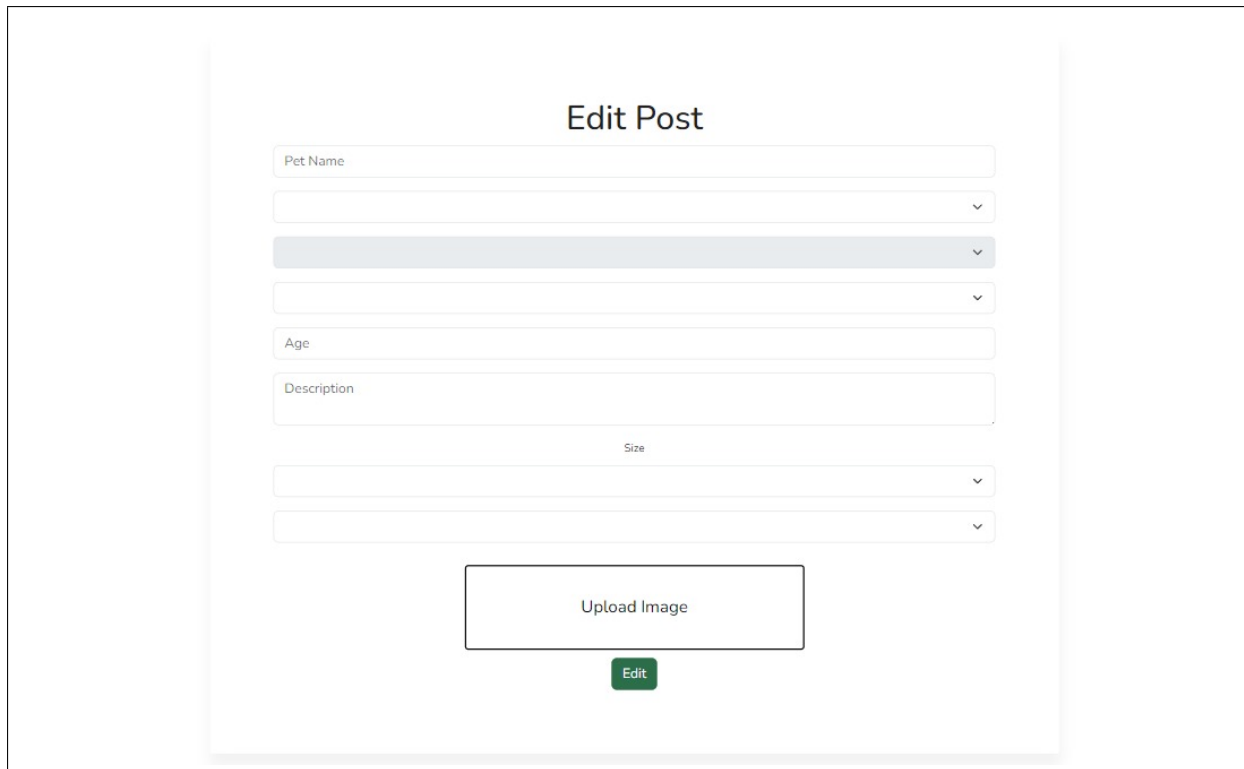
The image shows a 'Create Post' form with the following fields and buttons:

- Pet Name (text input)
- Pet to rehome (dropdown menu)
- Breed (dropdown menu)
- Spayed? (dropdown menu)
- Age (text input)
- Description (text input)
- Select a size (dropdown menu)
- Select a gender (dropdown menu)
- Upload Image (button)
- Upload document (button)
- Create (green button)

FIGURE 4.17 – Create Post interface

4.5.2 Update post

The figure 4.18 represent the update post interface. The user can edit his posts by filling the fields that he want to change. Then, click on edit to save the changes. If the data are not valid an error message will be shown .



The 'Edit Post' form is centered on a light gray background. It contains the following fields and controls:

- Pet Name:** A text input field.
- Dropdowns:** Four dropdown menus, the second of which is highlighted in light blue.
- Age:** A text input field.
- Description:** A larger text input field.
- Size:** A text input field.
- Buttons:** An 'Upload Image' button and a green 'Edit' button.

FIGURE 4.18 – Update post

4.5.3 Consult Post

The figure 4.19 present the consult post presentation. The user has the access to see multiple option and see the information provided with the pet. The user can like a post, comment it, save it and see if the pet is updated or not.

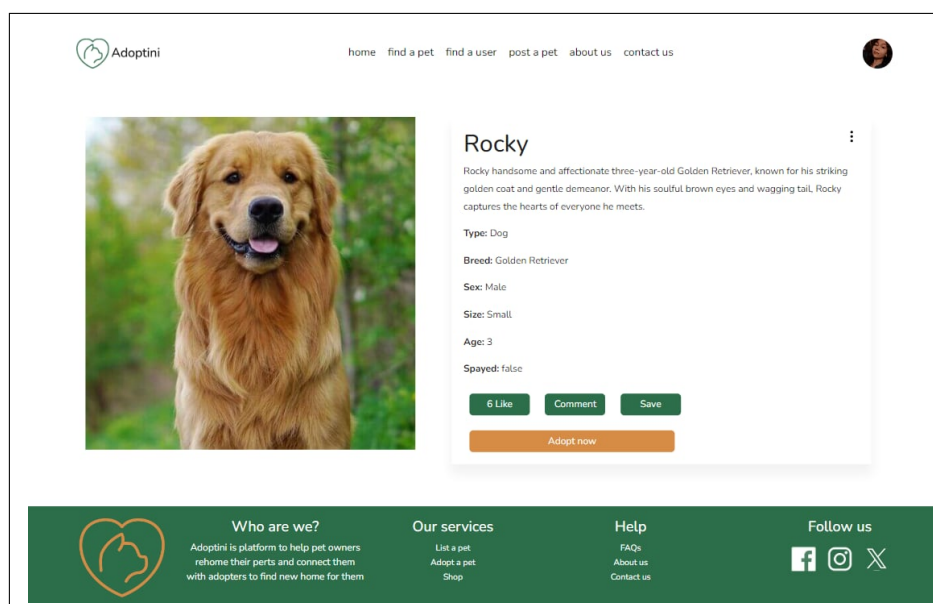


FIGURE 4.19 – Consult Post

4.5.4 Comment Post interface

as mentioned above the user can interact with the post by commenting the post and it's showed in the figure 4.20 bellow.

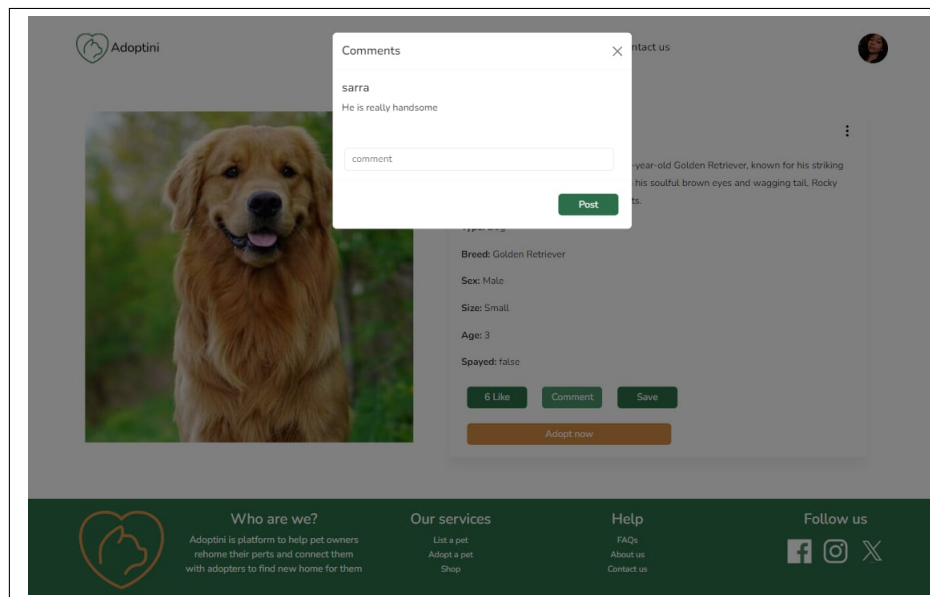


FIGURE 4.20 – Enter Caption

4.5.5 Save post Interface

Once the user click on save as the figure 4.19, the button of saved becomes unsaved , if the user is willing to delete it from his saved list as shown in the figure 4.21.

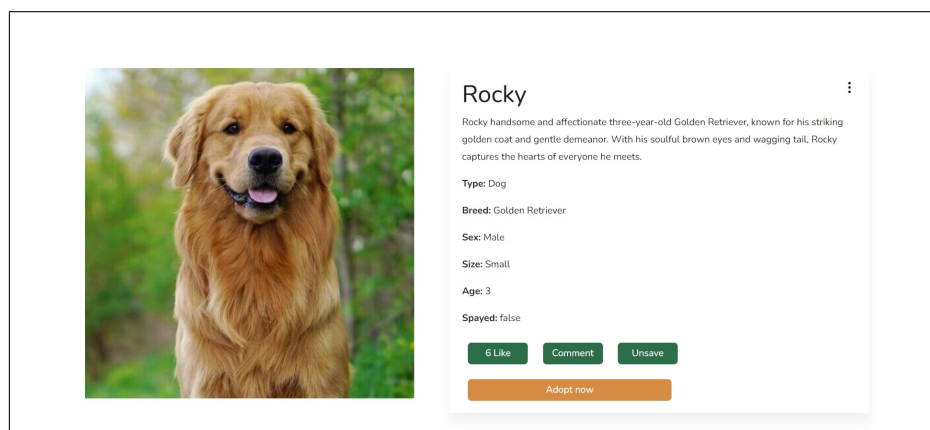


FIGURE 4.21 – Post already saved interface

4.5.6 Report Post interface

The user if is not the author of the post have the right to report it as shown in the figure 4.22 , once he click on report the user will be redirect to a report page where he should fill in the form in the figure 4.23 and click on the create to save the report .

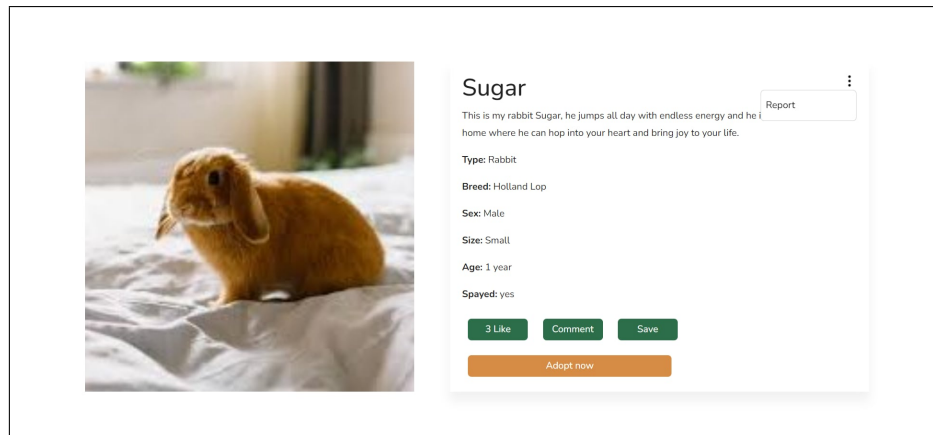


FIGURE 4.22 – Enter Caption

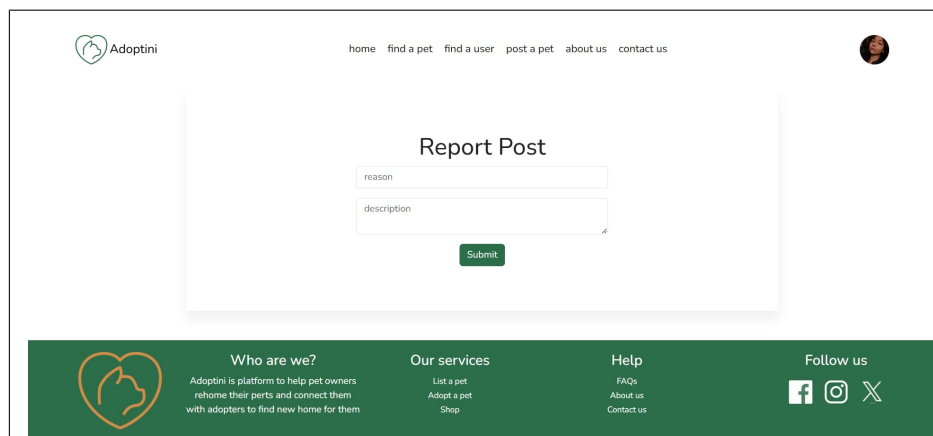


FIGURE 4.23 – Report Post interface

4.5.7 Consult feed interface

The user has the access to see all the pets presented in the feed as mentioned in the figure 4.24

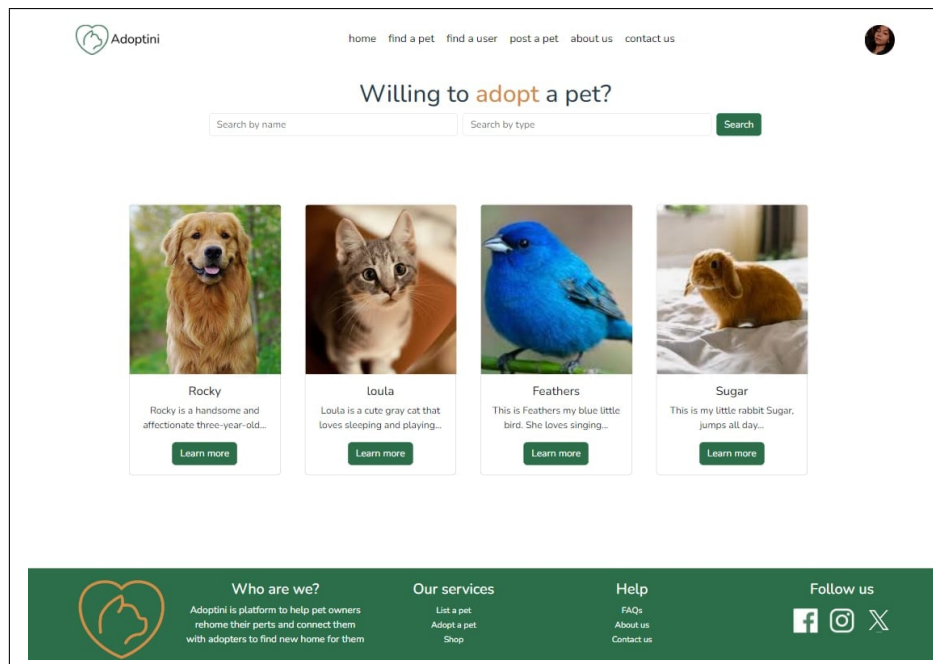


FIGURE 4.24 – Consult feed interface

4.5.8 Search for post interface

As discussed in the previous section the user can make a search by pet name or by type as shown in the figure 4.25.

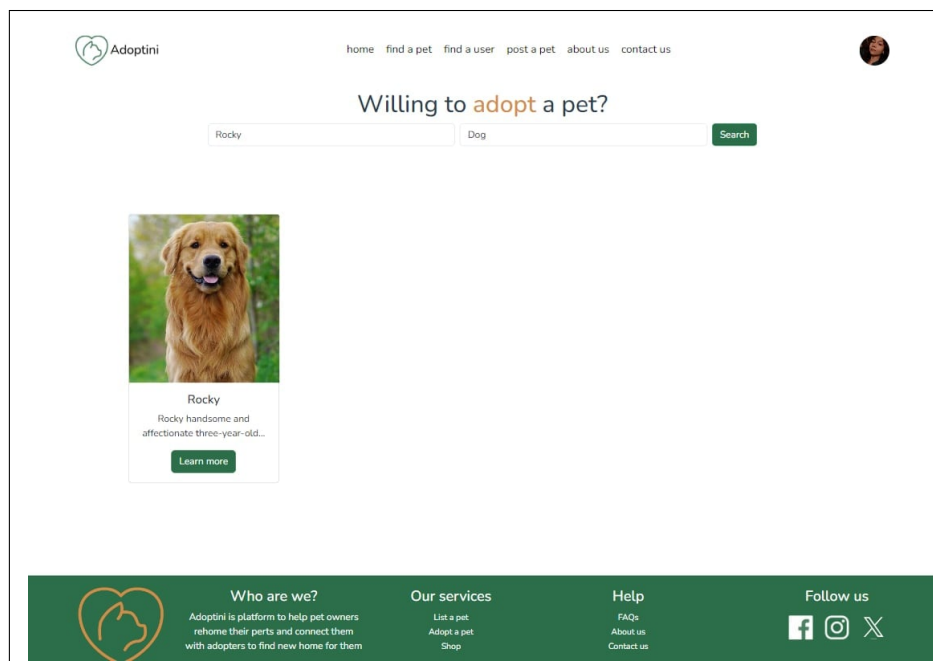


FIGURE 4.25 – Search for post interface

4.5.9 Mark as adoption

The user is willing to adopt the pet can enter into a discussion with the owner (which will be discussed in the upcoming sprint), But once the pet is adopted the admin can change the pet status to adopted, the button adopt now is disabled as shown in the figure 4.26 so he don't receive any more conversation about this pet.

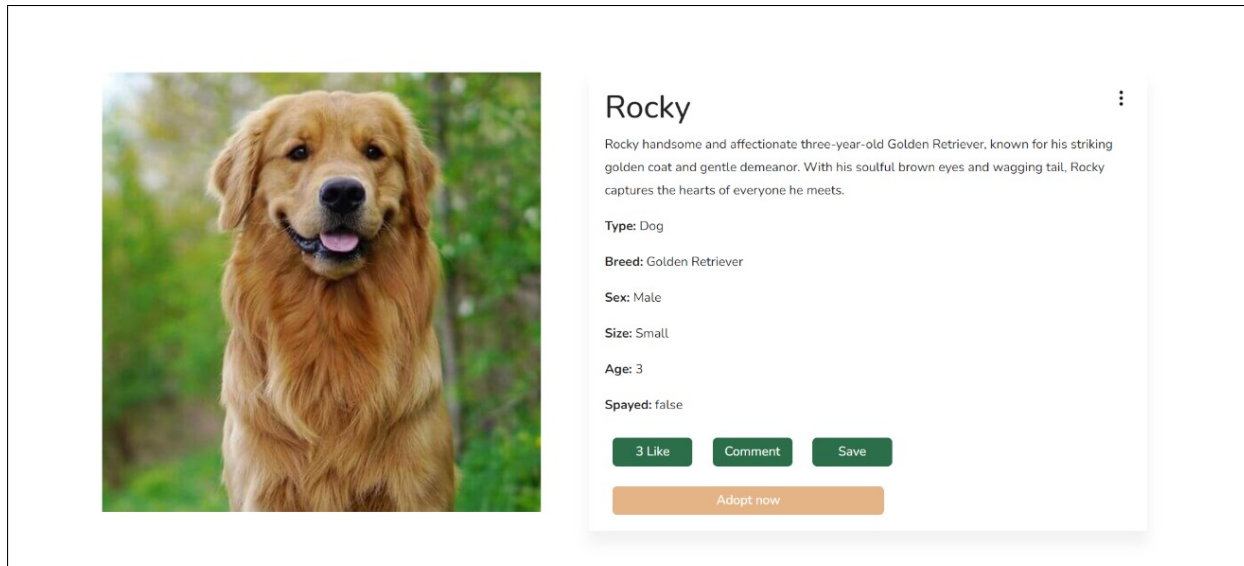


FIGURE 4.26 – Mark as adoption

4.6 Conclusion

In this chapter we discussed the third sprint of our project, which was concerned about the posts and process that you can do to interact with other's pets. we presented our backlog product, the conception and how did we think to realise our sprint. In the next Chapter, we will present the fourth sprint which will contain about admin.

4.7 Burn down chart sprint 3

In this sprint we have to develop a multiple functionalities and multiple design so this sprint was the longest sprint.

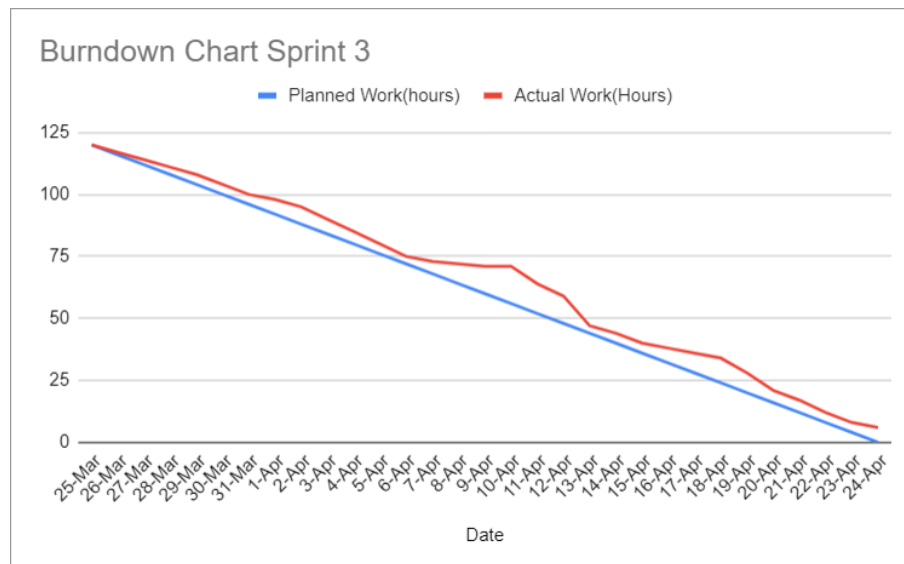


FIGURE 4.27 – Burn down chart sprint 3

Chapter5

SPRINT 4 : ADMIN MANAGEMENT

5.1 Introduction

In this chapter, we will present our fourth sprint which discuss the admin management system.

In this sprint we will start with the backlog of the sprint, we will present also the desired design interfaces, then we will continuing with the conception of the sprint, finally we will finish with a representation of the admin management system of our platform.

5.2 Backlog of the sprint 4

Id	User Story	Complexity
1	As an admin i can access to an admin dashbord.	4
2	As an admin i can consult all users.	2
3	As an admin i can assign a user as admin	3
4	As an admin i can assign a admin as user	3
5	As an admin i can change the status of a user account to inactive.	4
6	As an admin i can consult all posts.	2
7	As an admin i can reject a post.	3
8	As an admin i can validate a post.	3
9	As an admin i can consult all reports.	2
10	As an admin i can search for a report.	4
11	As an admin i can validate a report.	4
12	As an admin i can reject a report.	4

TABLE 5.1 – Sprint 4 backlog

5.3 Interface design : Modeling

5.3.1 User story : Access to an admin dashboard design

Once a user has the role of admin he had the access to an admin dashboard which will be represented in the figure 5.1.

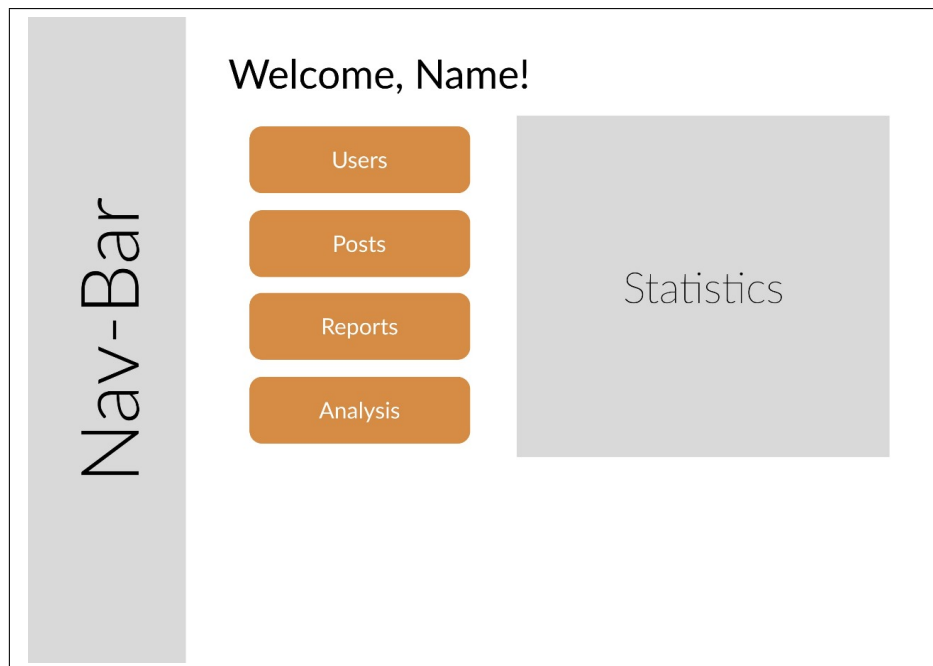


FIGURE 5.1 – Access to an admin dashboard design

5.3.2 User story :Consult all posts design

The admin has the right in the admin dashboard to consult all the post don't matter what is their status validated, rejected or under review as mentioned in the figure 5.2.

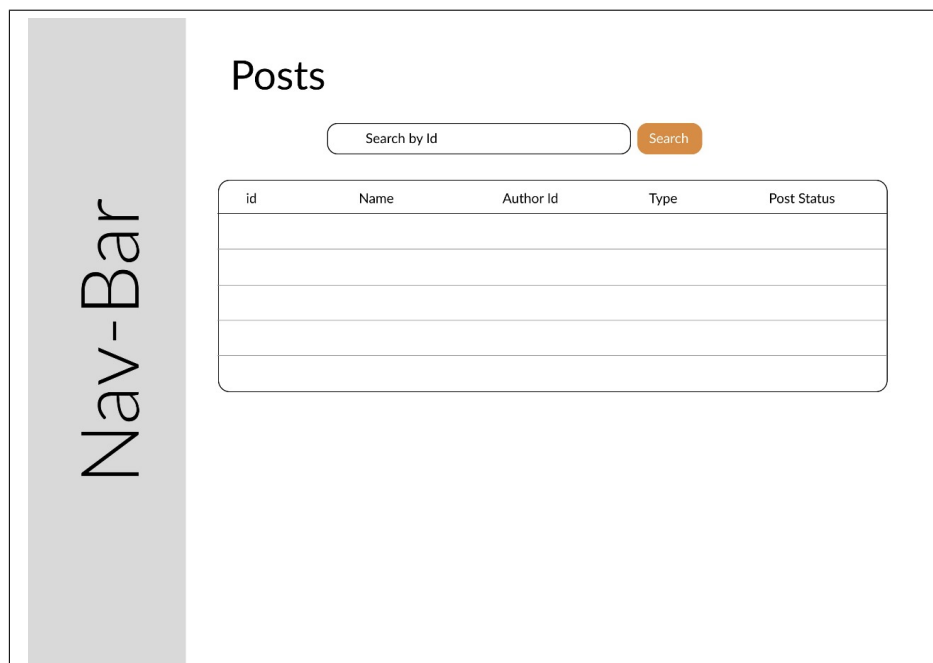


FIGURE 5.2 – Consult all posts design

5.3.3 Consult post details design

The admin can consult the post details as presented in the figure 5.3.



FIGURE 5.3 – Consult post details design

5.3.4 User story : Reject post design

The admin has the right to reject a post so it's not posted to the public if he suspect a problem like mentioned in the figure 5.3 by clicking i=on under review button.

5.3.5 User story : Validate Post design

As mentioned before the admin has also the right to validate post after review it as shown in the figure 5.3 by clicking on under review button.

5.3.6 User story : Consult all report design

The report created by the users are consulted by the admin in the admin dashboard as shown in the figure5.4.

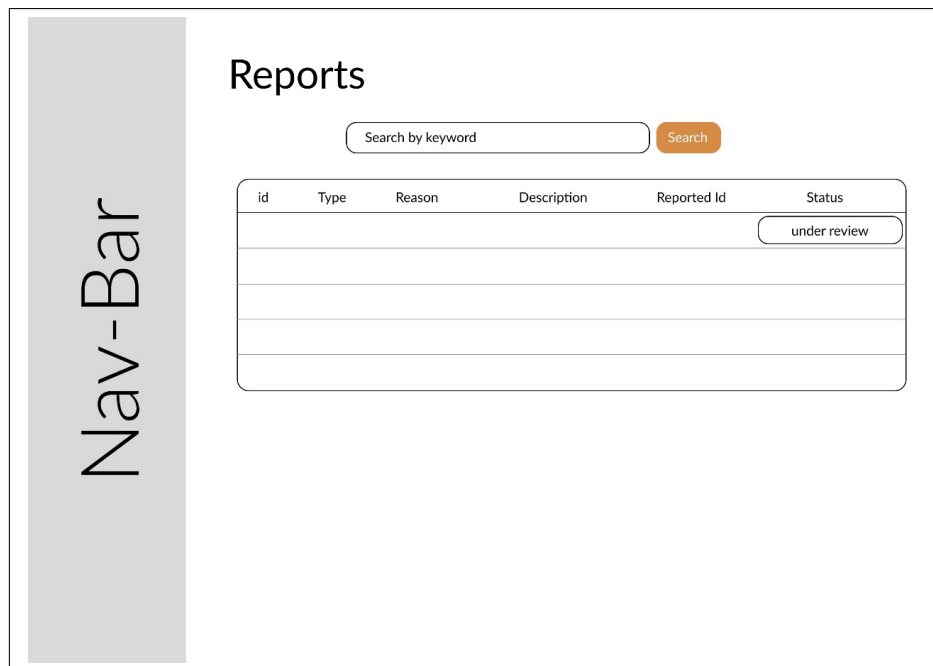


FIGURE 5.4 – Consult all report design

5.3.7 User story : Search for reports by keywords

The admin has the ability to search for reports by keywords in the description by typing the words in the nave bare like the design 5.4 in the nave bar.

5.3.8 User story : Validate and reject reports design

The reports received are either validated or rejected by the admin like the design 5.4 by clicking on the button under review.

5.3.9 User story : Consult all users

The admin by accessing to his dashboard can see the list of users no matter what is their account state as shown in the design 5.5.

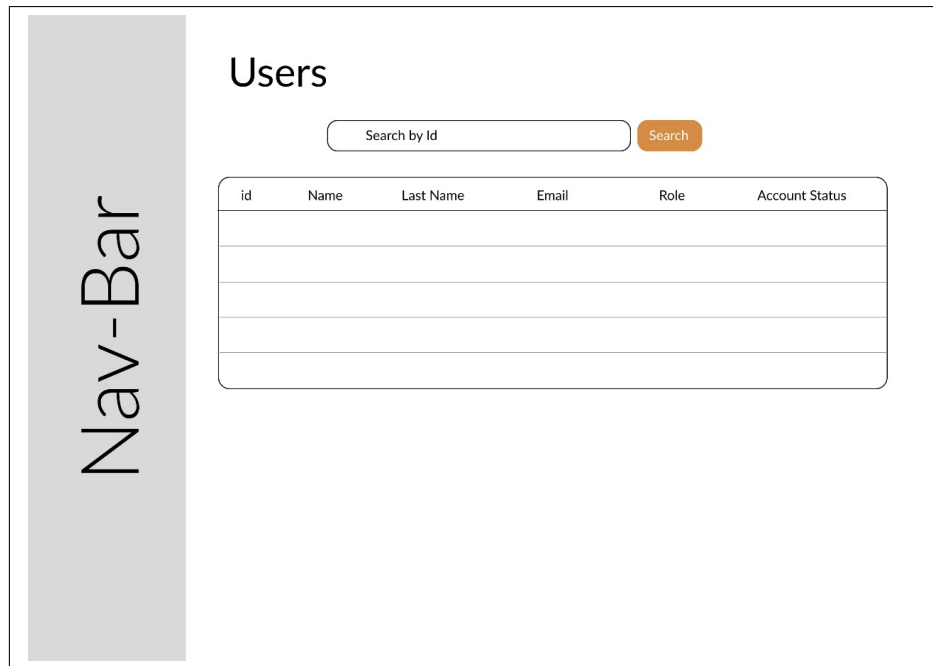


FIGURE 5.5 – Consult all users design

5.3.10 User story : Consult user details

The admin can consult the user details as shown in the figure 5.6.

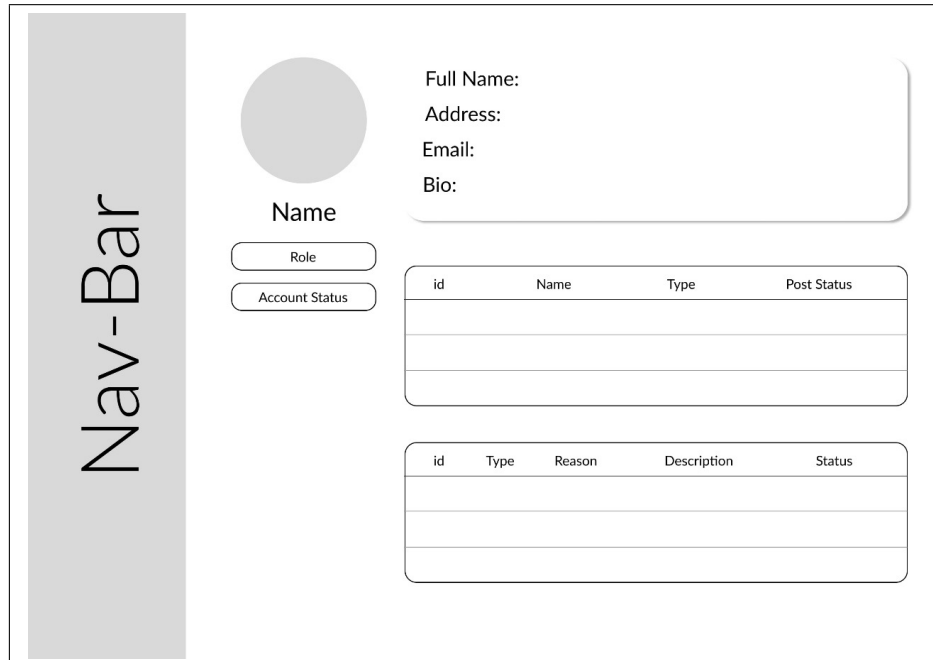


FIGURE 5.6 – Consult user details design

5.3.11 User story : Assign a role to a user

The admin has the ability to assign other users as admin so they have access to the admin dashboard also as indicated in the figure 5.6 and also and admin can return the

role of an admin to a user back again.

5.3.12 User story : Change account state

Some users has been attend a maximum number of alert or reporting can be banned by the admin so their account aren't available anymore as shown in the figure 5.6.

5.4 Conception of sprint 4

5.4.1 Admin use case diagram

Authenticated as an admin has multiple functionalities that the use don't have the figure 5.7 explain these functionalities.

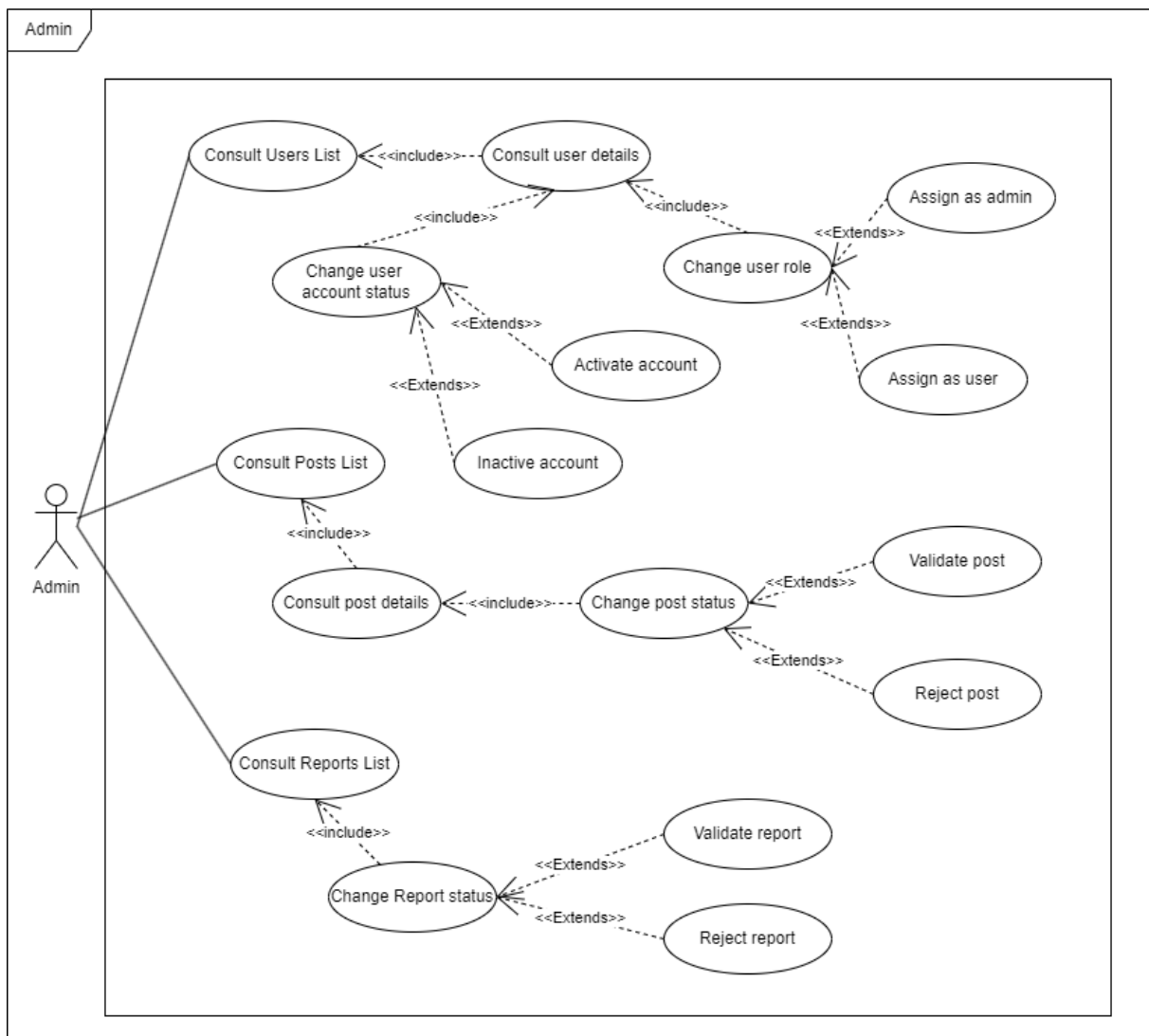


FIGURE 5.7 – Admin use case diagram

5.4.2 Access to a admin dashboard object sequence diagram

Once the user is authenticated as admin , he has the ability to access to multiple functionalities in the admin dashboard that isn't accessible by other role as shown in the figure 5.8.

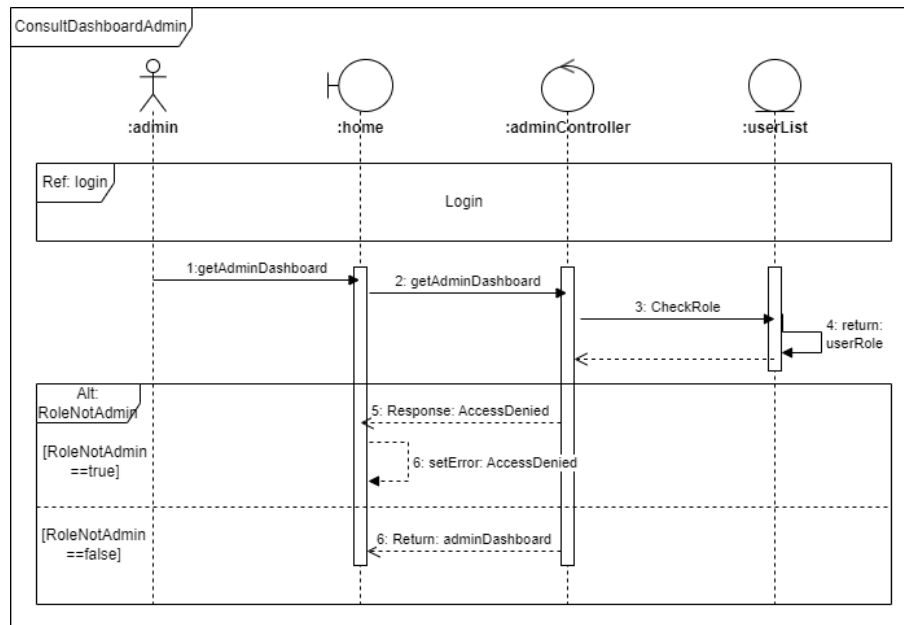


FIGURE 5.8 – Access to a admin dashboard object sequence diagram

5.4.3 Reject post object sequence diagram

The admin can consult the post with details and reject the post if he suspect the user or the information posted in as mentioned in the figure 5.9 , so the post is not displayed to other users.

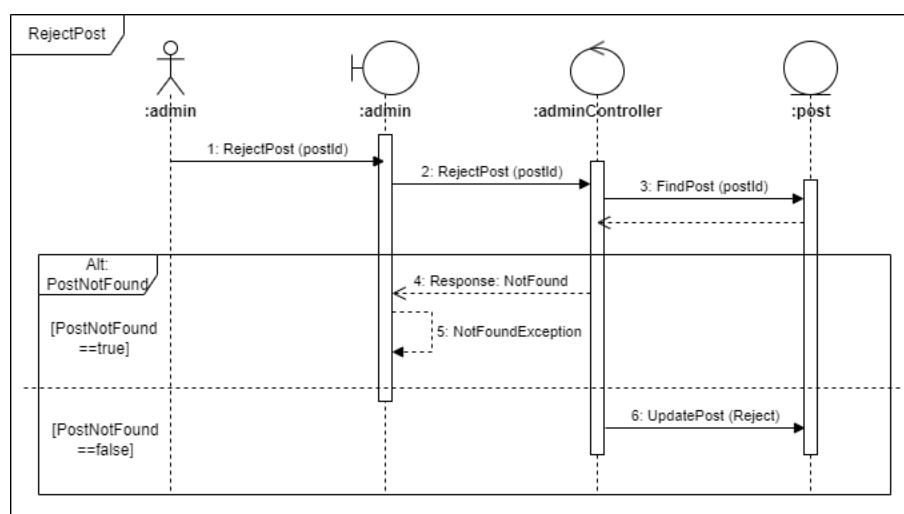


FIGURE 5.9 – Reject post object sequence diagram

5.4.4 Validate post object sequence diagram

The admin has also the ability to validate a post, the figure 5.10 show how the process is working .

Once he's authenticated as admin the user can change the post status to validated, a find post by Id will be launched by admin controller.

If the post is not found a not found exception will be returned to the admin, else the post will be validated and updated.

Once the post is validated then is now accessible to all other users.

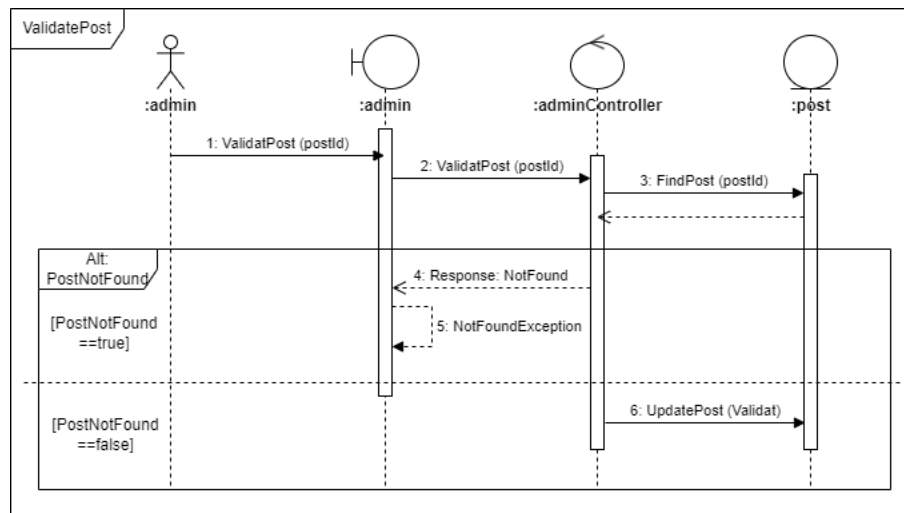


FIGURE 5.10 – Validate post object sequence diagram

5.4.5 Search for report by keyword object sequence diagram

The admin as shown in the figure 5.11 has the right to search for reports by keywords.

Once the admin access to consult all reports, he can search for the reports by keyword in the report description.

The admin controller launch a find reports by keyword, If one or many reports have the keyword will be returned, else a not found will be returned.

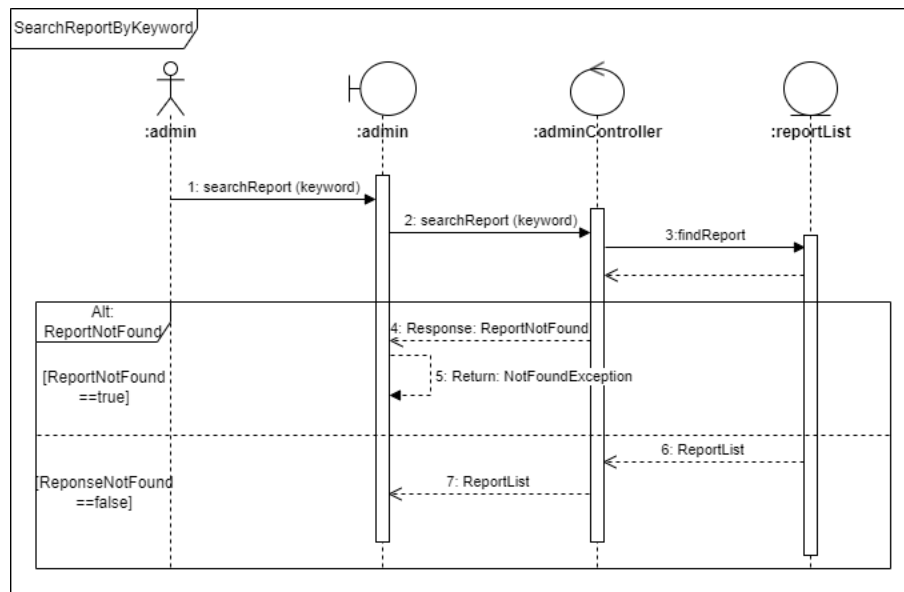


FIGURE 5.11 – Search for report by keyword object sequence diagram

5.4.6 Change the reports status

The figure 5.12 and 5.13 show the object sequence diagram of either validate or reject a post.

After consulting the reports details the admin have the right to either validate or reject a report.

Validate report object sequence diagram

if the report is find and get all details, the admin controller send a request to check the report status.

If the report is already validated, the response will be already validated.

Else the status of the report will be updated to validated, the warning count will be incremented by one and the report will be saved.

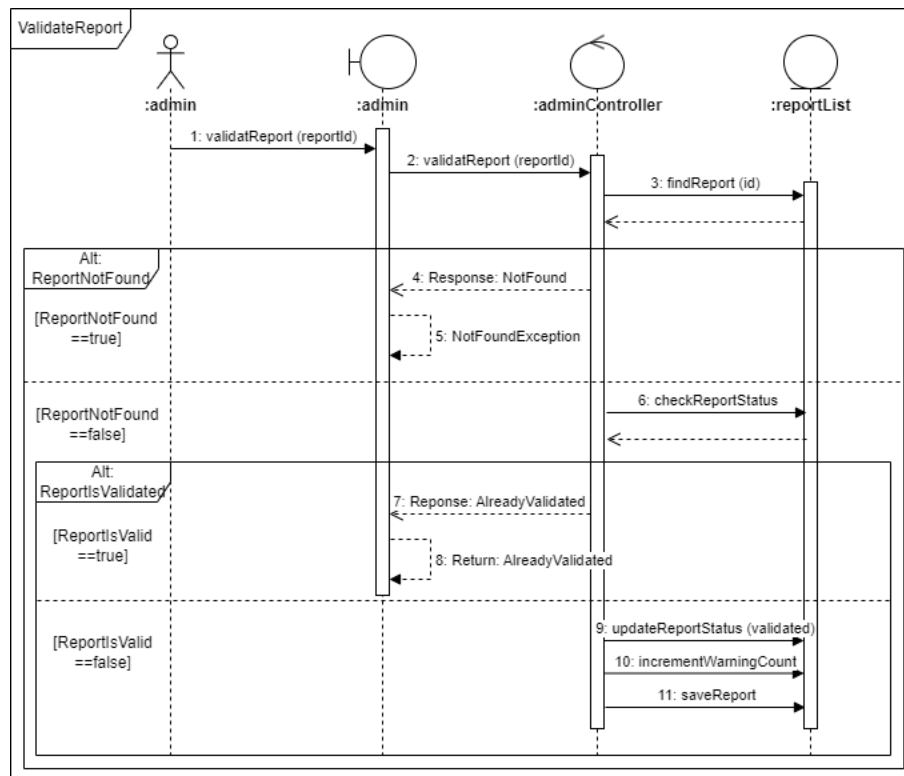


FIGURE 5.12 – Validate report object sequence diagram

Reject report sequence diagram

If after consulting the report details the admin decided to not validate the report the figure 5.13 show the object sequence diagram of how to reject the report.

As usual the admin controller will launch a find report by Id and depending on the return of the data base .

If the report is not found a not found exception will be returned, else the controller will update the report status to rejected and save the save the changes.

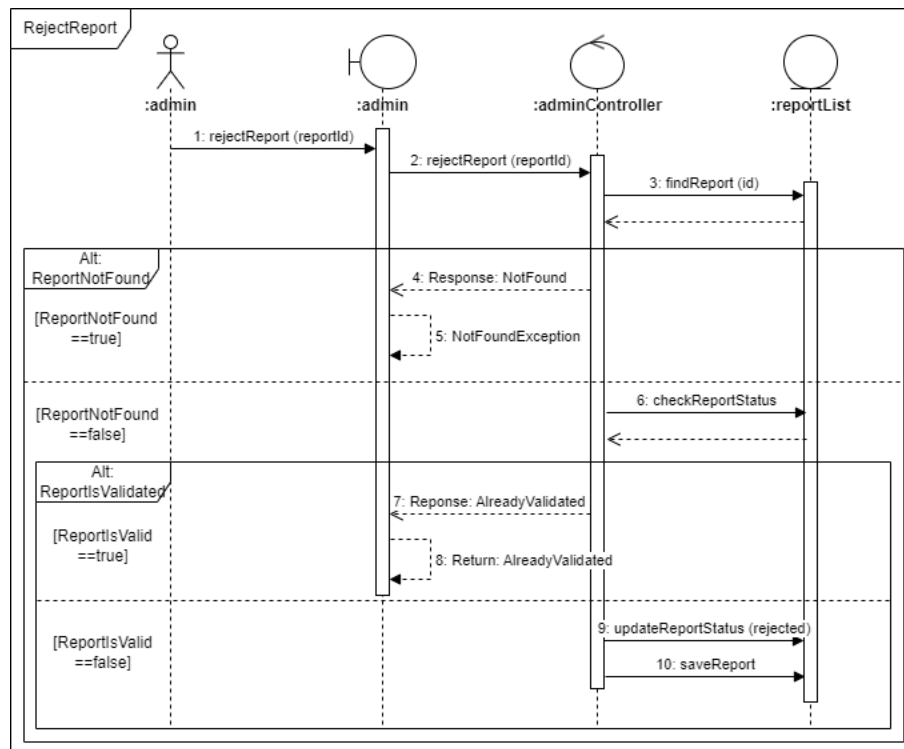


FIGURE 5.13 – Reject report sequence diagram

5.4.7 Assign role to user

The admin has the ability to change the role of a user of the platform the figure 5.14 and 5.15 show the conception of these user story.

Assign as admin object sequence diagram

The admin send a request to change the user role, the admin controller launch a find user by id to access to the user in the data base, if it's validated and the user is found the user role is updated to admin and the changes are saved .

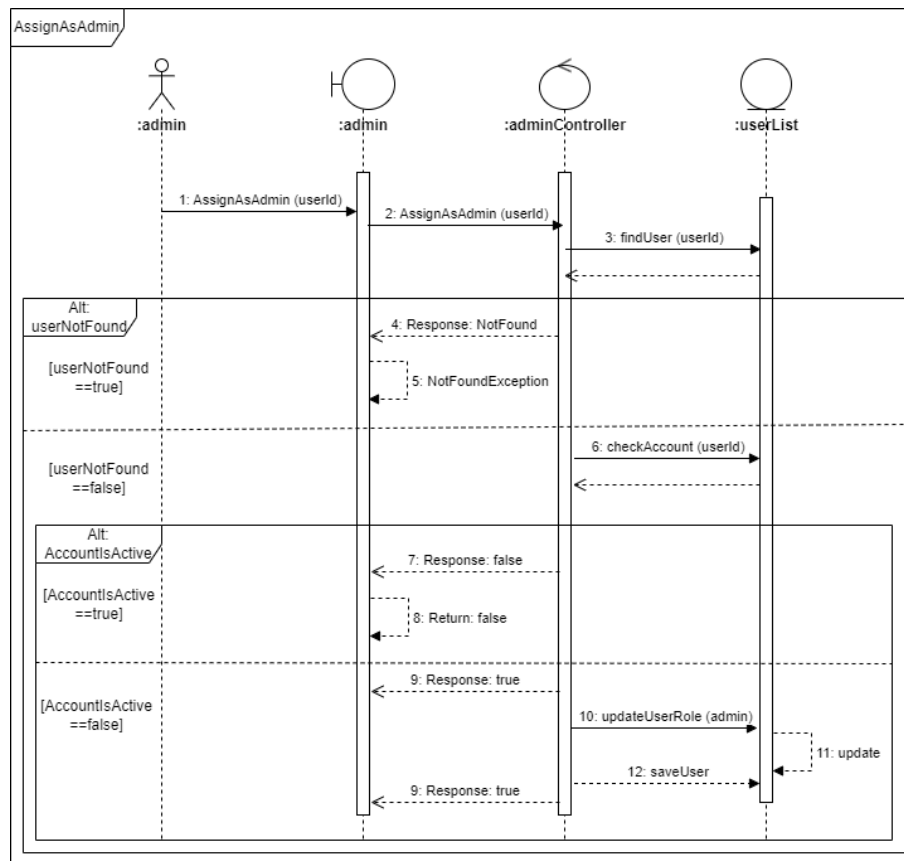


FIGURE 5.14 – Assign as admin object sequence diagram

Assign as user object sequence diagram

The admin has the ability also to change the role of an admin to a user, which presented in the figure 5.15.

And same steps as the assign user as admin, the only difference that the role given is a user role.

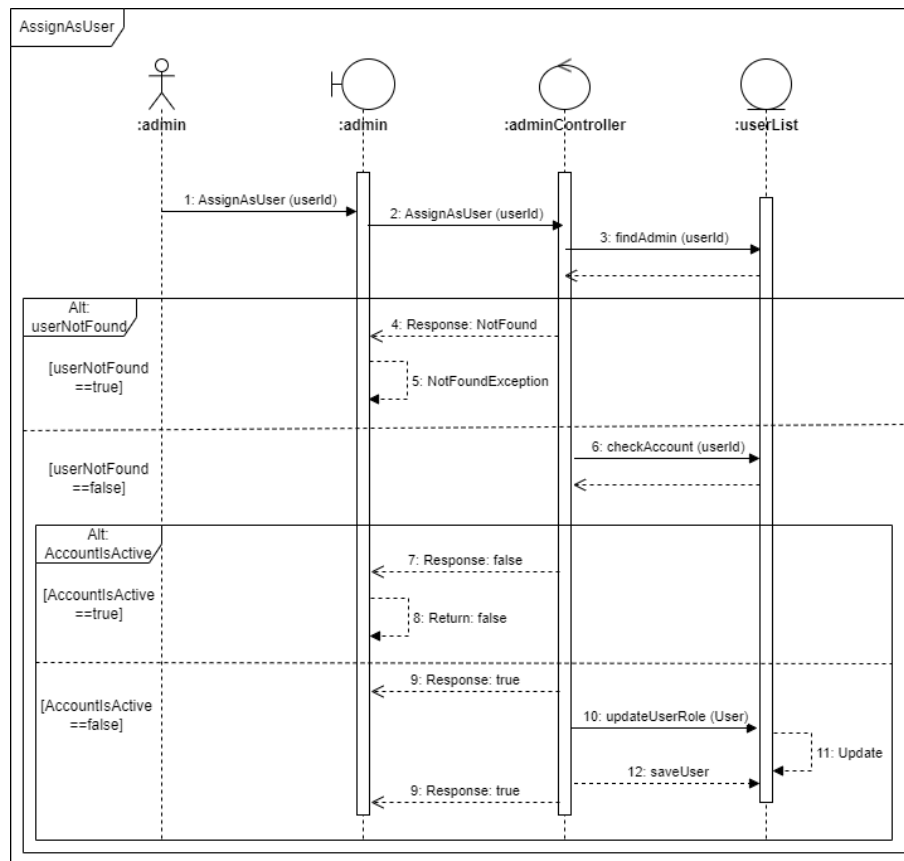


FIGURE 5.15 – Assign as user object sequence diagram

5.4.8 Change account state object sequence diagram

The admin has the right to change account state to inactive as explained in the figure 5.16. The admin controller start to find the user by id from the user's list, if the user is not found a not exception will be thrown, else the admin controller will check the role of that user found.

If the user is a Admin an error will be throw, else the admin controller will send a change account state to inactive.

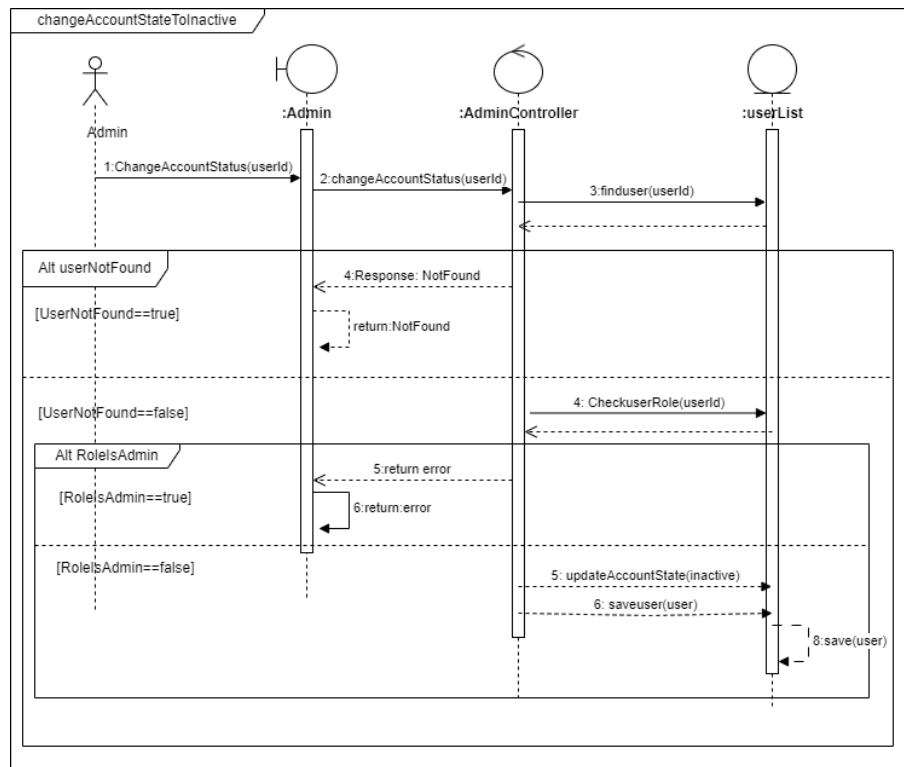


FIGURE 5.16 – change account state to inactive object sequence diagram

For the part of changing the account state to active, the user story start with a request from the admin, then retrieving the user from the users list, and finally updating the account state to active, as shown in the figure 5.17.

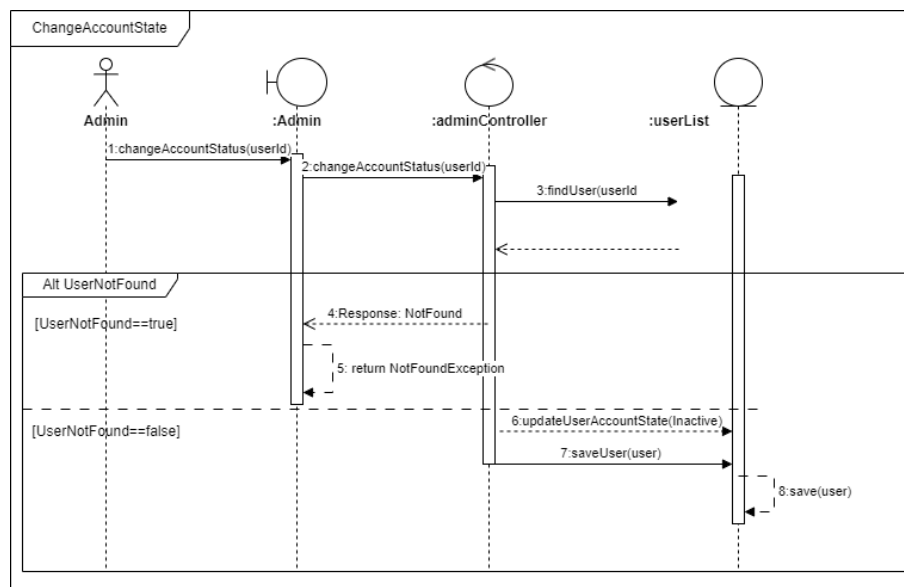


FIGURE 5.17 – changing the account state to active object sequence diagram

5.5 Realisation

In this section we will represent the work done in this sprint attached with screenshots with the completed tasks.

5.5.1 Posts list interface

The figure 5.18 represents the interface where an admin can consult all posts. The admin can search for a post by ID. By clicking on the ID of a one of the posts, an interface that shows that post's details which will be mentioned in later on.

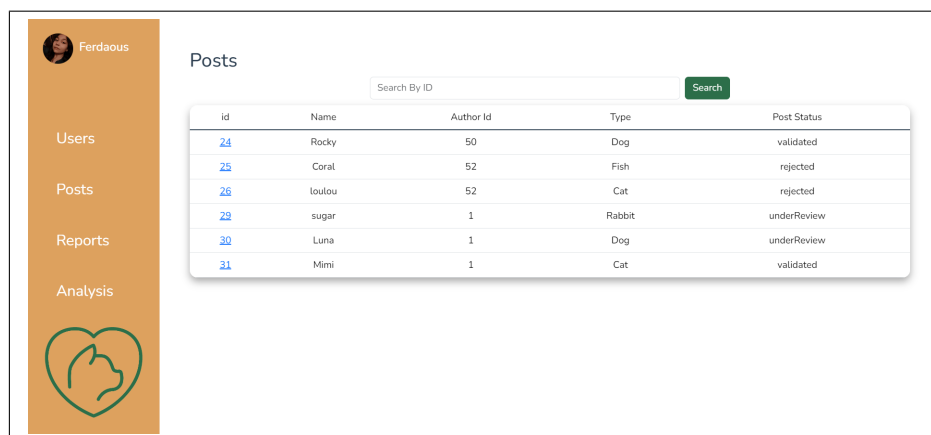


FIGURE 5.18 – Posts list interface

5.5.2 Post details interface

The figure 5.19 shows the consulting interface of the post details. An admin can validate or reject a post by clicking on the post status drop-down.

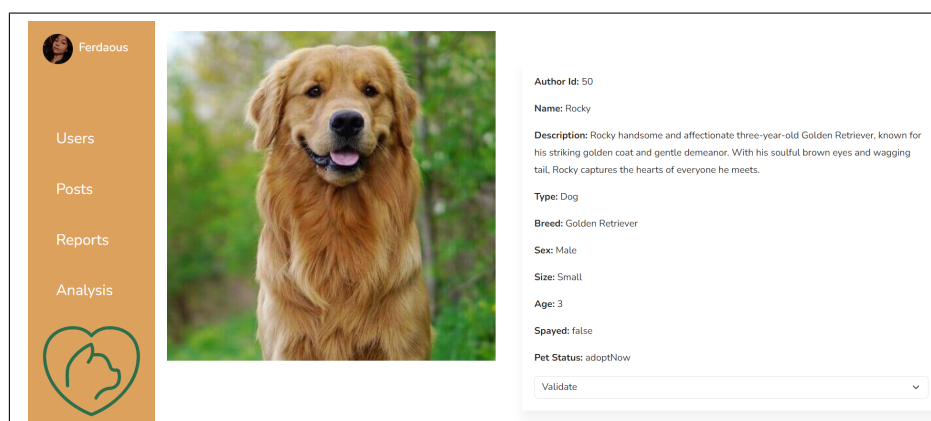
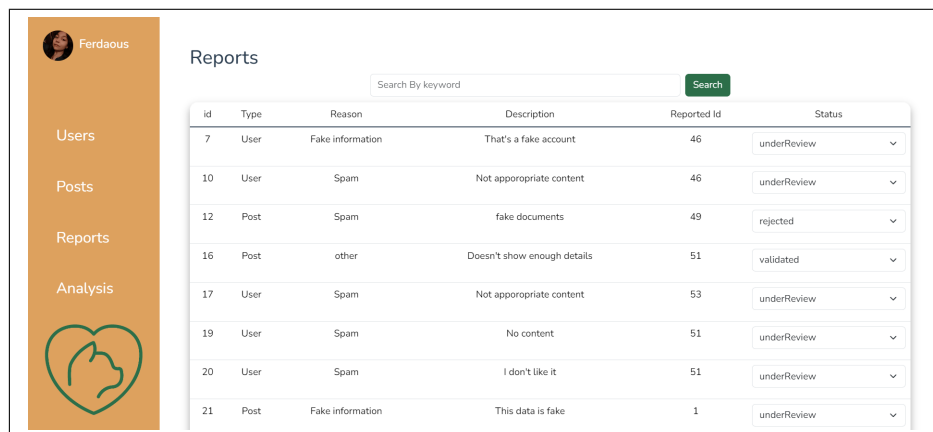


FIGURE 5.19 – Post details interface

5.5.3 Report list interface

An admin can consult the list of reports as it is represented in the figure 5.20. An admin has the accessibility to validate or reject a report by clicking on the report Status.

To search for a report, an admin can write a keyword and then click on search.

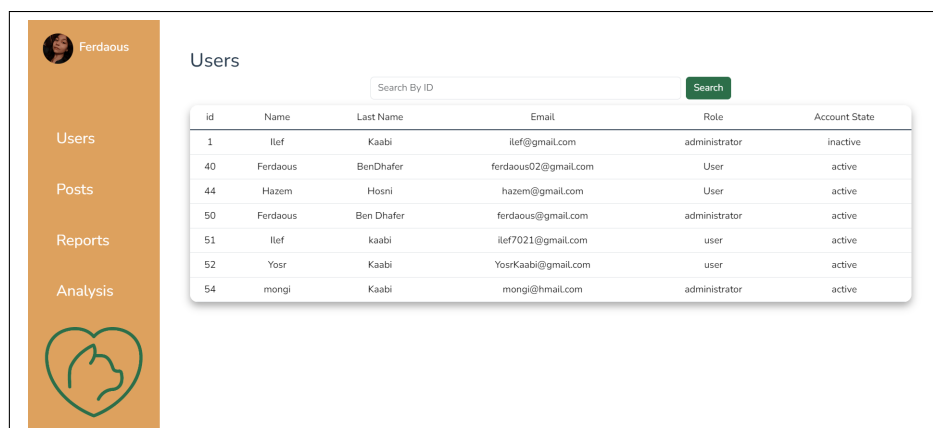


id	Type	Reason	Description	Reported Id	Status
7	User	Fake information	That's a fake account	46	underReview
10	User	Spam	Not appropriate content	46	underReview
12	Post	Spam	fake documents	49	rejected
16	Post	other	Doesn't show enough details	51	validated
17	User	Spam	Not appropriate content	53	underReview
19	User	Spam	No content	51	underReview
20	User	Spam	I don't like it	51	underReview
21	Post	Fake information	This data is fake	1	underReview

FIGURE 5.20 – Report list interface

5.5.4 Users list interface

This figure 5.21 displays the users list shown in the admin dashboard. An admin can search for a user by ID by writing it and clicking on search. By clicking on the ID of one of the users, an interface with all their details will be shown.



id	Name	Last Name	Email	Role	Account State
1	Ilef	Kaabi	ilef@gmail.com	administrator	inactive
40	Ferdaous	BenDhafer	ferdaous02@gmail.com	User	active
44	Hazem	Hosni	hazem@gmail.com	User	active
50	Ferdaous	Ben Dhafer	ferdaous@gmail.com	administrator	active
51	Ilef	kaabi	ilef7021@gmail.com	user	active
52	Yosr	Kaabi	YosrKaabi@gmail.com	user	active
54	mongi	Kaabi	mongi@hotmail.com	administrator	active

FIGURE 5.21 – Users list interface

5.5.5 User details interface

As mentioned above, an admin can consult the user details as it is shown in the figure 5.22. An admin has access to change the user's role from user to admin and vice versa by the role button and to activate and inactivate the user's account by the account status button.

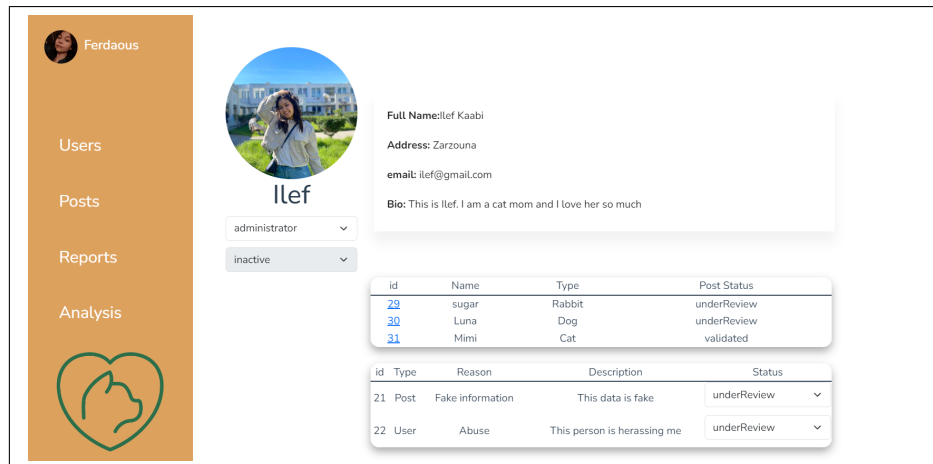


FIGURE 5.22 – User details interface

5.6 Conclusion

In this chapter, we presented the forth sprint our project which is admin management. We reviewed the sprint backlog and detailed the conception and implementation of the identified use case. In the following chapter, we will discuss the fifth and final sprint.

5.7 Burn down chart of sprint 4

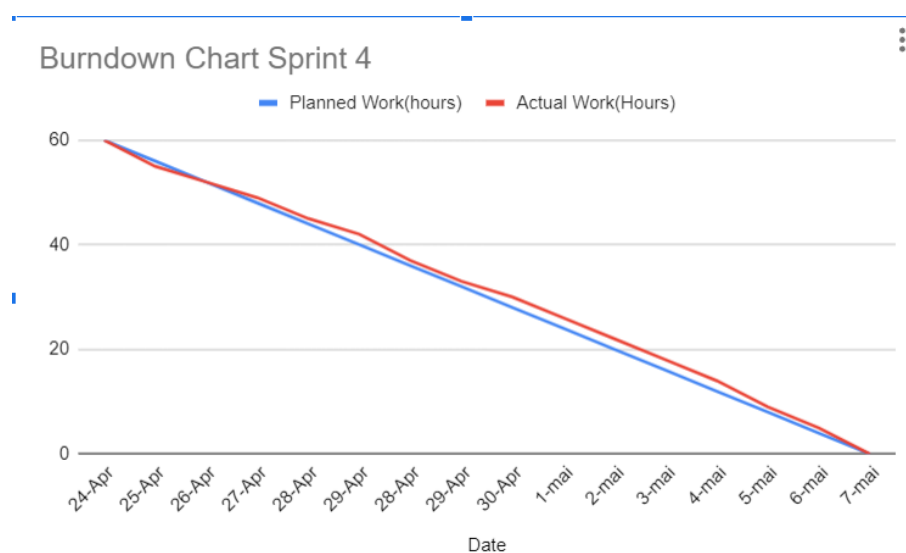


FIGURE 5.23 – Burn down chart of sprint 4

Chapter 6

SPRINT 5 : REAL TIME CHAT

6.1 Introduction

This chapter's objective is to present the fifth and final sprint that is real time chat. We will be representing the backlog of the sprint, the desired design and the conception, and finish with presenting the final interfaces for the chat.

6.2 Backlog of sprint 5

Id	User Story	Complexity
1	As a user I can send messages	5
2	As a user I can receive messages	5
3	As a user I can consult my messages	5

TABLE 6.1 – Sprint 4 backlog

6.3 Interface design : Modeling

6.3.1 User story : Chat widget

Once a user click on "Adopt Now", widget will appear and they would be able to send the owner a message as it is shown in the figure 6.1.

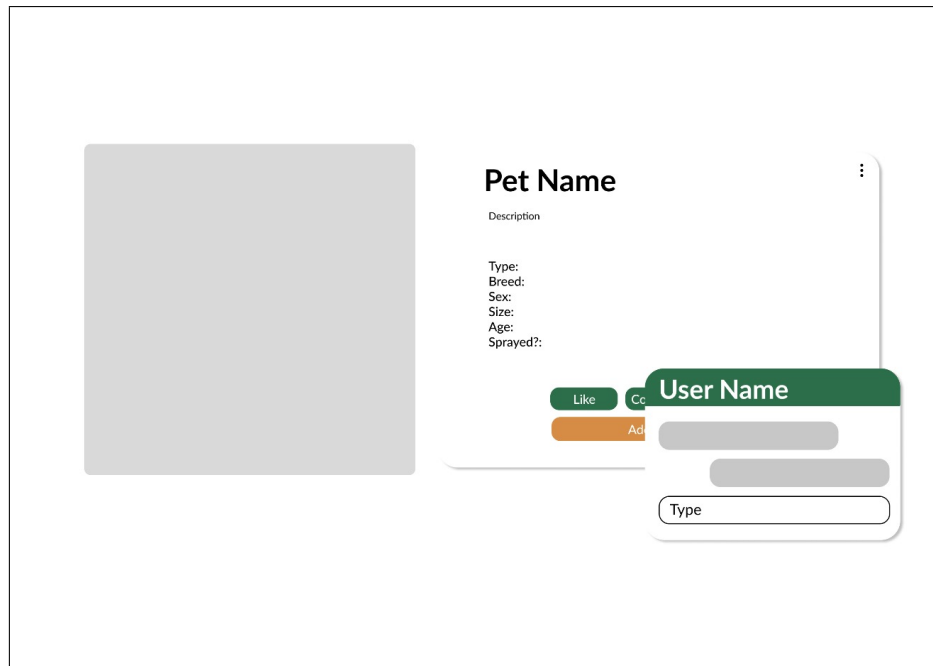


FIGURE 6.1 – Chat widget

6.3.2 User story : Consult my messages

The chats created by users are always stored in the data base, so the user can refer to any time he want as showed in the figure 6.2.

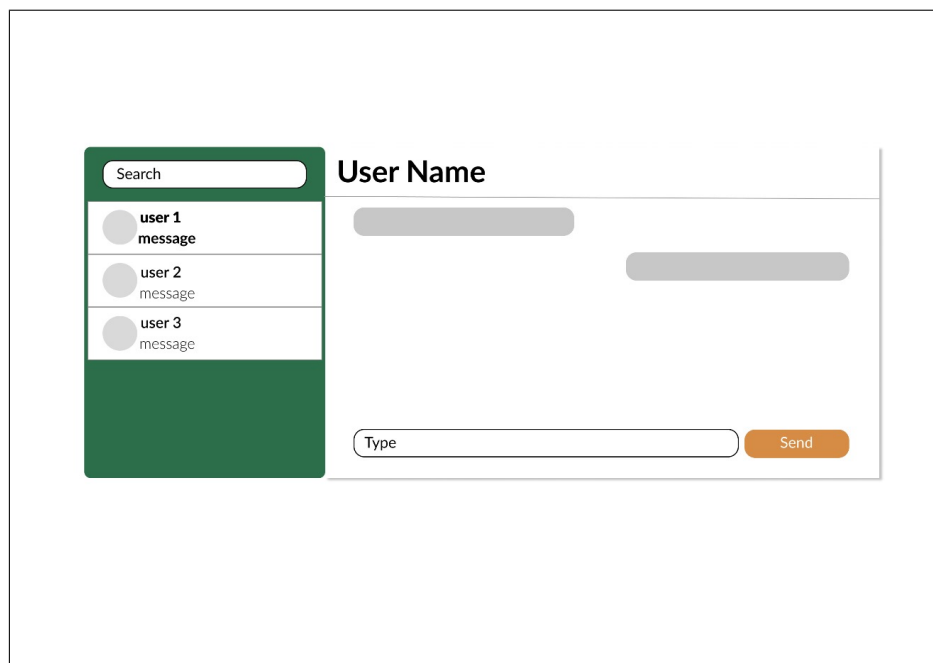


FIGURE 6.2 – Chat page

6.4 Conception of sprint 5 : Real time chat

In this part we will show the working context of real time chat implemented by nest vue js .

6.4.1 Web socket implementation

Web socket definition

The WebSocket protocol is a TCP-based network protocol. It defines how data is exchanged between networks. Because it is very reliable and efficient, it is used by almost all clients. TCP establishes communication between two **endpoints**, which are referred to as **sockets**. This allows data to be transferred in both directions. [<https://www.ionos.com/digitalguides/development/what-is-websocket/>]

Overall the websocket is used to handle connection between two or many endpoints to ensure an efficient connection.

6.4.2 Event Handler

After Establishing the connection the rooms creation are started to be implemented. the a room contain to members the one who created the room and the desired user to communicate with.

Once all the data is retrieved the previous chats are here and the web socket is ready to emit or receive new messages.

6.4.3 Object sequence diagram : Real time chat

The figure 6.3 present the object sequence diagram of a real time chat.

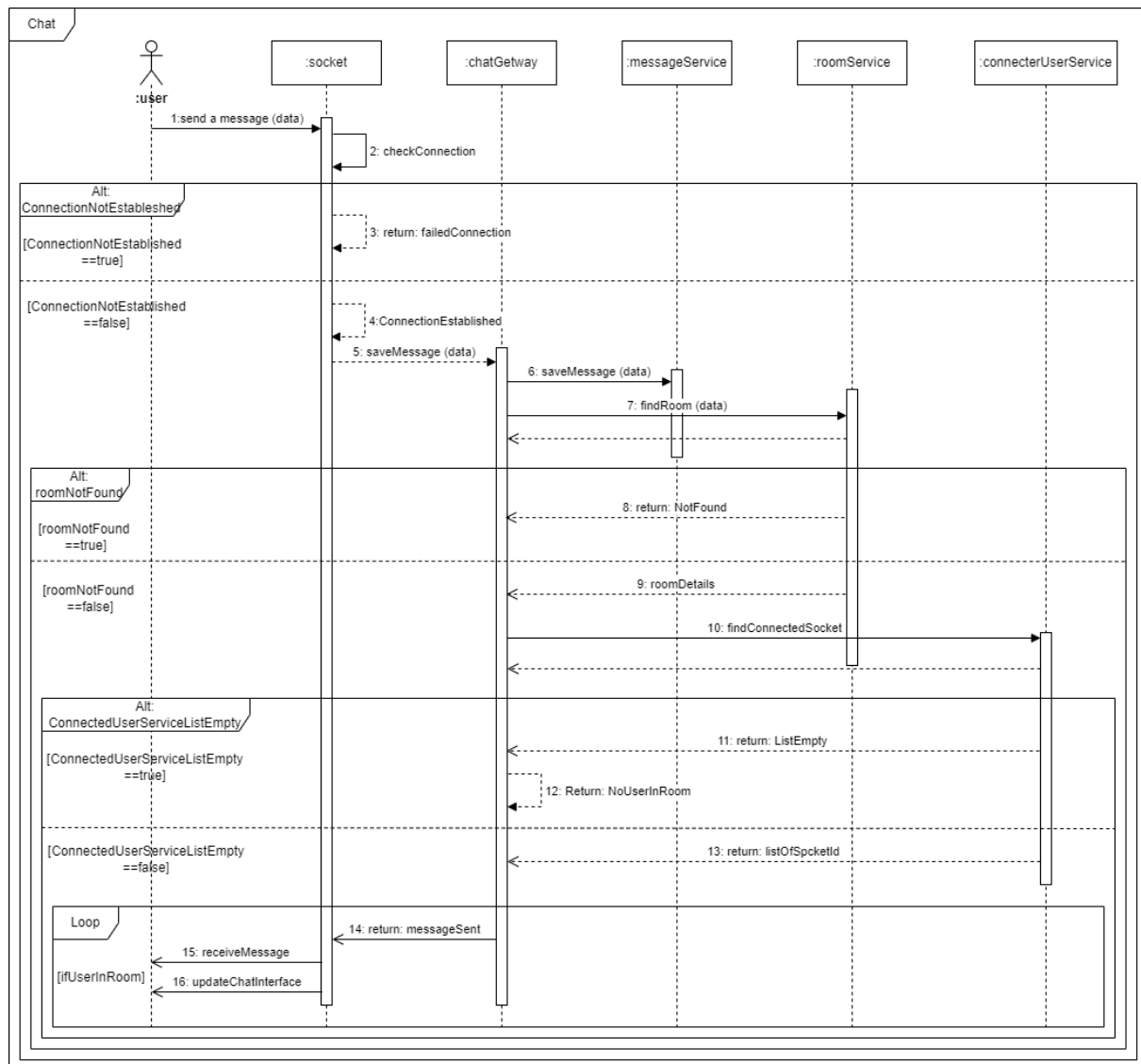


FIGURE 6.3 – object sequence diagram of a real time chat

6.5 Realisation

In this section, we will showcase the work completed during this sprint, including screenshots of the finished interfaces.

6.5.1 Chat Widget interface

As soon as the user clicks on "Adopt now" a widget pops up in the bottom right of the page to open a conversation with the owner as it is shown in the figure 6.4.

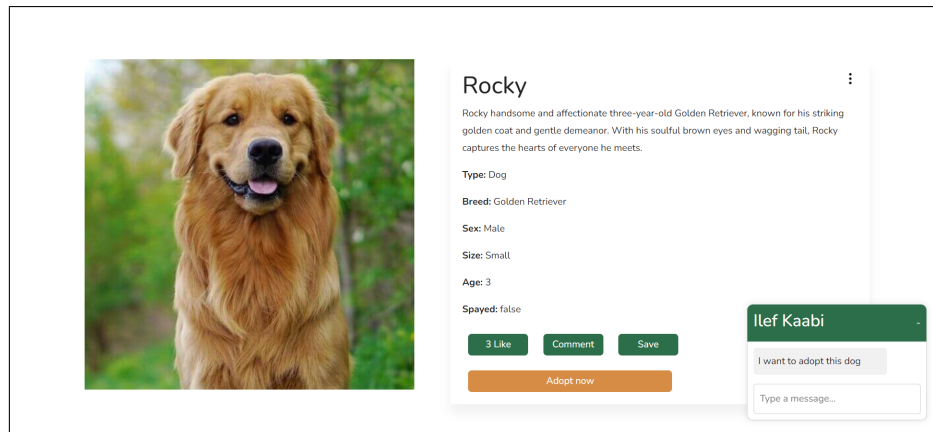


FIGURE 6.4 – Chat widget interface

6.5.2 Chat page interface

The figure 6.5 represent the chat page interface where the user can consult their messages, send and receive messages.

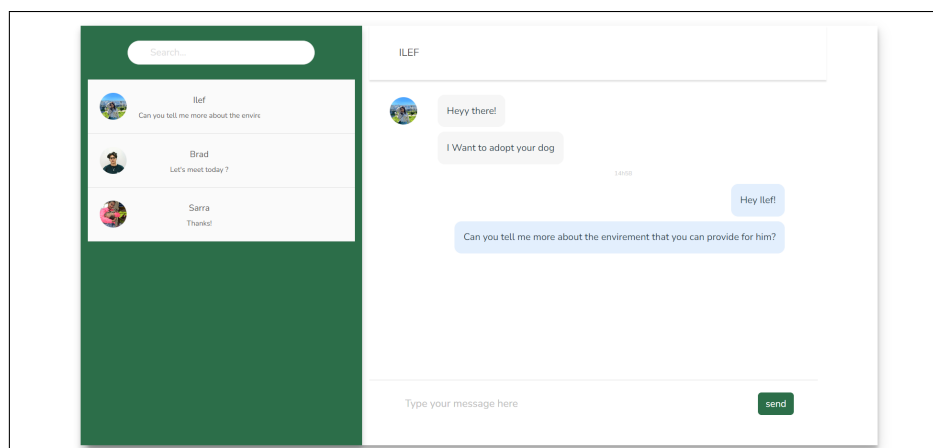


FIGURE 6.5 – Chat page interface

6.6 Conclusion

In the last chapter we give our attention to the real time chat implementation in the project, So by this final functionality the users can get into a direct communication with the pet owner to adopt their pet.

In this sprint we presented the sprint backlog, we got to know a new module in our application which is the websocket and how to implement it, the conception and implementation of the object sequence diagram of the chat and finally the realisation of a real time chat.

6.7 Burn down chart of sprint 5

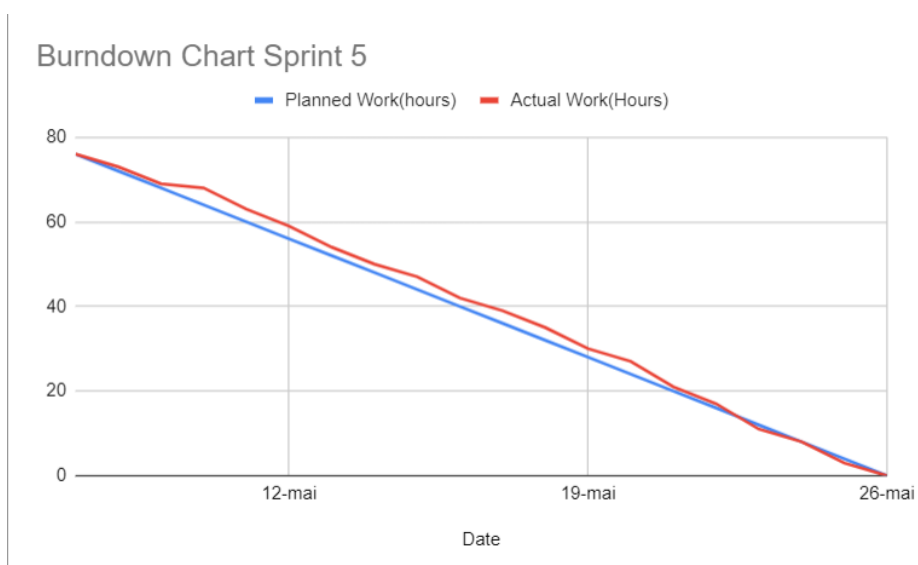


FIGURE 6.6 – Burn down chart of sprint 5

General conclusion

In this report, we discussed the different phases that we went through to realize our project with our company INTUITIV INTELECTIV group. In this project, we managed to create and develop a platform for pet adoption which links between pet owners that want to rehome their pets and adopters in safe, secure space. Our website gives the user the space to post their pets, interact with other posts by liking, commenting and saving them, besides the ability to follow other pet owners and communicate with them.

The first step was the identification of the different functional, non-functional and technical requirements of our project. Next, we developed the product backlog which allowed the construction of the sprint backlogs. Once the planification phase ended, we started realizing sprint by sprint. For each sprint, we prepared a detailed conception as we created sequence diagrams and object diagrams, and interface modeling using Figma which were realized lately using VueJs, Nest Js and MySQL.

During this experience, we had the chance to develop and enrich our knowledge from languages to business logic skills. Teamwork taught us how to manage and optimize our tasks to ensure smooth running of projects and professional work environment.

From our perspective, as much as Adoptini is rich with diverse special functionalities such as the document verification by the admin to ensure security and the real time chat, yet it can be developed more. We can improve the verification and use AI to generate the documents automatically. Moreover, adding an e-shop to the platform that specializes in selling pet stuff such as food, toys, accessories.. For the admin part, In addition, we plan to implement detailed statistical graphs to provide in-depth visualization of expenses related to posts interactions and number of pets successfully adopted.

Finally, we hope that this project will meet up with the expectations of both society and the jury members.

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Abstract

This project is realized as a part of an internship with INTUITIV INTERACTIV group. It consists of developing a platform which links pet owners to find a new home to their pets and adopters to find a new friend taking into consideration security and safety of all its transactions.

Key words : pet adoption, VueJs, NestJs, MySQL

7.1 Résumé

Ce projet est réalisé dans le cadre d'un stage au sein du groupe INTUITIV INTERACTIV. Il consiste à développer une plateforme qui aide les propriétaires d'animaux à trouver un nouveau maison pour leurs animaux et les adoptants à trouver un nouvel ami en tenant compte de la sécurité et de la sûreté de toutes leurs transactions.

Mots clés : adoption de pets, VueJs, NestJs, MySQL